

Quantum Entanglement

Information in the Whole, and What We Can Do with It

Ayan Mukhopadhyay

Lecture notes: from global quantum information to teleportation, networks, and cryptography

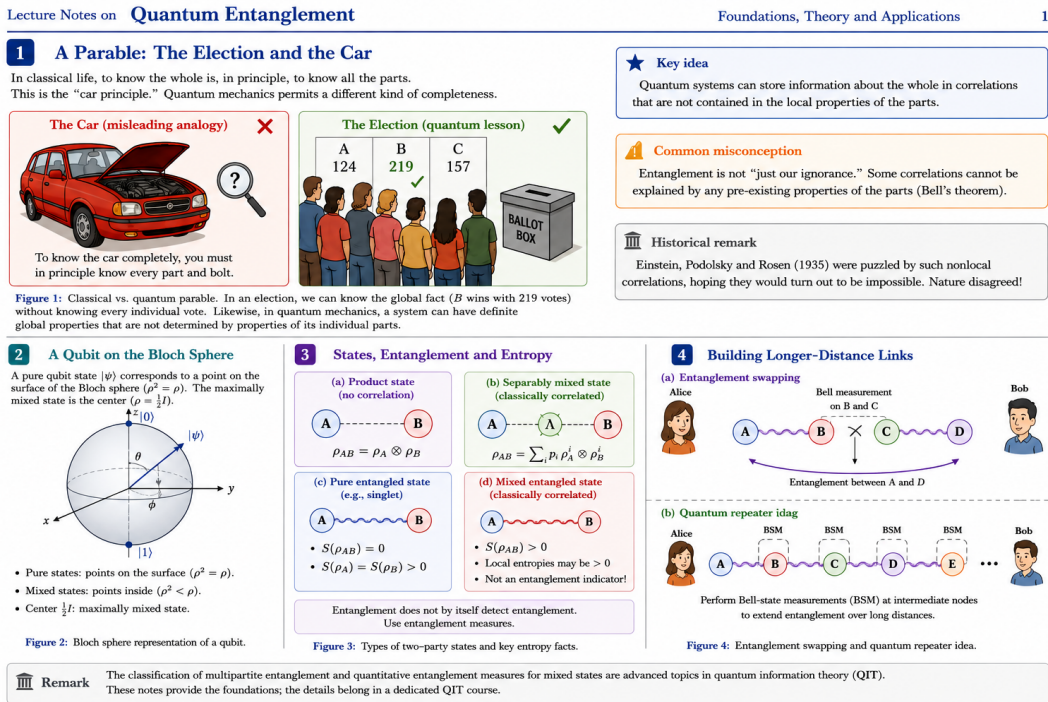


Figure 1: Conceptual Summary of the notes

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1 A Parable: Knowing the Whole Without Knowing the Parts

The phenomenon that we shall later define formally as quantum entanglement begins with a statement that sounds almost impossible: we may know the state of a complete system perfectly, while being unable to assign a complete pure state to each of its parts separately.

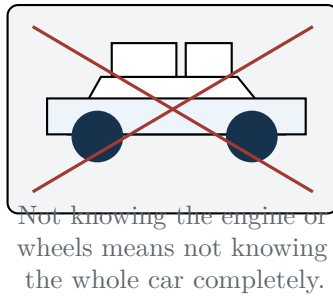
How can knowledge of the whole be complete if knowledge of the parts is not? Before introducing the mathematical language needed to answer this question, let us begin with a parable. The contrast is summarized playfully in figure 2: the car picture fails because missing information about a component is simply missing information about the car, whereas the election picture allows a definite collective fact to coexist with ignorance about individual contributions.

1.1 The car analogy—and why it fails

One might try to explain entanglement by saying:

“I know everything about the car, but I do not know everything about its wheels, its seats, or its engine.”

The car analogy fails



The election parable helps

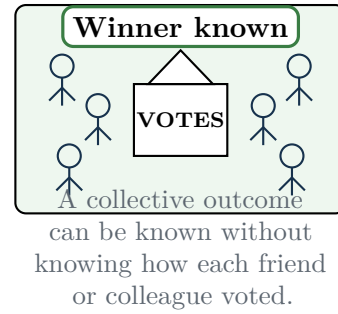


Figure 2: Two ways of trying to picture “knowledge of the whole without knowledge of every part.” The car analogy is misleading; the election parable captures global certainty together with local uncertainty, though it remains classical.

This makes no sense as an account of complete knowledge. A car is built from parts, each of which has its own properties. If we do not know which engine the car has, whether a wheel is damaged, or how the seats are arranged, then we do not know everything about the car.

The car analogy therefore fails. It treats the whole as nothing more than a collection of independently existing parts. Any missing information about a part is simply missing information about the whole.

Entanglement is interesting precisely because quantum systems need not be like this. Some information may belong to the relation among the parts and not to any part considered by itself.

1.2 A national election

Imagine that a national election is taking place. All eligible citizens and permanent residents cast their votes. Suppose that you know with certainty which candidate will win after all the votes are counted.

At the same time, you do not know how your friends voted. You do not know how your colleagues voted. You may not know how any particular individual voted. Nevertheless, you know the collective result.

Here the information you possess concerns the population as a whole. It does not tell you the contribution made by each individual voter.

This is much closer to the strange logic of quantum entanglement:

Key idea

The whole can possess a definite property even when the corresponding properties of its individual parts are not definite for us.

The election result is a global fact. It is determined by the votes taken together. Knowledge of that global fact does not require knowledge of every individual vote.

This is the first idea that we want to retain: information about a composite system need not be distributed among its parts in an obvious way. Some of it may be visible only when the parts are considered together.

1.3 Why this is still only a parable

An ordinary election is, of course, classical. Every person has cast a definite vote. Even if we do not know the vote, it already exists as an individual fact. Our uncertainty is merely ignorance.

Quantum entanglement goes further. In the quantum states that motivate the notion of entanglement, it is generally not correct to imagine that every subsystem carries a complete set of hidden, predetermined answers and that we simply do not know them. What can be definite is the relation between the outcomes, even when the individual outcomes are not definite in advance.

The election story therefore captures only the first surprise:

We may know something about the whole with certainty while being unable to predict what we will find in each part.

In section 4 we shall return to this question and make the statement experimentally precise: Bell's theorem will show why the quantum correlations cannot be understood simply as an election whose ballots have already been filled in and hidden from us, provided the hidden description is required to be local.

1.4 The simplest quantum example

Consider two spin- $\frac{1}{2}$ particles, labelled A and B , prepared in the state

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle_A |\downarrow\rangle_B - |\downarrow\rangle_A |\uparrow\rangle_B). \quad (1)$$

For the moment, we need not analyse this formula in detail. We only need to read what it says qualitatively.

There are two alternatives in the state. In one, particle A is found with spin up and particle B with spin down. In the other, particle A is found with spin down and particle B with spin up. Quantum mechanics combines these two alternatives into one state of the pair.

If the spins of both particles are measured along the same direction, the two answers are always opposite. If A is found to be up, B is found to be down; if A is found to be down, B is found to be up.

Thus the relation between the outcomes is perfectly predictable. The individual outcome is not. Before the measurement, we cannot say with certainty whether particle A will give “up” or “down.” The same is true of particle B .

We therefore know the pair in a way in which we do not know either particle separately. The certainty resides in the correlation:

The two answers will be opposite.

This statement belongs to the pair. It is not a property that can be assigned to particle A alone or to particle B alone.

1.5 Information in the relation

The central message of the parable can now be stated more carefully:

In the phenomenon we are trying to understand, some physical information is contained in the relations among the parts and cannot be attributed to any one part by itself.

This does not mean that the individual parts contain no information at all. It means that the information contained in the complete state is not exhausted by listing separate properties of the individual parts.

For an ordinary car, we expect complete knowledge of the whole to be assembled from complete knowledge of the engine, the wheels, the seats, and their arrangement. For the two

quantum particles in Eq. (1), this way of thinking fails. The pair has a sharply defined collective structure even though neither particle has a sharply defined individual spin outcome.

The election parable captures the idea of global certainty together with local uncertainty. The quantum state adds the crucial new element: the global correlation is not merely a summary of definite but unknown local facts.

1.6 The question that will guide us

We have not yet explained how quantum mechanics describes this situation. We have only identified the puzzle:

How can a complete system be in a perfectly definite state while its parts, considered separately, are not?

To answer this, we shall not begin by postulating a new kind of state. We shall start from the familiar description of the complete system by a pure state vector. We will then ask a concrete operational question: what probabilities can an observer obtain by measuring only one part of the system? The need to sum over the unobserved part will lead us naturally to the partial trace. Only then will the reduced density matrix emerge as the object that contains all information accessible through measurements on the chosen subsystem.

For now, the essential idea is simple:

Entanglement is the possibility that the most precise information available about a system is information about the whole and about the relations among its parts, rather than a collection of separate facts about each part.

2 From the Pure State of the Whole to the State of a Part

The parable has left us with a precise question. Suppose that the composite system AB is in a pure state that we know completely. What should we use to predict the results of experiments performed only on subsystem A ?

We shall answer this without assuming the density-matrix formalism. We begin only with state vectors, the Born rule, and the standard description of a composite system by a tensor-product Hilbert space.

2.1 A pure state of a composite system

Let $\mathcal{H}_A$ and $\mathcal{H}_B$ be the Hilbert spaces of the two subsystems. The Hilbert space of the complete system is

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B. \quad (2)$$

Choose orthonormal bases $\{|a_i\rangle\}$ for A and $\{|b_\mu\rangle\}$ for B . A completely general pure state of the joint system can be written as

$$|\Psi\rangle_{AB} = \sum_{i,\mu} c_{i\mu} |a_i\rangle_A |b_\mu\rangle_B, \quad \sum_{i,\mu} |c_{i\mu}|^2 = 1. \quad (3)$$

The coefficients $c_{i\mu}$ contain the complete information about the pure state of the whole system.

Suppose that both subsystems are measured in the chosen bases. The Born rule gives the joint probability

$$p(i, \mu) = |c_{i\mu}|^2. \quad (4)$$

Now imagine that we measure only A . We record the result a_i but do not measure B , or we simply ignore its result. The outcome a_i can then occur with any of the mutually exclusive outcomes b_μ on B . Therefore

$$p_A(i) = \sum_{\mu} p(i, \mu) = \sum_{\mu} |c_{i\mu}|^2. \quad (5)$$

This is the first appearance of the basic idea: to describe A alone, we must sum over the alternatives belonging to the part that we do not observe.

Equation (5), however, gives the probabilities only for one particular measurement basis. We want a single mathematical object that predicts the result of every possible measurement on A . To construct it, we first introduce an operation on operators called the partial trace.

2.2 The partial trace

For an ordinary operator X_A acting on $\mathcal{H}_A$, its trace is the sum of its diagonal matrix elements in any orthonormal basis:

$$\text{Tr}_A X_A = \sum_i \langle a_i | X_A | a_i \rangle. \quad (6)$$

The partial trace is a closely related operation. It takes an operator on the joint system AB and traces over only one of the two subsystems.

For any operator X_{AB} on $\mathcal{H}_A \otimes \mathcal{H}_B$, we define the partial trace over B by

$$\text{Tr}_B X_{AB} = \sum_{\mu} (I_A \otimes \langle b_{\mu} |) X_{AB} (I_A \otimes | b_{\mu} \rangle). \quad (7)$$

The result is an operator acting only on $\mathcal{H}_A$.

The notation in Eq. (7) has a simple meaning. We insert the same state $|b_{\mu}\rangle$ on the right and $\langle b_{\mu}|$ on the left, and then sum over a complete basis of the subsystem that is being ignored. Thus the partial trace is the operator version of summing over all unobserved alternatives.

Its action is especially transparent on an elementary product operator:

$$\begin{aligned} \text{Tr}_B (|a_i\rangle\langle a_k| \otimes |b_{\mu}\rangle\langle b_{\nu}|) \\ = \langle b_{\nu} | b_{\mu} \rangle |a_i\rangle\langle a_k| = \delta_{\mu\nu} |a_i\rangle\langle a_k|. \end{aligned} \quad (8)$$

The partial trace keeps the terms that are diagonal in the unobserved subsystem and sums them. Terms connecting two orthogonal states of B vanish. Although we used a basis to write Eq. (7), the result does not depend on which orthonormal basis of B is chosen.

Checkpoint: trace before reading on

Consider

$$|\Omega\rangle_{AB} = \frac{|00\rangle + |01\rangle}{\sqrt{2}}.$$

Construct $|\Omega\rangle\langle\Omega|$ and trace over B using Eq. (8). What is ρ_A ? Only afterward notice that the state can also be written as $|0\rangle_A \otimes |+\rangle_B$. The two routes must of course give the same answer.

The defining operational property of the partial trace is

$$\text{Tr}_{AB} [X_{AB}(O_A \otimes I_B)] = \text{Tr}_A [(\text{Tr}_B X_{AB})O_A] \quad (9)$$

for every operator O_A acting only on A . In words, if the other operator acts only on A , we may first trace out B and then perform the remaining trace on A . This identity is the reason the partial trace is exactly the operation needed for local measurements.

2.3 The reduced density matrix emerges

Return now to the pure state $|\Psi\rangle_{AB}$. Associated with this state is the projector

$$\Pi_\Psi = |\Psi\rangle\langle\Psi|. \quad (10)$$

At this stage, Π_Ψ is simply a convenient operator constructed from the state vector. For any operator X_{AB} ,

$$\langle\Psi|X_{AB}|\Psi\rangle = \text{Tr}_{AB}(\Pi_\Psi X_{AB}). \quad (11)$$

We now trace the projector over subsystem B and define

$$\boxed{\rho_A \equiv \text{Tr}_B |\Psi\rangle\langle\Psi|} \quad (12)$$

This operator is called the *reduced density matrix* of subsystem A . It has not been introduced as an independent postulate. It has emerged from a pure state of the whole system by applying the operation that discards the unobserved subsystem.

Using Eq. (3), we find

$$|\Psi\rangle\langle\Psi| = \sum_{i,k} \sum_{\mu,\nu} c_{i\mu} c_{k\nu}^* |a_i\rangle\langle a_k| \otimes |b_\mu\rangle\langle b_\nu|, \quad (13)$$

$$\rho_A = \sum_{i,k} \left(\sum_{\mu} c_{i\mu} c_{k\mu}^* \right) |a_i\rangle\langle a_k|. \quad (14)$$

Thus the matrix elements of the reduced density matrix are

$$(\rho_A)_{ik} = \sum_{\mu} c_{i\mu} c_{k\mu}^*. \quad (15)$$

The diagonal element $(\rho_A)_{ii}$ is precisely the probability found in Eq. (5). The off-diagonal elements contain the coherence information needed to predict measurements in bases other than $\{|a_i\rangle\}$.

The operator ρ_A has three immediate properties:

- (i) it is Hermitian, $\rho_A^\dagger = \rho_A$;
- (ii) it has unit trace, $\text{Tr}_A \rho_A = 1$;
- (iii) it is positive, meaning $\langle\chi|\rho_A|\chi\rangle \geq 0$ for every $|\chi\rangle \in \mathcal{H}_A$.

For example, positivity follows directly from

$$\langle\chi|\rho_A|\chi\rangle = \sum_{\mu} \left| \sum_i \langle\chi|a_i\rangle c_{i\mu} \right|^2 \geq 0. \quad (16)$$

Later we shall use these three properties to define density matrices more generally, including situations in which the complete system itself is not known to be in a pure state.

2.4 Why the reduced density matrix contains all local measurement information

Let a projective measurement on subsystem A have possible outcomes labelled by r , with projectors P_r satisfying

$$P_r P_s = \delta_{rs} P_r, \quad \sum_r P_r = I_A. \quad (17)$$

When this measurement is performed on A while nothing is measured on B , the corresponding projector on the full Hilbert space is $P_r \otimes I_B$. The Born rule gives

$$p(r) = \langle \Psi | (P_r \otimes I_B) | \Psi \rangle \quad (18)$$

$$= \text{Tr}_{AB} [| \Psi \rangle \langle \Psi | (P_r \otimes I_B)] \quad (19)$$

$$= \boxed{\text{Tr}_A(\rho_A P_r)}. \quad (20)$$

In the last step we used the key property of the partial trace, Eq. (9).

Similarly, for any observable

$$O_A = \sum_r o_r P_r, \quad (21)$$

its expectation value is

$$\boxed{\langle O_A \rangle = \langle \Psi | (O_A \otimes I_B) | \Psi \rangle = \text{Tr}_A(\rho_A O_A)}. \quad (22)$$

Therefore ρ_A determines the probability distribution of every projective measurement on A . Since ordinary observables are measured through their spectral projectors, it determines every measurement statistic available to an observer confined to A .

For the present discussion, projective measurements are sufficient. Chapter 3 will define density matrices and entanglement using this familiar measurement language. Chapter 5 will then return to the measurement problem and prove the stronger statement for arbitrary generalized measurements.

The reduced density matrix contains all information required to predict the outcomes of every local projective measurement and ordinary observable introduced so far. Chapter 5 will show that the statement remains true for arbitrary generalized measurements. The reduced state does not, however, contain all information about the joint system or about correlations with the subsystem that has been traced out.

An important consequence is that if two different pure states of AB produce the same reduced density matrix ρ_A , no experiment performed only on A can distinguish those global states. They may nevertheless be distinguished by joint measurements involving both A and B .

2.5 Example 1: a product state

Consider

$$| \Psi \rangle_{AB} = | \psi \rangle_A \otimes | \phi \rangle_B, \quad | \psi \rangle_A = \alpha | 0 \rangle + \beta | 1 \rangle, \quad (23)$$

where $|\alpha|^2 + |\beta|^2 = 1$. Then

$$\rho_A = \text{Tr}_B (| \psi \rangle \langle \psi | \otimes | \phi \rangle \langle \phi |) \quad (24)$$

$$= | \psi \rangle \langle \psi | \text{Tr}_B (| \phi \rangle \langle \phi |) \quad (25)$$

$$= | \psi \rangle \langle \psi |. \quad (26)$$

Nothing is lost from the description of A : it still has its own pure state. In the basis $\{|0\rangle, |1\rangle\}$,

$$\rho_A = \begin{pmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{pmatrix}. \quad (27)$$

The diagonal entries give the probabilities of obtaining 0 or 1. The off-diagonal entries are essential for predicting measurements in another basis. For example, with

$$| \pm \rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}, \quad (28)$$

we obtain

$$p(\pm) = \langle \pm | \rho_A | \pm \rangle = \frac{1}{2} |\alpha \pm \beta|^2. \quad (29)$$

Thus a list of probabilities in only one basis would not be enough; the full matrix is required.

2.6 Example 2: the singlet state

Checkpoint: predict before calculating

The singlet has no preferred spatial direction. Before performing the partial trace explicitly, ask what one-qubit density matrix could possibly describe a subsystem if every spin direction is to be statistically equivalent. This symmetry argument should already suggest the answer that the calculation below will produce.

Return to the singlet state introduced in the parable,

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle_A |\downarrow\rangle_B - |\downarrow\rangle_A |\uparrow\rangle_B). \quad (30)$$

Its projector contains four terms:

$$\begin{aligned} |\Psi^-\rangle\langle\Psi^-| &= \frac{1}{2} (|\uparrow\downarrow\rangle\langle\uparrow\downarrow| + |\downarrow\uparrow\rangle\langle\downarrow\uparrow| \\ &\quad - |\uparrow\downarrow\rangle\langle\downarrow\uparrow| - |\downarrow\uparrow\rangle\langle\uparrow\downarrow|). \end{aligned} \quad (31)$$

When we trace over B , the two cross terms vanish because $\langle\uparrow|\downarrow\rangle = 0$. The remaining terms give

$$\rho_A = \frac{1}{2} (|\uparrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow|) = \frac{I_A}{2}. \quad (32)$$

Therefore a measurement of the spin of A along the z direction gives “up” and “down” with equal probabilities. In fact, because $\rho_A = I_A/2$, the same is true along every direction. If

$$P_{\pm}(\mathbf{n}) = \frac{1}{2} (I_A \pm \mathbf{n} \cdot \boldsymbol{\sigma}), \quad (33)$$

then

$$p_{\pm}(\mathbf{n}) = \text{Tr}_A \left[\frac{I_A}{2} P_{\pm}(\mathbf{n}) \right] = \frac{1}{2}. \quad (34)$$

The state of the pair is perfectly known and pure, but every spin measurement on either particle separately is maximally unpredictable. The definite information lies in the correlation between the two particles, not in the local outcomes.

Notice also that

$$\rho_A^2 = \frac{I_A}{4} \neq \rho_A. \quad (35)$$

Thus the reduced state is not represented by a one-dimensional projector and cannot be written as a pure state vector for A . In Chapter 3 we shall define a *mixed state* formally. The present example will then show that a subsystem can have a mixed reduced state even when the complete system is in a pure state.

2.7 Example 3: a continuous family of correlated pure states

Consider the family of pure states

$$|\Psi(p, \varphi)\rangle = \sqrt{p} |0\rangle_A |0\rangle_B + e^{i\varphi} \sqrt{1-p} |1\rangle_A |1\rangle_B, \quad 0 \leq p \leq 1. \quad (36)$$

Checkpoint: where can the phase go?

Before tracing over B , decide whether an observer restricted to subsystem A should be able to determine the phase φ . Then use the partial trace to test the prediction. If the phase disappears locally, ask what kind of joint observable might still reveal it.

Tracing over B gives

$$\rho_A = p|0\rangle\langle 0| + (1-p)|1\rangle\langle 1|. \quad (37)$$

A measurement in the $\{|0\rangle, |1\rangle\}$ basis gives probabilities p and $1-p$. A measurement in the $\{|+\rangle, |-\rangle\}$ basis gives

$$p(+) = p(-) = \frac{1}{2}. \quad (38)$$

Both results follow from the same reduced density matrix.

The phase φ has disappeared from ρ_A . This does not mean that the phase was physically meaningless. It remains part of the joint state and can be detected through suitable measurements involving both particles. For example,

$$\langle \sigma_x \otimes \sigma_x \rangle = 2\sqrt{p(1-p)} \cos \varphi. \quad (39)$$

No measurement on A alone can reveal φ , because all values of φ give the same ρ_A . This illustrates precisely what is retained and what is discarded by the partial trace.

2.8 Example 4: a general two-qubit pure state

For

$$|\Psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle, \quad |a|^2 + |b|^2 + |c|^2 + |d|^2 = 1, \quad (40)$$

tracing over the second qubit gives

$$\rho_A = \begin{pmatrix} |a|^2 + |b|^2 & ac^* + bd^* \\ a^*c + b^*d & |c|^2 + |d|^2 \end{pmatrix}. \quad (41)$$

For example,

$$p_A(0) = |a|^2 + |b|^2, \quad (42)$$

$$p_A(1) = |c|^2 + |d|^2, \quad (43)$$

$$\langle \sigma_x \rangle_A = 2\operatorname{Re}(ac^* + bd^*), \quad (44)$$

$$\langle \sigma_z \rangle_A = |a|^2 + |b|^2 - |c|^2 - |d|^2. \quad (45)$$

The diagonal and off-diagonal entries together determine measurements in every direction. This is the concrete sense in which ρ_A contains the complete local measurement information.

A pure whole can have a mixed part

We began this chapter with the complete system AB in a pure state $|\Psi\rangle_{AB}$. One might therefore have expected each subsystem to have its own pure state vector. The examples above show that this expectation is wrong. A product state is special: then ρ_A is a one-dimensional projector. For an entangled pure state, however, the reduced state is generically not a projector at all. The singlet gives the sharpest example,

$$|\Psi^-\rangle_{AB} \langle \Psi^-| \quad \text{is pure,} \quad \rho_A = \frac{I_A}{2} \quad \text{is mixed.}$$

Thus mixed states are not introduced merely because an experimentalist may be ignorant about a preparation. They are forced upon us by the subsystem structure of quantum

mechanics itself: a perfectly known pure state of a larger system can give a mixed state for one of its parts. Chapter 3 will make this statement precise and the Schmidt decomposition will show that, for a pure bipartite state, local mixedness is exactly the signature of entanglement.

2.9 What has been achieved

The logical route of this section is important:

1. The complete system was assumed to be in an ordinary pure state $|\Psi\rangle_{AB}$.
2. To ask only about subsystem A , we had to sum over unobserved alternatives of B .
3. The partial trace was introduced as the operator operation that performs this elimination.
4. Applying it to the pure-state projector produced the reduced density matrix ρ_A .
5. The Born rule then showed that ρ_A predicts every local projective measurement and ordinary observable introduced so far.

We are now ready for the conceptual step that Chapter 2 has forced upon us. The complete system may be in a perfectly known pure state while a subsystem cannot be described by any state vector at all. To understand that possibility we must enlarge the notion of a quantum state.

3 Density Matrices, Entanglement, and Entropy

Chapter 2 began with the most complete description available in elementary quantum mechanics: a pure state $|\Psi\rangle_{AB}$ of the whole system. Yet when we asked only about one part, the answer was often not another state vector. For the singlet, for example,

$$|\Psi^-\rangle_{AB}\langle\Psi^-| \text{ is pure, } \quad \rho_A = \rho_B = \frac{I}{2} \text{ is mixed.} \quad (46)$$

This is the first and most important reason density matrices are unavoidable. Mixed states are not merely bookkeeping devices for an experimentalist who has forgotten how a system was prepared. They arise even when the state of a larger isolated system is known exactly: entanglement can make the whole pure while each part is mixed.

This chapter has one main purpose: to make that statement precise and thereby to make the meaning of entanglement precise. We shall first understand what pure and mixed states mean operationally, then distinguish classical mixtures from entanglement, derive the Schmidt decomposition rather than quote it, and finally connect the Schmidt coefficients to Shannon and von Neumann entropy. Throughout this chapter ordinary projective measurements are enough. The more general theory of measurement will be developed only after Bell's theorem, when we are ready to ask how a realistic apparatus can extract and manipulate quantum information.

3.1 Why mixed states are unavoidable

Measurement uncertainty is not the same as mixedness

A pure state can certainly give random measurement outcomes. Consider the qubit

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}. \quad (47)$$

A measurement in the $\{|0\rangle, |1\rangle\}$ basis gives the two outcomes with probability $1/2$. This randomness does not mean that the state is mixed. If instead we measure in the basis $\{|+\rangle, |-\rangle\}$, the result $+$ occurs with certainty. Equivalently, the rank-one projector

$$P_+ = |+\rangle\langle +| \quad (48)$$

obeys

$$\langle +|P_+|+\rangle = 1. \quad (49)$$

The uncertainty of a pure state is therefore *measurement dependent*. This is the familiar quantum uncertainty associated with measuring an observable for which the state is not an eigenstate; uncertainty relations quantify the corresponding spreads for incompatible observables. For every pure state $|\psi\rangle$ there is nevertheless a rank-one question, $P_\psi = |\psi\rangle\langle\psi|$, whose answer is certain.

The reduced state of one particle in the singlet behaves differently. Since

$$\rho_A = \frac{I}{2}, \quad (50)$$

for *every* normalized qubit state $|\phi\rangle$,

$$\text{Tr}(\rho_A |\phi\rangle\langle\phi|) = \frac{1}{2}. \quad (51)$$

No change of basis reveals a rank-one outcome that was secretly certain all along. There is an additional kind of uncertainty here: the local state itself is mixed. Later in this chapter we shall prove the general statement that a pure state always admits a rank-one outcome of probability one, whereas a genuinely mixed state admits none.

Two different sources of unpredictability

A pure state may give uncertain outcomes because we choose to measure an observable for which it is not an eigenstate. That uncertainty can be removed by choosing a measurement containing the state itself as an eigenvector. A mixed state contains a further, basis-independent lack of definiteness: even in its own eigenbasis several outcomes remain possible. For a reduced state of a pure entangled system, this local mixedness is produced by entanglement with the other subsystem.

A second route to mixed states: uncertainty in the preparation

Mixed states can also arise for a completely different reason. Suppose a preparation device produces the normalized state $|\psi_\alpha\rangle$ with probability p_α , where

$$p_\alpha \geq 0, \quad \sum_\alpha p_\alpha = 1. \quad (52)$$

The label α is not revealed to the observer. If a projective measurement has outcomes labelled by r and projectors P_r , the Born rule must be applied to each possible preparation and then averaged:

$$p(r) = \sum_\alpha p_\alpha \langle \psi_\alpha | P_r | \psi_\alpha \rangle. \quad (53)$$

Using

$$\langle \psi_\alpha | P_r | \psi_\alpha \rangle = \text{Tr}(|\psi_\alpha\rangle\langle\psi_\alpha| P_r), \quad (54)$$

we may rewrite Eq. (53) as

$$p(r) = \text{Tr}(\rho P_r), \quad \rho \equiv \sum_\alpha p_\alpha |\psi_\alpha\rangle\langle\psi_\alpha|. \quad (55)$$

The single operator ρ therefore contains all the projective-measurement statistics accessible to an observer who does not know α .

For example, a qubit prepared as $|0\rangle$ half the time and $|1\rangle$ half the time has

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{I}{2}. \quad (56)$$

Exactly the same one-qubit density matrix arose in Chapter 2 from one particle of a pure singlet. The two global stories are very different. In the first, the mixedness comes from classical uncertainty in the preparation; in the second, it comes from entanglement with another quantum system. Yet an observer with access only to the qubit sees the same density matrix and hence the same local measurement statistics. A central lesson of density-matrix quantum mechanics is that the local state records what can be predicted locally, not which global story produced it.

3.2 The density matrix

A *density matrix*, or *density operator*, on a Hilbert space $\mathcal{H}$ is an operator ρ satisfying

$$\boxed{\rho^\dagger = \rho, \quad \rho \geq 0, \quad \text{Tr } \rho = 1.} \quad (57)$$

Positivity means

$$\langle \chi | \rho | \chi \rangle \geq 0 \quad (58)$$

for every $|\chi\rangle \in \mathcal{H}$. It guarantees non-negative probabilities. Unit trace guarantees that the probabilities of a complete set of mutually exclusive outcomes sum to one.

For an ordinary observable O , its expectation value is

$$\boxed{\langle O \rangle = \text{Tr}(\rho O).} \quad (59)$$

For a projective measurement with projectors P_r , the probability of outcome r is therefore

$$p(r) = \text{Tr}(\rho P_r). \quad (60)$$

At this point it is useful to separate the operator ρ from a particular matrix representation. In an arbitrary orthonormal basis $\{|i\rangle\}$,

$$\rho_{ij} = \langle i | \rho | j \rangle. \quad (61)$$

A change of basis by a unitary matrix U changes the matrix entries according to

$$\rho' = U^\dagger \rho U, \quad (62)$$

but does not change the physical state. Because ρ is Hermitian, the spectral theorem guarantees an orthonormal eigenbasis $\{|n\rangle\}$ in which

$$\boxed{\rho = \sum_n \lambda_n |n\rangle\langle n|, \quad \lambda_n \geq 0, \quad \sum_n \lambda_n = 1.} \quad (63)$$

The eigenvalues of a density matrix therefore form an ordinary probability distribution.

A trace identity we shall use repeatedly

For finite-dimensional operators A and B ,

$$\boxed{\text{Tr}(AB) = \text{Tr}(BA)}. \quad (64)$$

To see this directly, choose any orthonormal basis $\{|i\rangle\}$:

$$\text{Tr}(AB) = \sum_i \langle i|AB|i\rangle \quad (65)$$

$$= \sum_{i,j} \langle i|A|j\rangle \langle j|B|i\rangle \quad (66)$$

$$= \sum_{j,i} \langle j|B|i\rangle \langle i|A|j\rangle \quad (67)$$

$$= \text{Tr}(BA). \quad (68)$$

Repeating the same argument gives the cyclic identity

$$\boxed{\text{Tr}(ABC) = \text{Tr}(BCA) = \text{Tr}(CAB)}. \quad (69)$$

The word *cyclic* matters: in general one may rotate the order of the factors, but one may not exchange arbitrary neighbouring factors.

Pure and mixed states: an operational distinction

A pure state $|\psi\rangle$ is represented by

$$\rho_\psi = |\psi\rangle\langle\psi|. \quad (70)$$

Its purity can be recognized directly by a measurement. If we test the rank-one projector

$$P_\psi = |\psi\rangle\langle\psi|, \quad (71)$$

then

$$\text{Tr}(\rho_\psi P_\psi) = 1. \quad (72)$$

Thus every pure state belongs with certainty to some one-dimensional subspace, even though measurements in other bases may have probabilistic outcomes.

Now consider a genuinely mixed state in its spectral basis,

$$\rho = \sum_n \lambda_n |n\rangle\langle n|, \quad (73)$$

with at least two non-zero eigenvalues. Let

$$\lambda_{\max} = \max_n \lambda_n. \quad (74)$$

Because at least two eigenvalues are non-zero and their sum is one, $\lambda_{\max} < 1$. For any normalized vector $|\phi\rangle$,

$$\text{Tr}(\rho|\phi\rangle\langle\phi|) = \langle\phi|\rho|\phi\rangle = \sum_n \lambda_n |\langle n|\phi\rangle|^2 \leq \lambda_{\max} \sum_n |\langle n|\phi\rangle|^2 = \lambda_{\max} < 1.$$

Therefore no rank-one projective outcome is certain for a genuinely mixed state. Even the eigenbasis of ρ , which is the basis best adapted to the state, produces the irreducible probability distribution $\{\lambda_n\}$. This is the precise sense in which mixedness contains an uncertainty beyond the basis-dependent randomness of measuring a pure state in an incompatible basis.

The same distinction has a compact algebraic form. A pure-state projector obeys

$$\rho_\psi^2 = \rho_\psi, \quad \text{Tr}(\rho_\psi^2) = 1. \quad (75)$$

For the general spectral form,

$$\text{Tr}(\rho^2) = \sum_n \lambda_n^2 \leq 1, \quad (76)$$

with equality only when one eigenvalue is one and all the others vanish. Thus, in finite dimensions,

$$\boxed{\rho \text{ pure} \iff \rho^2 = \rho \iff \text{Tr}(\rho^2) = 1.} \quad (77)$$

The quantity $\text{Tr}(\rho^2)$ is called the *purity*. Its name is now physically motivated: it detects whether there exists a one-dimensional state of the system that can be affirmed with certainty.

3.3 Reduced states for a general density matrix

We now return to a bipartite system with

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B. \quad (78)$$

The global state may itself be mixed and is described by ρ_{AB} . The reduced states are

$$\boxed{\rho_A = \text{Tr}_B \rho_{AB}, \quad \rho_B = \text{Tr}_A \rho_{AB}.} \quad (79)$$

Using an orthonormal basis $\{|b_\mu\rangle\}$ of $\mathcal{H}_B$,

$$\rho_A = \sum_\mu (I_A \otimes \langle b_\mu |) \rho_{AB} (I_A \otimes |b_\mu\rangle). \quad (80)$$

This is the same partial trace introduced in Chapter 2, now applied to an arbitrary density matrix.

Section 2 already established the operational meaning of ρ_A for local projective measurements. If Alice measures a projector P_r^A while Bob is ignored, then

$$p(r) = \text{Tr}_{AB} [\rho_{AB} (P_r^A \otimes I_B)] \quad (81)$$

$$= \boxed{\text{Tr}_A(\rho_A P_r^A)}. \quad (82)$$

Likewise, for any observable O_A acting only on Alice,

$$\langle O_A \rangle = \text{Tr}_{AB} [\rho_{AB} (O_A \otimes I_B)] = \text{Tr}_A(\rho_A O_A). \quad (83)$$

Thus ρ_A contains everything needed to predict every ordinary local observable on A . It need not contain the correlations between measurements on A and B ; those belong to the joint state ρ_{AB} .

For a product state,

$$\rho_{AB} = \rho_A \otimes \rho_B, \quad (84)$$

we recover

$$\text{Tr}_B(\rho_A \otimes \rho_B) = \rho_A \text{Tr} \rho_B = \rho_A. \quad (85)$$

By contrast, the classically correlated state

$$\rho_{AB}^{\text{cc}} = \frac{1}{2} |00\rangle\langle 00| + \frac{1}{2} |11\rangle\langle 11| \quad (86)$$

has

$$\rho_A = \rho_B = \frac{I}{2}. \quad (87)$$

Each subsystem is maximally mixed, yet measurements in the computational basis are perfectly correlated. Local mixedness and correlation therefore need not mean entanglement. We now make that distinction precise.

3.4 What entanglement means

Before stating the definition, figure 3 separates three notions that must not be confused: absence of correlation, classical correlation, and entanglement.

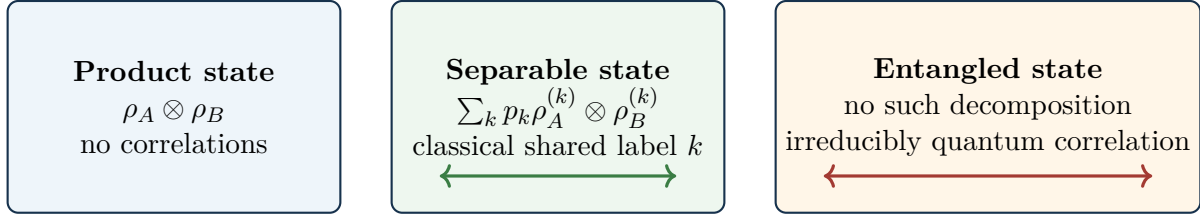


Figure 3: Product, separable, and entangled bipartite states. Every product state is separable, but a separable state may still contain classical correlations. Entanglement is the failure of every convex decomposition into product states.

3.4.1 Pure bipartite states

A pure state $|\Psi\rangle_{AB}$ is a *product state* if

$$|\Psi\rangle_{AB} = |\psi\rangle_A \otimes |\phi\rangle_B. \quad (88)$$

A pure state is *entangled* if no such factorization exists.

This definition is short, but by itself it does not yet tell us how much of the global state is genuinely shared between the two subsystems. The Schmidt decomposition supplies exactly that structure.

Deriving the Schmidt decomposition from the reduced state

The Schmidt decomposition is sufficiently important that we should not treat it as a formula that appears from nowhere. In fact, the reduced density matrix constructed in Chapter 2 almost gives the decomposition to us.

Let $|\Psi\rangle_{AB}$ be any pure bipartite state and diagonalize its reduced state on A :

$$\rho_A = \text{Tr}_B |\Psi\rangle\langle\Psi| = \sum_n \lambda_n |u_n\rangle\langle u_n|, \quad \lambda_n \geq 0. \quad (89)$$

We keep only the eigenvectors with $\lambda_n > 0$. For each of them define a vector in $\mathcal{H}_B$ by partially projecting the global state onto $|u_n\rangle_A$:

$$|v_n\rangle_B \equiv \frac{1}{\sqrt{\lambda_n}} {}_A\langle u_n | \Psi \rangle_{AB}. \quad (90)$$

Why is the normalization chosen this way? Because

$$\langle v_m | v_n \rangle = \frac{1}{\sqrt{\lambda_m \lambda_n}} \langle \Psi | (|u_m\rangle\langle u_n| \otimes I_B) | \Psi \rangle \quad (91)$$

$$= \frac{1}{\sqrt{\lambda_m \lambda_n}} \langle u_n | \rho_A | u_m \rangle \quad (92)$$

$$= \delta_{mn}. \quad (93)$$

Thus the vectors $\{|v_n\rangle_B\}$ are orthonormal automatically.

Now insert the identity on subsystem A into the global state. Components in the kernel of ρ_A give zero when projected onto $|\Psi\rangle$, so

$$|\Psi\rangle_{AB} = \sum_n |u_n\rangle_A {}_A\langle u_n| \Psi\rangle_{AB} \quad (94)$$

$$= \sum_n \sqrt{\lambda_n} |u_n\rangle_A |v_n\rangle_B. \quad (95)$$

This is the Schmidt decomposition. Nothing new had to be postulated: we simply diagonalized the local state ρ_A and used the global pure state to construct the matching vectors on B .

Tracing Eq. (95) over A gives

$$\rho_B = \sum_n \lambda_n |v_n\rangle\langle v_n|. \quad (96)$$

We have therefore proved at the same time that ρ_A and ρ_B possess exactly the same non-zero eigenvalues. This fact, which would otherwise look like an accidental symmetry of two partial traces, is built into the structure of every bipartite pure state.

Remark

In matrix language the same construction is the singular-value decomposition of the coefficient matrix $c_{i\mu}$ introduced in Chapter 2. The derivation above is often more transparent physically because it begins with the reduced density matrix, an object whose operational meaning we have already established.

If R of the Schmidt coefficients are non-zero, R is called the *Schmidt rank*. The reduced states are

$$\rho_A = \sum_{n=1}^R \lambda_n |u_n\rangle\langle u_n|, \quad \rho_B = \sum_{n=1}^R \lambda_n |v_n\rangle\langle v_n|, \quad (97)$$

and therefore

$$|\Psi\rangle_{AB} \text{ is a product state} \iff R = 1 \iff \rho_A \text{ is pure} \iff \rho_B \text{ is pure}, \quad (98)$$

$$|\Psi\rangle_{AB} \text{ is entangled} \iff R > 1 \iff \rho_A \text{ is mixed} \iff \rho_B \text{ is mixed}. \quad (99)$$

3.5 The Schmidt basis: a direct view of local uncertainty

The Schmidt decomposition gives a particularly simple operational picture of a pure entangled state. If Alice measures in the Schmidt basis $\{|u_n\rangle_A\}$, the probability of obtaining result n is

$$p_A(n) = \langle u_n | \rho_A | u_n \rangle = \lambda_n. \quad (100)$$

Bob obtains the same probability distribution in the matching Schmidt basis $\{|v_n\rangle_B\}$.

For a product state one coefficient equals one and the rest vanish. Alice's Schmidt-basis outcome is then certain. For an entangled state several λ_n are non-zero, and the state of either part is correspondingly less definite. At maximal entanglement in a $d \times d$ system,

$$\lambda_1 = \cdots = \lambda_d = \frac{1}{d}, \quad \rho_A = \rho_B = \frac{I}{d}, \quad (101)$$

so no Schmidt-basis result is preferred locally, even though the global state is perfectly pure and perfectly known.

More information in the whole, less definite information in the parts

For a bipartite pure state, entanglement and local mixedness are two sides of the same fact. The more broadly the Schmidt weights are distributed, the less predictable the individual subsystem is in its Schmidt basis. The missing local definiteness has not disappeared from quantum theory: it resides in the correlations of the global pure state.

We now need a quantitative measure of how spread out the probabilities $\{\lambda_n\}$ are. That question leads first to Shannon entropy and then, almost inevitably, to von Neumann entropy.

3.6 Shannon entropy: uncertainty of an ordinary probability distribution

For a classical probability distribution $\{p_i\}$, the *Shannon entropy* is

$$H(\{p_i\}) = - \sum_i p_i \log p_i. \quad (102)$$

We use the convention $0 \log 0 = 0$. With logarithms to base two, entropy is measured in bits.

The formula has the behaviour we want from a measure of uncertainty. If one outcome is certain, say $p_1 = 1$, then

$$H = 0. \quad (103)$$

If d outcomes are equally likely,

$$p_i = \frac{1}{d}, \quad (104)$$

then

$$H = \log d, \quad (105)$$

which is the largest possible value for d outcomes.

For a two-outcome distribution $\{p, 1 - p\}$,

$$H_2(p) = -p \log_2 p - (1 - p) \log_2 (1 - p). \quad (106)$$

It vanishes at $p = 0, 1$ and reaches one bit at $p = 1/2$. Thus Shannon entropy quantifies how uncertain we are about the result of a classical random variable.

For a bipartite pure state in Schmidt form, the local Schmidt-basis outcomes have probabilities $\{\lambda_n\}$. Their Shannon entropy is therefore

$$H(\{\lambda_n\}) = - \sum_n \lambda_n \log \lambda_n. \quad (107)$$

We now show that quantum mechanics contains exactly this quantity in a basis-independent form.

3.7 Von Neumann entropy: the basis-independent entropy of a quantum state

The *von Neumann entropy* of a density matrix is

$$S(\rho) = - \text{Tr}(\rho \log \rho). \quad (108)$$

To understand this expression, diagonalize ρ as in Eq. (63). Since

$$\log \rho = \sum_n (\log \lambda_n) |n\rangle \langle n|, \quad (109)$$

we obtain

$$S(\rho) = -\text{Tr} \left[\sum_n \lambda_n \log \lambda_n |n\rangle\langle n| \right] \quad (110)$$

$$= -\sum_n \lambda_n \log \lambda_n. \quad (111)$$

Thus

$$\boxed{S(\rho) = H(\{\lambda_n\})}, \quad (112)$$

where the λ_n are the eigenvalues of ρ .

This is the conceptual bridge between the two entropies. Shannon entropy measures uncertainty in an ordinary probability distribution. Von Neumann entropy applies the same function to the intrinsic spectrum of a quantum state, so the answer does not depend on which basis we use to write the matrix. The eigenbasis is especially revealing: it removes uncertainty that comes merely from choosing a measurement basis badly and exposes the probabilities $\{\lambda_n\}$ that remain intrinsic to the mixed state. A pure state has one eigenvalue equal to one, so this irreducible distribution is certain; a mixed state has several non-zero eigenvalues, so uncertainty remains even in its own eigenbasis.

A pure state has spectrum $\{1, 0, \dots\}$ and therefore

$$S(\rho_{\text{pure}}) = 0. \quad (113)$$

The maximally mixed state in d dimensions has spectrum $\{1/d, \dots, 1/d\}$ and therefore

$$S(I/d) = \log d. \quad (114)$$

In this sense von Neumann entropy quantifies the mixedness, or local uncertainty, of a quantum state. One must nevertheless be careful with the word “ignorance”: a mixed density matrix need not mean that the system is secretly in one definite pure state unknown to us. When the mixed state is a reduced state of an entangled pure system, the mixedness is produced by entanglement itself.

3.8 Entanglement entropy for a bipartite pure state

For a pure bipartite state, the Schmidt decomposition gives

$$\rho_A = \sum_n \lambda_n |u_n\rangle\langle u_n|, \quad \rho_B = \sum_n \lambda_n |v_n\rangle\langle v_n|. \quad (115)$$

The two reduced states therefore have the same non-zero spectrum, and hence

$$S(\rho_A) = S(\rho_B). \quad (116)$$

The *entanglement entropy* of the pure state is defined by

$$\boxed{S_{\text{ent}}(|\Psi\rangle_{AB}) = S(\rho_A) = S(\rho_B)}. \quad (117)$$

In Schmidt variables,

$$\boxed{S_{\text{ent}} = -\sum_n \lambda_n \log \lambda_n = H(\{\lambda_n\})}. \quad (118)$$

This equation admits an unusually transparent interpretation. The global state $|\Psi\rangle_{AB}$ is pure, so there is no uncertainty about the state of the whole. Nevertheless either subsystem is mixed. For a pure global state, that local mixedness has only one source: entanglement with the

other subsystem. The same set of numbers $\{\lambda_n\}$ therefore measures both how uncertain the local Schmidt-basis outcomes are and how entangled the two parts are.

For the family

$$|\Psi(p, \varphi)\rangle = \sqrt{p}|00\rangle + e^{i\varphi}\sqrt{1-p}|11\rangle, \quad (119)$$

the Schmidt weights are p and $1-p$, so

$$S_{\text{ent}} = -p \log_2 p - (1-p) \log_2 (1-p). \quad (120)$$

It vanishes at $p = 0$ and $p = 1$, where the state is a product, and reaches one bit at $p = 1/2$, where the two Schmidt outcomes are equally likely. The phase φ does not change the Schmidt weights and therefore does not change the entanglement entropy.

For the singlet,

$$\lambda_1 = \lambda_2 = \frac{1}{2}, \quad S_{\text{ent}} = 1 \text{ bit}. \quad (121)$$

The state of the pair is completely known and pure; the state of either particle is maximally mixed. This is perhaps the cleanest example of the principle

for a pure bipartite state:
 more entanglement $\iff$ greater local mixedness in the entropy sense

(122)

3.9 Mixed bipartite states: classical correlation is not entanglement

For a mixed state, failure to factorize as one product density matrix is not enough to establish entanglement. A state can fail to be a product while containing only classical correlations.

A bipartite density matrix is called *separable* if it can be written as

$$\rho_{AB} = \sum_k p_k \rho_A^{(k)} \otimes \rho_B^{(k)}, \quad p_k \geq 0, \quad \sum_k p_k = 1. \quad (123)$$

A mixed state is *entangled* if no such decomposition exists.

The meaning of Eq. (123) is operational. A separable state can be prepared by choosing a classical label k with probability p_k , preparing $\rho_A^{(k)}$ in laboratory A , and preparing $\rho_B^{(k)}$ in laboratory B . Shared randomness and classical communication are sufficient; no entangled resource is required.

The state in Eq. (86) is already displayed in separable form:

$$\rho_{AB}^{\text{cc}} = \frac{1}{2}(|0\rangle\langle 0|)_A \otimes (|0\rangle\langle 0|)_B + \frac{1}{2}(|1\rangle\langle 1|)_A \otimes (|1\rangle\langle 1|)_B. \quad (124)$$

It is strongly correlated but not entangled.

A useful contrasting family is the two-qubit Werner state

$$\rho_W(p) = p|\Psi^-\rangle\langle\Psi^-| + (1-p)\frac{I_{AB}}{4}, \quad 0 \leq p \leq 1. \quad (125)$$

It is separable for $0 \leq p \leq 1/3$ and entangled for $p > 1/3$. We quote this threshold as a useful fact; deriving practical mixed-state entanglement criteria belongs to a more advanced treatment.

3.10 Why reduced entropy is not an entanglement measure for mixed states

It is tempting to use $S(\rho_A)$ as an entanglement measure for every bipartite state. This is correct when ρ_{AB} is pure, but false in general, because local entropy does not know why the local state is mixed.

Consider the product state

$$\rho_{AB} = \frac{I_A}{2} \otimes |0\rangle_B \langle 0|. \quad (126)$$

It contains no correlation and no entanglement, yet

$$S(\rho_A) = 1 \text{ bit}. \quad (127)$$

The entropy comes entirely from the mixedness of subsystem A .

The classically correlated state

$$\rho_{AB}^{cc} = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11| \quad (128)$$

also satisfies

$$S(\rho_A) = S(\rho_B) = 1 \text{ bit}, \quad (129)$$

even though it is separable. The singlet has the same reduced density matrices and the same reduced entropies, but is entangled. Therefore

For a pure state of AB , all local mixedness is caused by entanglement. For a mixed state of AB , local mixedness can come from ordinary preparation uncertainty, classical correlation, quantum entanglement, or a combination of these.

Consequently,

$$\boxed{S(\rho_A) \text{ is an entanglement measure only when } \rho_{AB} \text{ is pure.}} \quad (130)$$

Mixed-state entanglement measures such as entanglement of formation, distillable entanglement, concurrence, and logarithmic negativity belong to a more advanced course.

3.11 A brief glimpse of multipartite entanglement

With three or more subsystems, entanglement is no longer characterized by a single bipartition or a single number. A pure tripartite state is fully separable if

$$|\Psi\rangle_{ABC} = |\psi\rangle_A |\phi\rangle_B |\chi\rangle_C, \quad (131)$$

and biseparable if, for example,

$$|\Psi\rangle_{ABC} = |\psi\rangle_A \otimes |\Phi\rangle_{BC}. \quad (132)$$

It has genuine tripartite entanglement if it is not separable across any bipartition.

Two famous examples are

$$|\text{GHZ}\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}}, \quad (133)$$

$$|W\rangle = \frac{|001\rangle + |010\rangle + |100\rangle}{\sqrt{3}}. \quad (134)$$

They distribute entanglement differently. If one qubit of the GHZ state is discarded, the remaining pair is separable,

$$\frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|, \quad (135)$$

whereas tracing one qubit from the W state leaves bipartite entanglement in the remaining pair. The subject becomes considerably richer for multipartite and mixed states; we mention it here only to prevent the misleading impression that a single reduced entropy classifies all forms of entanglement.

3.12 What we have learned, and the question Bell asks next

The conceptual chain is now short enough to see in one view:

1. A density matrix describes all ordinary measurement statistics of a system, whether its mixedness comes from classical preparation uncertainty or from being part of a larger quantum system.
2. A pure bipartite state is entangled exactly when its Schmidt rank is larger than one, equivalently when either reduced state is mixed.
3. The Schmidt weights $\{\lambda_n\}$ are ordinary probabilities for Schmidt-basis outcomes. Their Shannon entropy is exactly the von Neumann entropy of either reduced state.
4. For a globally pure bipartite state this local entropy is the entanglement entropy: greater entanglement, in this entropy measure, means less locally definite information and more information stored in the correlations of the whole.
5. For globally mixed states, local entropy alone cannot distinguish classical uncertainty from quantum entanglement; separability is the correct structural definition.

We now know what entanglement *is*. A deeper question remains. Could its correlations nevertheless be explained by some ordinary, locally carried information that quantum mechanics has omitted? Bell's theorem turns that question into an experimentally testable inequality.

4 Bell Correlations: Why Entanglement Is Not Classical Ignorance

The first chapter began with an election parable and then warned us not to take that parable too literally. In an ordinary election every voter has cast a definite ballot; our uncertainty about an individual vote is merely ignorance. For the singlet state, by contrast, we said that it is generally not legitimate to imagine that every possible spin measurement already has a hidden answer waiting to be uncovered.

Until now that distinction has remained a promise. The formal definition of entanglement in Chapter 3 tells us that the singlet is not a product state and cannot be generated as a classical mixture of product states. But a natural question remains:

Could quantum mechanics nevertheless be an incomplete statistical theory in which the particles carry additional hidden information that helps determine (or determine only probabilistically) the measurement outcomes, while the quantum state merely describes our ignorance of that information?

Bell's theorem turns this philosophical-looking question into an experimental one [1, 2]. It identifies a quantitative limit obeyed by a large class of classical explanations and shows that quantum mechanics can violate that limit.

The point of this chapter is not merely to derive the number $2\sqrt{2}$. We shall separate carefully four statements that are often run together:

1. entangled states need not admit a classical mixture of product states;
2. Bell correlations cannot be reproduced by a *local* hidden-variable model;
3. Bell nonlocality does not permit faster-than-light signalling;
4. quantum theory also exhibits a more general obstruction called *contextuality*, which does not fundamentally rely on spatial separation.

The first three belong to the main line of the course. The final topic, contextuality, is presented as an optional extension at the end of the chapter.

4.1 From ordinary common causes to a local hidden-variable model

Before introducing the symbol λ , it is useful to see why hidden variables are not an exotic idea. They are simply the familiar idea of an *unobserved common cause*.

A classical common cause: two market forecasts

Suppose Alice in Santiago and Bob in London are each asked one of two yes/no questions about tomorrow's markets. Alice's question is labelled $x \in \{0, 1\}$, Bob's is labelled $y \in \{0, 1\}$, and their answers are recorded as $a, b = \pm 1$. Their answers may be correlated even if Alice and Bob never communicate. Both may be reacting to the same underlying economic situation: interest rates, inflation news, oil prices, earnings announcements, risk sentiment, and many other variables.

Call the complete relevant market information λ . We may not know λ in detail, but there is nothing mysterious about supposing that it exists. It is simply a possible common cause of the two correlated answers. On different days the market state is different, so λ varies from one run of the experiment to another according to some probability distribution.

Bell asks whether the correlations of two quantum systems can be explained in this same classical way. Imagine a source that emits two systems, one toward Alice and one toward Bob. On each run Alice chooses one of two measurements, labelled

$$x \in \{0, 1\}, \quad (136)$$

and Bob independently chooses

$$y \in \{0, 1\}. \quad (137)$$

Their measurement outcomes are

$$a, b \in \{+1, -1\}. \quad (138)$$

Thus throughout this section

$$\boxed{x, y = \text{measurement choices}, \quad a, b = \text{measurement outcomes}.} \quad (139)$$

The experiment is summarized schematically in figure 4.

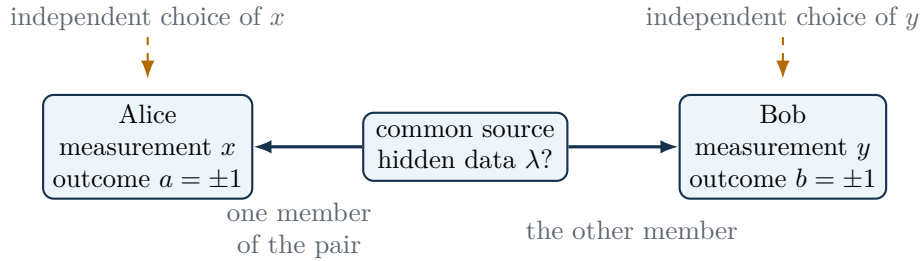


Figure 4: The Bell scenario. A common source may distribute shared information λ , but Alice and Bob choose their measurements independently. A local hidden-variable model permits the common cause λ but does not permit Alice's later measurement choice to enter Bob's local response, or vice versa.

A hidden-variable explanation supposes that each emitted pair possesses some additional information λ . The symbol may represent a number, a vector, microscopic properties of the two systems, a long instruction record, or any other classical data. Bell's argument places no restriction on how complicated λ may be.

It is important to understand what it means for Alice and Bob to have the *same* λ . It does not mean that the two particles must carry an identical physical number. Rather, λ denotes the complete common state of that particular pair. It could contain separate pieces,

$$\lambda = (\lambda_A, \lambda_B, \lambda_{\text{shared}}), \quad (140)$$

with different pieces relevant to the two laboratories. One symbol is used because all of this information belongs to the same preparation event in the common past of Alice and Bob.

Nor is λ the same on every run. The source may emit different hidden states on different runs. We describe this by an ordinary probability density

$$\mu(\lambda) \geq 0, \quad \int d\lambda \mu(\lambda) = 1. \quad (141)$$

Thus a particular run has one value of λ , shared as the complete preparation of that pair, while repeated runs sample different λ 's from $\mu(\lambda)$.

What makes the model local?

Suppose λ is a complete specification of all common causes established before the two measurements. Alice's local outcome may then depend on her own choice x and on λ , but not on Bob's independently chosen y . Likewise Bob's outcome may depend on y and λ , but not on x . Therefore the local response probabilities have the form

$$P_A(a|x, \lambda), \quad P_B(b|y, \lambda). \quad (142)$$

Once the complete common cause has been specified, there is by assumption no additional shared randomness left unaccounted for. The conditional joint probability therefore factorizes:

$$P(a, b|x, y, \lambda) = P_A(a|x, \lambda)P_B(b|y, \lambda). \quad (143)$$

If some further common random variable were still correlating the two sides, then λ was not complete; that variable should simply be included in an enlarged λ .

There is one further assumption. Alice and Bob choose their measurements independently of the hidden state emitted by the source. Equivalently,

$$\mu(\lambda|x, y) = \mu(\lambda). \quad (144)$$

This is usually called *measurement independence*. Averaging over the unknown hidden state gives the most general local hidden-variable model used in the Bell argument:

$$P(a, b|x, y) = \int d\lambda \mu(\lambda) P_A(a|x, \lambda) P_B(b|y, \lambda). \quad (145)$$

Equation (145) is the precise mathematical meaning of the classical local-common-cause explanation that CHSH will test.

What Bell locality allows and forbids

Bell locality does *not* say that Alice and Bob cannot be correlated. They may share arbitrarily complicated information prepared by the common source. Locality says that, once this common cause has been fully specified, Alice's response uses only (x, λ) and Bob's response uses only (y, λ) . The settings are also assumed to be chosen independently of λ .

4.2 Stochastic models and deterministic instruction tables

Nothing in Eq. (145) assumes determinism. Even when x and λ are fixed, Alice may still have, for example,

$$P_A(+1|x, \lambda) = 0.7, \quad P_A(-1|x, \lambda) = 0.3. \quad (146)$$

This is still an ordinary classical probability model. Classical does not mean deterministic: a classical coin toss is stochastic.

There is, however, a useful fact. Any such stochastic local model can be written as a probabilistic mixture of deterministic local response rules. Introduce an additional local random number r_A , uniformly distributed in $[0, 1]$, and define

$$A(x, \lambda, r_A) = \begin{cases} +1, & r_A \leq P_A(+1|x, \lambda), \\ -1, & r_A > P_A(+1|x, \lambda). \end{cases} \quad (147)$$

Averaging over r_A reproduces exactly Alice's original probabilities. Similarly introduce a local random number r_B for Bob. If we enlarge the hidden description to

$$\Lambda = (\lambda, r_A, r_B), \quad (148)$$

then, once Λ is specified, the two responses are deterministic:

$$a = A_x(\Lambda) = \pm 1, \quad b = B_y(\Lambda) = \pm 1. \quad (149)$$

Alice's response need not use r_B and Bob's need not use r_A ; including both in Λ is simply a convenient mathematical bookkeeping device. To an observer who does not know the extra random numbers, the model remains effectively stochastic.

Thus

every stochastic local model is a classical mixture of deterministic local instruction tables.

(150)

Consequently, if an inequality is obeyed by every deterministic instruction table, averaging over such tables cannot make a stochastic local model violate it.

A concrete local model that reaches the CHSH bound

Imagine that the source randomly chooses one of two instruction cards, each with probability $1/2$. A row tells Alice what to answer for either possible measurement x and tells Bob what to answer for either possible measurement y :

hidden state Λ	A_0	A_1	B_0	B_1
red card	+1	-1	+1	-1
blue card	-1	+1	-1	+1

The source sends Alice her two local instructions and Bob his two local instructions before either measurement is chosen. Alice later reads only the entry selected by x ; Bob reads only the entry selected by y . No communication is required.

Because the two cards are equally likely, every individual measurement result is random: for either setting Alice obtains $+1$ half the time and -1 half the time, and similarly for Bob. Nevertheless the outcomes are correlated. The four correlations are

$$E(0,0) = +1, \quad E(0,1) = -1, \quad E(1,0) = -1, \quad E(1,1) = +1. \quad (151)$$

For the CHSH combination

$$\mathcal{S} = E(0,0) + E(0,1) + E(1,0) - E(1,1), \quad (152)$$

this model gives

$$\mathcal{S} = 1 - 1 - 1 - 1 = -2, \quad |\mathcal{S}| = 2. \quad (153)$$

Thus random local outcomes and strong correlations are perfectly compatible with an ordinary local common-cause model. What Bell's theorem constrains is the *pattern of all four correlations together*.

4.3 Why the CHSH combination bounds every local hidden-variable model

Why choose precisely the combination in Eq. (152)? We want an experimentally measurable quantity that is constrained for *every* possible local instruction table, however the source chooses those tables. The plus-plus-plus-minus pattern has exactly this property.

Fix one enlarged hidden state Λ . Before averaging over anything, its four deterministic local responses are

$$A_0(\Lambda), \quad A_1(\Lambda), \quad B_0(\Lambda), \quad B_1(\Lambda), \quad \text{each} = \pm 1. \quad (154)$$

For this one instruction table form

$$S_\Lambda = A_0 B_0 + A_0 B_1 + A_1 B_0 - A_1 B_1 \quad (155)$$

$$= A_0(B_0 + B_1) + A_1(B_0 - B_1), \quad (156)$$

where the common argument Λ has been suppressed in the second line. Now B_0 and B_1 can only be ± 1 . There are only two cases:

1. If $B_0 = B_1$, then

$$B_0 + B_1 = \pm 2, \quad B_0 - B_1 = 0. \quad (157)$$

2. If $B_0 = -B_1$, then

$$B_0 + B_1 = 0, \quad B_0 - B_1 = \pm 2. \quad (158)$$

Exactly one bracket survives, and the multiplying A_x is itself ± 1 . Therefore every deterministic local instruction table obeys

$$\boxed{S_\Lambda = \pm 2.} \quad (159)$$

The experimentally observed correlations are averages over the hidden states. For deterministic responses,

$$E(x, y) = \int d\Lambda \nu(\Lambda) A_x(\Lambda) B_y(\Lambda), \quad (160)$$

where $\nu(\Lambda)$ denotes the probability distribution of the enlarged hidden variable. Hence

$$|\mathcal{S}| = \left| \int d\Lambda \nu(\Lambda) S_\Lambda \right| \quad (161)$$

$$\leq \int d\Lambda \nu(\Lambda) |S_\Lambda| \quad (162)$$

$$\leq 2 \int d\Lambda \nu(\Lambda) \quad (163)$$

$$= 2. \quad (164)$$

We have proved the CHSH inequality

$$\boxed{|\mathcal{S}| \leq 2.} \quad (165)$$

Because every stochastic local model can be represented as a mixture of the deterministic tables just considered, this bound applies to the full stochastic model in Eq. (145), not merely to models in which outcomes were predetermined from the outset.

What CHSH has proved

The logical statement is

$$\text{local hidden-variable model} \implies |\mathcal{S}| \leq 2.$$

The hidden state may be arbitrarily complicated, may vary randomly from run to run, and may generate outcomes stochastically. Therefore an observed value $|\mathcal{S}| > 2$ rules out the entire class of models described by Eq. (145), provided the measurement-independence assumption is maintained.

Checkpoint: where did locality enter?

The numerical step $S_\Lambda = \pm 2$ is elementary. The physical content came earlier: locality allowed one and the same local response $A_x(\Lambda)$ to be used irrespective of Bob's choice y , and one and the same $B_y(\Lambda)$ irrespective of Alice's choice x . Without that local structure

there is no single four-entry instruction table to which the CHSH algebra can be applied.

4.4 The singlet prediction from the reduced-state formalism

We now return to the state that has accompanied us since Chapter 1,

$$|\Psi^-\rangle = \frac{|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle}{\sqrt{2}}. \quad (166)$$

The physical spin component along a unit vector $\mathbf{a}$ is

$$S_{\mathbf{a}} = \frac{\hbar}{2} \mathbf{a} \cdot \boldsymbol{\sigma}. \quad (167)$$

Its eigenvalues are $\pm\hbar/2$. Dividing the recorded result by $\hbar/2$ gives the dimensionless dichotomic observable

$$A(\mathbf{a}) = \mathbf{a} \cdot \boldsymbol{\sigma}, \quad (168)$$

whose possible outcomes are ± 1 . Similarly, Bob measures

$$B(\mathbf{b}) = \mathbf{b} \cdot \boldsymbol{\sigma}. \quad (169)$$

Thus the choices $x = 0, 1$ in the Bell experiment will correspond to two vectors $\mathbf{a}_0, \mathbf{a}_1$ for Alice, and $y = 0, 1$ to two vectors $\mathbf{b}_0, \mathbf{b}_1$ for Bob.

For fixed directions $\mathbf{a}, \mathbf{b}$, the correlation is

$$E(\mathbf{a}, \mathbf{b}) = \langle \Psi^- | (\mathbf{a} \cdot \boldsymbol{\sigma}) \otimes (\mathbf{b} \cdot \boldsymbol{\sigma}) | \Psi^- \rangle. \quad (170)$$

There is a useful way to derive this correlation that reconnects directly with our earlier discussion of reduced states. The singlet has zero total spin, so for each Cartesian component $i = x, y, z$,

$$(\sigma_i \otimes I + I \otimes \sigma_i) |\Psi^-\rangle = 0. \quad (171)$$

Equivalently,

$$(\sigma_i \otimes I) |\Psi^-\rangle = -(I \otimes \sigma_i) |\Psi^-\rangle. \quad (172)$$

Using

$$\sigma_i \otimes \sigma_j = (\sigma_i \otimes I)(I \otimes \sigma_j), \quad (173)$$

we can now proceed one step at a time:

$$\langle \Psi^- | \sigma_i \otimes \sigma_j | \Psi^- \rangle = \langle \Psi^- | (\sigma_i \otimes I)(I \otimes \sigma_j) | \Psi^- \rangle \quad (174)$$

$$= -\langle \Psi^- | (I \otimes \sigma_i)(I \otimes \sigma_j) | \Psi^- \rangle \quad (175)$$

$$= -\langle \Psi^- | I \otimes (\sigma_i \sigma_j) | \Psi^- \rangle. \quad (176)$$

For any operator O_B acting only on Bob's subsystem, the definition of the reduced density matrix gives

$$\langle \Psi^- | I \otimes O_B | \Psi^- \rangle = \text{Tr}(\rho_B O_B), \quad \rho_B = \text{Tr}_A |\Psi^-\rangle \langle \Psi^-|. \quad (177)$$

Taking $O_B = \sigma_i \sigma_j$ therefore yields

$$\langle \Psi^- | \sigma_i \otimes \sigma_j | \Psi^- \rangle = -\text{Tr}(\rho_B \sigma_i \sigma_j). \quad (178)$$

Chapter 2 showed that the reduced state of either particle is maximally mixed,

$$\rho_B = \frac{I}{2}. \quad (179)$$

The Pauli matrices obey

$$\sigma_i \sigma_j = \delta_{ij} I + i \epsilon_{ijk} \sigma_k. \quad (180)$$

Since $\text{Tr } I = 2$ and $\text{Tr } \sigma_k = 0$,

$$\text{Tr}(\sigma_i \sigma_j) = 2 \delta_{ij}. \quad (181)$$

Hence

$$\langle \Psi^- | \sigma_i \otimes \sigma_j | \Psi^- \rangle = -\delta_{ij}. \quad (182)$$

Finally expand the two dot products in Eq. (170):

$$E(\mathbf{a}, \mathbf{b}) = \sum_{i,j} a_i b_j \langle \Psi^- | \sigma_i \otimes \sigma_j | \Psi^- \rangle \quad (183)$$

$$= - \sum_{i,j} a_i b_j \delta_{ij} \quad (184)$$

$$= - \sum_i a_i b_i \quad (185)$$

$$= -\mathbf{a} \cdot \mathbf{b}. \quad (186)$$

Therefore

$$E(\mathbf{a}, \mathbf{b}) = -\mathbf{a} \cdot \mathbf{b}. \quad (187)$$

This derivation is worth noticing. The maximally mixed local state did not mean that the pair possessed no information. The local Bloch vectors vanish, while the nontrivial information survives in the joint correlator.

4.5 Quantum mechanics violates the CHSH bound

Choose Alice's two measurement directions to be

$$\mathbf{a}_0 = \hat{\mathbf{z}}, \quad \mathbf{a}_1 = \hat{\mathbf{x}}, \quad (188)$$

and Bob's to be

$$\mathbf{b}_0 = \frac{\hat{\mathbf{z}} + \hat{\mathbf{x}}}{\sqrt{2}}, \quad \mathbf{b}_1 = \frac{\hat{\mathbf{z}} - \hat{\mathbf{x}}}{\sqrt{2}}. \quad (189)$$

The four scalar products are

	$\mathbf{b}_0$	$\mathbf{b}_1$	
$\mathbf{a}_0$	$1/\sqrt{2}$	$1/\sqrt{2}$	
$\mathbf{a}_1$	$1/\sqrt{2}$	$-1/\sqrt{2}$	

(190)

so Eq. (187) gives

$$E(\mathbf{a}_0, \mathbf{b}_0) = E(\mathbf{a}_0, \mathbf{b}_1) = E(\mathbf{a}_1, \mathbf{b}_0) = -\frac{1}{\sqrt{2}}, \quad (191)$$

$$E(\mathbf{a}_1, \mathbf{b}_1) = +\frac{1}{\sqrt{2}}. \quad (192)$$

Substituting these four experimentally accessible correlations into CHSH,

$$\mathcal{S} = E(\mathbf{a}_0, \mathbf{b}_0) + E(\mathbf{a}_0, \mathbf{b}_1) + E(\mathbf{a}_1, \mathbf{b}_0) - E(\mathbf{a}_1, \mathbf{b}_1) \quad (193)$$

$$= -\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \quad (194)$$

$$= -\frac{4}{\sqrt{2}} = -2\sqrt{2}. \quad (195)$$

Thus

$$|\mathcal{S}| = 2\sqrt{2} > 2. \quad (196)$$

The local hidden-variable bound is violated.

The violation does not arise because the individual outcomes cease to be ordinary numbers ± 1 . Alice and Bob still record ordinary binary outcomes. The impossibility lies in explaining *all four possible correlation experiments at once* by one underlying local common-cause model.

The force of Bell's theorem

Bell's theorem does not say merely that quantum mechanics contains correlations. Classical systems can be strongly correlated. It says that there exist experimentally accessible quantum correlations whose complete statistics cannot be reproduced by any model satisfying the local factorization condition in Eq. (143), together with measurement independence in Eq. (144). Equivalently, the observed statistics cannot have the local-common-cause form of Eq. (145).

4.6 Why quantum mechanics stops at $2\sqrt{2}$: the Tsirelson bound

The CHSH expression has an algebraic maximum of 4: if four correlation numbers could be chosen independently from $[-1, 1]$, the combination in Eq. (152) could reach that value. Local hidden variables are restricted to 2, while the singlet reaches $2\sqrt{2}$. Why can quantum mechanics not go all the way to 4?

Let Alice's two dichotomic observables be A_0, A_1 and Bob's be B_0, B_1 . Each is Hermitian and has eigenvalues ± 1 . Squaring either eigenvalue gives 1, so

$$A_x^2 = I_A, \quad B_y^2 = I_B. \quad (197)$$

In particular their operator norms are

$$\|A_x\| = \|B_y\| = 1. \quad (198)$$

Introduce the CHSH operator

$$\hat{\mathcal{B}} = A_0 \otimes (B_0 + B_1) + A_1 \otimes (B_0 - B_1). \quad (199)$$

Its expectation value is exactly the experimentally defined CHSH combination:

$$\langle \hat{\mathcal{B}} \rangle_\rho \equiv \text{Tr}(\rho \hat{\mathcal{B}}) \quad (200)$$

$$= \langle A_0 \otimes B_0 \rangle_\rho + \langle A_0 \otimes B_1 \rangle_\rho + \langle A_1 \otimes B_0 \rangle_\rho - \langle A_1 \otimes B_1 \rangle_\rho \quad (201)$$

$$= \mathcal{S}. \quad (202)$$

Thus bounding $\|\hat{\mathcal{B}}\|$ bounds every quantum value of CHSH.

The square of $\hat{\mathcal{B}}$ is the crucial step, so we do not skip the algebra. Define

$$C = B_0 + B_1, \quad D = B_0 - B_1. \quad (203)$$

Then

$$\hat{\mathcal{B}} = A_0 \otimes C + A_1 \otimes D, \quad (204)$$

and hence

$$\hat{\mathcal{B}}^2 = A_0^2 \otimes C^2 + A_1^2 \otimes D^2 \quad (205)$$

$$+ A_0 A_1 \otimes CD + A_1 A_0 \otimes DC. \quad (206)$$

Using $B_0^2 = B_1^2 = I_B$,

$$C^2 + D^2 = (B_0 + B_1)^2 + (B_0 - B_1)^2 \quad (207)$$

$$= 2B_0^2 + 2B_1^2 \quad (208)$$

$$= 4I_B. \quad (209)$$

For the cross terms,

$$CD = (B_0 + B_1)(B_0 - B_1) \quad (210)$$

$$= -B_0B_1 + B_1B_0 \quad (211)$$

$$= -[B_0, B_1], \quad (212)$$

$$DC = (B_0 - B_1)(B_0 + B_1) \quad (213)$$

$$= B_0B_1 - B_1B_0 \quad (214)$$

$$= [B_0, B_1]. \quad (215)$$

Substitution into Eq. (206), together with $A_0^2 = A_1^2 = I_A$, gives

$$\hat{\mathcal{B}}^2 = 4I_{AB} - A_0A_1 \otimes [B_0, B_1] + A_1A_0 \otimes [B_0, B_1] \quad (216)$$

$$= 4I_{AB} - [A_0, A_1] \otimes [B_0, B_1]. \quad (217)$$

Therefore

$$\boxed{\hat{\mathcal{B}}^2 = 4I_{AB} - [A_0, A_1] \otimes [B_0, B_1].} \quad (218)$$

We now use only elementary properties of the operator norm:

$$\|X + Y\| \leq \|X\| + \|Y\|, \quad \|XY\| \leq \|X\| \|Y\|. \quad (219)$$

For Alice,

$$\|[A_0, A_1]\| = \|A_0A_1 - A_1A_0\| \quad (220)$$

$$\leq \|A_0A_1\| + \|A_1A_0\| \quad (221)$$

$$\leq \|A_0\| \|A_1\| + \|A_1\| \|A_0\| \quad (222)$$

$$= 2, \quad (223)$$

and similarly

$$\|[B_0, B_1]\| \leq 2. \quad (224)$$

The norm of a tensor product satisfies

$$\|X \otimes Y\| = \|X\| \|Y\|, \quad (225)$$

so

$$\|[A_0, A_1] \otimes [B_0, B_1]\| \leq 4. \quad (226)$$

Consequently Eq. (218) gives

$$\|\hat{\mathcal{B}}^2\| \leq \|4I_{AB}\| + \|[A_0, A_1] \otimes [B_0, B_1]\| \quad (227)$$

$$\leq 4 + 4 = 8. \quad (228)$$

Finally, $\hat{\mathcal{B}}$ is Hermitian. If its eigenvalues are β_n , then

$$\|\hat{\mathcal{B}}\| = \max_n |\beta_n|, \quad \|\hat{\mathcal{B}}^2\| = \max_n \beta_n^2 = \|\hat{\mathcal{B}}\|^2. \quad (229)$$

Therefore

$$\|\hat{\mathcal{B}}\| \leq 2\sqrt{2}. \quad (230)$$

For any density matrix ρ , the expectation value of a Hermitian operator lies between its smallest and largest eigenvalues, and hence

$$|\mathcal{S}| = |\text{Tr}(\rho\hat{\mathcal{B}})| \leq \|\hat{\mathcal{B}}\| \leq 2\sqrt{2}. \quad (231)$$

The settings in Eq. (189) and the singlet saturate this bound.

The structure of Eq. (218) is revealing. A Bell violation requires incompatible local choices. If Alice's observables commute, $[A_0, A_1] = 0$, or Bob's commute, $[B_0, B_1] = 0$, then

$$\hat{\mathcal{B}}^2 = 4I_{AB}, \quad \|\hat{\mathcal{B}}\| = 2. \quad (232)$$

Thus the extra quantum room between 2 and $2\sqrt{2}$ is tied directly to local measurement incompatibility. The value $2\sqrt{2}$ is not an arbitrary numerical curiosity: it is the largest CHSH correlation permitted by the operator algebra of quantum theory.

4.7 What Bell's theorem does not say

Because Bell's theorem is conceptually powerful, it is also easy to overstate. Several distinctions are essential.

Bell does not rule out every conceivable hidden-variable theory

The contradiction is with *local* hidden-variable models satisfying the assumptions used in deriving the inequality. Bell's theorem is therefore not a proof that the phrase "hidden variable" is meaningless. It tells us that any deeper description reproducing the observed Bell correlations cannot keep all of the classical locality structure encoded in Eq. (143).

Entanglement and Bell nonlocality are not identical notions

For a pure bipartite state, entanglement means failure of factorization. For a mixed state, it means failure of every separable decomposition. Bell nonlocality asks a different operational question: whether the statistics of specified local measurements can be reproduced by a local hidden-variable model.

Consequently,

Entanglement versus Bell nonlocality

Bell-nonlocal correlations imply entanglement, but entanglement need not imply a CHSH violation.

There exist entangled mixed states whose correlations do not violate the CHSH inequality for the measurements under consideration. Entanglement is thus a broader state property; Bell nonlocality is a stronger statement about measurement statistics.

This distinction is one reason the formal definition of entanglement in Chapter 3 was needed before Bell's theorem. Bell violation is a spectacular witness of nonclassical correlation, but it is not the definition of entanglement.

A Bell experiment is statistical

No single pair of particles violates an inequality. Each run produces only a pair of outcomes $a, b = \pm 1$. The correlations $E(x, y)$ are estimated from many trials with the four different setting pairs, and the resulting value of $\mathcal{S}$ is compared with the local bound. Real experiments must therefore control finite statistics and ensure that the assumptions connecting the data to the Bell scenario are actually justified.

Bell nonlocality is not a signalling protocol

A violation of the CHSH inequality does not by itself provide a way to send a controllable message faster than light. Bell nonlocality concerns the structure of joint correlations; signalling concerns whether one observer can change the other observer's local statistics by a freely chosen local action. We postpone the proof until Chapter 5, where the theory of generalized local measurements lets us establish the stronger statement for an arbitrary local measurement, not merely for the projective measurements used in the CHSH experiment.

4.8 Optional advanced topic: contextuality and the Peres–Mermin square

Bell's theorem uses spatial separation to formulate a particularly sharp notion of classicality: Alice's outcome should not depend on Bob's distant measurement choice, and vice versa. Quantum mechanics contains an even more general obstruction to a classical hidden catalogue of properties. It is called *contextuality* [3, 4, 5].

This subsection is optional. Nothing later in the main development requires it. It is included because the Peres–Mermin square gives one of the cleanest ways to see how deeply the quantum algebra resists the idea that every observable simply carries a pre-existing value.

What is a measurement context?

Two observables that commute can be measured compatibly. A collection of mutually commuting observables is called a *measurement context*. The same observable may belong to more than one such compatible collection.

A noncontextual hidden-value picture tries to assign to every observable A a value $v(A)$ that is independent of which other commuting observables happen to be measured alongside it. For the observables below the eigenvalues are ± 1 , so

$$v(A) \in \{+1, -1\}. \quad (233)$$

The assignment must also respect functional relations among compatible observables. In particular, if A and B commute and

$$C = AB, \quad (234)$$

then a pre-existing assignment intended to reproduce the corresponding joint measurement must satisfy

$$\boxed{v(C) = v(A)v(B)}. \quad (235)$$

Similarly,

$$v(I) = +1, \quad v(-I) = -1. \quad (236)$$

The question is whether one can assign such values consistently to all observables at once.

The square

Consider two qubits and the following nine observables:

$$\boxed{\begin{array}{ccc} \sigma_x \otimes I & I \otimes \sigma_x & \sigma_x \otimes \sigma_x \\ I \otimes \sigma_y & \sigma_y \otimes I & \sigma_y \otimes \sigma_y \\ \sigma_x \otimes \sigma_y & \sigma_y \otimes \sigma_x & \sigma_z \otimes \sigma_z \end{array}} \quad (237)$$

Every row consists of mutually commuting observables, and every column does as well. Thus each row and each column defines a legitimate measurement context. Some of the commutators

vanish for an elementary reason: operators acting on different qubits commute. The less obvious cases work because two minus signs cancel. For example, although $\sigma_x \sigma_y = -\sigma_y \sigma_x$ on one qubit,

$$(\sigma_x \otimes \sigma_y)(\sigma_y \otimes \sigma_x) = (\sigma_y \otimes \sigma_x)(\sigma_x \otimes \sigma_y). \quad (238)$$

Now multiply the operators within each row. Using the Pauli algebra gives

$$R_1 = +I, \quad R_2 = +I, \quad R_3 = +I. \quad (239)$$

The third row is worth checking explicitly:

$$(\sigma_x \otimes \sigma_y)(\sigma_y \otimes \sigma_x) \quad (240)$$

$$= (i\sigma_z) \otimes (-i\sigma_z) = \sigma_z \otimes \sigma_z, \quad (241)$$

so multiplying by the final $\sigma_z \otimes \sigma_z$ gives $+I$.

For the columns,

$$C_1 = +I, \quad C_2 = +I, \quad C_3 = -I. \quad (242)$$

Again the exceptional sign can be seen directly:

$$(\sigma_x \otimes \sigma_x)(\sigma_y \otimes \sigma_y) \quad (243)$$

$$= (i\sigma_z) \otimes (i\sigma_z) = -\sigma_z \otimes \sigma_z, \quad (244)$$

and the last factor therefore produces $-I$.

Why a noncontextual value assignment is impossible

Suppose nevertheless that every one of the nine observables possesses a context-independent predetermined value ± 1 . Because the three observables in a row commute, the product rule Eq. (235) requires the product of their assigned values to equal the eigenvalue of the row product. The three rows therefore impose

$$v(R_1) = +1, \quad v(R_2) = +1, \quad v(R_3) = +1. \quad (245)$$

The columns require

$$v(C_1) = +1, \quad v(C_2) = +1, \quad v(C_3) = -1. \quad (246)$$

Now multiply all six constraints. On the left, every one of the nine assigned values occurs exactly twice—once in its row and once in its column. Since $v(A)^2 = 1$, the product of the left-hand sides must be

$$+1. \quad (247)$$

On the right, however, the five positive context products and the one negative context product give

$$(+1)^5(-1) = -1. \quad (248)$$

Thus the same supposed assignment demands

$$+1 = -1, \quad (249)$$

a contradiction.

The contextuality lesson

The Peres–Mermin contradiction does not arise because we chose a peculiar quantum state. It follows from operator identities among compatible observables. There is no single assignment of context-independent values to all nine observables that can respect the quantum-mechanical relations in every commuting row and column.

This is called a *state-independent* contextuality proof. Unlike the CHSH violation above, we did not choose an entangled state and calculate a special correlation function. The obstruction is already encoded in the algebra of the observables on the four-dimensional Hilbert space of two qubits.

What contextuality does not mean

It is tempting to summarize contextuality by saying “measurement changes the answer.” That is too crude. Ordinary measurement disturbance is not the logical issue. The contradiction appears before specifying a dynamical model of disturbance. What fails is the attempt to imagine one globally consistent catalogue of values that is independent of which compatible set is used to reveal them.

Nor does contextuality say that nothing is objective or that experimental records are ambiguous. In a particular measurement context the apparatus produces definite outcomes, and the quantum theory gives precise probability rules for them. The impossibility concerns the attempt to regard all possible contexts as merely uncovering different portions of one pre-existing, context-independent assignment.

Bell nonlocality and contextuality

The two no-go results are related but answer different questions:

	Bell	Contextuality
Physical organization	separated laboratories	compatible measurement contexts
Classical idea tested	local hidden causes	context-independent value assignments
Key restriction	Alice’s response is independent of Bob’s setting, and vice versa	the value assigned to an observable is independent of which compatible observables accompany it
Representative test	CHSH inequality	Peres–Mermin square

Bell’s theorem uses spacelike separation to turn the notion of context into a particularly compelling locality requirement. Contextuality shows that the tension with a classical catalogue of predetermined properties is more general than spatial nonlocality alone.

4.9 What we have learned, and why the next chapter follows

We can now close the conceptual loop opened by the election parable.

1. A classical shared variable can certainly correlate distant systems, but every local hidden-variable model obeys the CHSH bound $|\mathcal{S}| \leq 2$.
2. The singlet predicts $E(\mathbf{a}, \mathbf{b}) = -\mathbf{a} \cdot \mathbf{b}$ and reaches $|\mathcal{S}| = 2\sqrt{2}$ for suitable settings.
3. The violation rules out the idea that all the observed correlations arise from a shared local hidden description of the Bell type.

4. Bell nonlocality is a statement about correlations, not a recipe for controllable faster-than-light communication. The general no-signalling proof will follow once we have developed arbitrary local measurements.
5. Optionally, the Peres–Mermin square shows a still broader obstruction: quantum observables do not admit one globally consistent, noncontextual assignment of predetermined values respecting all compatible functional relations.

Chapters 1–3 told us how information can reside in a composite quantum state and how entanglement is quantified for pure bipartite systems. Chapter 4 has shown that entangled correlations can resist every local hidden-variable explanation of the Bell type. Before turning to the historical experiments, we now return to a question that has so far been deliberately postponed:

What is the most general quantum measurement, what state does it leave behind, and how can local measurements manipulate entanglement without enabling superluminal signalling?

The next chapter develops that measurement theory from the projective rule the student already knows, and only then returns to Bell tests in the laboratory.

5 Measurements Beyond Ideal Projectors: POVMs, Measurement Operators, and Entanglement

Until now, ideal projective measurements have been enough. They allowed us to define reduced states, characterize entanglement, and formulate Bell's theorem. We now return to measurement itself and ask a more general question:

What is the most general way a quantum apparatus can produce classical outcomes, and what happens to the quantum state when one of those outcomes is recorded?

The answer will not replace projective measurement. It will grow naturally out of it. We shall therefore begin with the familiar case and make every state-update step explicit before introducing anything new.

5.1 Projective measurement revisited: why the projector is the basic object

Let an observable have spectral decomposition

$$O = \sum_r o_r P_r, \quad (250)$$

where P_r projects onto the eigenspace with eigenvalue o_r . The projectors satisfy

$$\boxed{P_r P_s = \delta_{rs} P_r, \quad \sum_r P_r = I.} \quad (251)$$

The use of projectors rather than individual eigenvectors matters whenever an eigenvalue is degenerate.

A concrete degenerate example: measuring S_z^2 for spin one

For a spin-one particle, choose the usual S_z basis

$$|+1\rangle, \quad |0\rangle, \quad |-1\rangle, \quad (252)$$

with

$$S_z |m\rangle = \hbar m |m\rangle. \quad (253)$$

Then

$$S_z^2 |0\rangle = 0, \quad (254)$$

while

$$S_z^2 |+1\rangle = \hbar^2 |+1\rangle, \quad S_z^2 |-1\rangle = \hbar^2 |-1\rangle. \quad (255)$$

There are therefore only two distinct measurement outcomes, 0 and $\hbar^2$, even though the Hilbert space has dimension three. Their projectors are

$$\boxed{P_0 = |0\rangle\langle 0|, \quad P_{\hbar^2} = |+1\rangle\langle +1| + |-1\rangle\langle -1|.} \quad (256)$$

Thus

$$S_z^2 = 0 P_0 + \hbar^2 P_{\hbar^2}. \quad (257)$$

The outcome $\hbar^2$ tells us that the state lies in the two-dimensional subspace spanned by $|+1\rangle$ and $|-1\rangle$; it does not tell us which of those two basis vectors is present.

It is worth checking the projector relations explicitly:

$$P_0^2 = P_0, \quad (258)$$

$$P_{\hbar^2}^2 = P_{\hbar^2}, \quad (259)$$

$$P_0 P_{\hbar^2} = 0, \quad (260)$$

$$P_0 + P_{\hbar^2} = I. \quad (261)$$

The algebra simply states that the two outcomes correspond to orthogonal subspaces that together fill the whole Hilbert space.

5.2 Projective measurement of a pure state

Take the normalized pure state

$$|\psi\rangle = \alpha|+1\rangle + \beta|0\rangle + \gamma|-1\rangle, \quad |\alpha|^2 + |\beta|^2 + |\gamma|^2 = 1. \quad (262)$$

Consider first the outcome 0. Applying the corresponding projector gives

$$P_0|\psi\rangle = \beta|0\rangle. \quad (263)$$

This vector is not normalized, and that is useful rather than inconvenient: its squared norm is precisely the probability of the outcome,

$$p(0) = \|P_0|\psi\rangle\|^2 \quad (264)$$

$$= \langle\psi|P_0^\dagger P_0|\psi\rangle \quad (265)$$

$$= \langle\psi|P_0|\psi\rangle \quad (266)$$

$$= |\beta|^2. \quad (267)$$

Here we used $P_0^\dagger = P_0$ and $P_0^2 = P_0$.

If outcome 0 is actually obtained, the state must be normalized. The post-measurement state is therefore

$$|\psi_0\rangle = \frac{P_0|\psi\rangle}{\sqrt{p(0)}}. \quad (268)$$

Up to an irrelevant overall phase, this is simply $|0\rangle$.

The degenerate outcome is more revealing:

$$P_{\hbar^2}|\psi\rangle = \alpha|+1\rangle + \gamma|-1\rangle. \quad (269)$$

Hence

$$p(\hbar^2) = |\alpha|^2 + |\gamma|^2, \quad (270)$$

and, conditioned on observing $S_z^2 = \hbar^2$,

$$|\psi_{\hbar^2}\rangle = \frac{\alpha|+1\rangle + \gamma|-1\rangle}{\sqrt{|\alpha|^2 + |\gamma|^2}}. \quad (271)$$

The relative phase between the $|+1\rangle$ and $|-1\rangle$ amplitudes has survived. An ideal measurement of S_z^2 does not resolve the degeneracy; it projects onto the whole eigenspace.

For the ideal projective measurements used in these notes we adopt the *Lüders update rule*. For any pure input state,

$$|\tilde{\psi}_r\rangle = P_r|\psi\rangle, \quad p(r) = \langle\tilde{\psi}_r|\tilde{\psi}_r\rangle = \langle\psi|P_r|\psi\rangle, \quad |\psi_r\rangle = \frac{P_r|\psi\rangle}{\sqrt{p(r)}}. \quad (272)$$

The tilde reminds us that $|\tilde{\psi}_r\rangle$ is unnormalized. This single vector already contains two pieces of information: its norm gives the probability of the outcome, while its direction gives the state left behind when that outcome occurs.

5.3 Why the projective update has the form $P_r \rho P_r$ for a mixed state

Now suppose the initial state is mixed,

$$\rho = \sum_{\alpha} q_{\alpha} |\psi_{\alpha}\rangle \langle \psi_{\alpha}|. \quad (273)$$

For each possible pure preparation, the unnormalized post-measurement vector for outcome r is $P_r |\psi_{\alpha}\rangle$. Therefore the corresponding unnormalized density matrix, after averaging over the unrevealed preparation label α , is

$$\tilde{\rho}_r = \sum_{\alpha} q_{\alpha} P_r |\psi_{\alpha}\rangle \langle \psi_{\alpha}| P_r \quad (274)$$

$$= P_r \left(\sum_{\alpha} q_{\alpha} |\psi_{\alpha}\rangle \langle \psi_{\alpha}| \right) P_r \quad (275)$$

$$= \boxed{P_r \rho P_r}. \quad (276)$$

The probability of outcome r is the trace of this unnormalized state:

$$p(r) = \text{Tr}(\tilde{\rho}_r) \quad (277)$$

$$= \text{Tr}(P_r \rho P_r) \quad (278)$$

$$= \text{Tr}(\rho P_r^2) \quad (279)$$

$$= \boxed{\text{Tr}(\rho P_r)}. \quad (280)$$

Finally,

$$\boxed{\rho_r = \frac{P_r \rho P_r}{\text{Tr}(\rho P_r)}}. \quad (281)$$

Thus the pure- and mixed-state rules have exactly the same structure:

Pure input	Mixed input
$ \tilde{\psi}_r\rangle = P_r \psi\rangle$	$\tilde{\rho}_r = P_r \rho P_r$
$p(r) = \ \tilde{\psi}_r\ ^2$	$p(r) = \text{Tr} \tilde{\rho}_r$
$ \psi_r\rangle = \tilde{\psi}_r\rangle / \sqrt{p(r)}$	$\rho_r = \tilde{\rho}_r / p(r)$

This pattern is the one we shall generalize. The projector P_r will be replaced by a more general measurement operator M_r .

5.4 Why projectors are not enough: a detector that sometimes fails to click

Suppose we intend to measure a qubit in the basis $\{|0\rangle, |1\rangle\}$. An ideal apparatus has two records, “0” and “1”. A realistic detector may have a third possibility:

$$0, \quad 1, \quad \emptyset \text{ (no click)}. \quad (282)$$

Assume that when the detector works it identifies the basis state correctly, and that it clicks with efficiency $0 \leq \eta \leq 1$. Then

$$p(0) = \eta \langle 0 | \rho | 0 \rangle, \quad (283)$$

$$p(1) = \eta \langle 1 | \rho | 1 \rangle, \quad (284)$$

$$p(\emptyset) = 1 - \eta. \quad (285)$$

All three probabilities can be written in the form

$$p(r) = \text{Tr}(\rho E_r) \quad (286)$$

with

$$E_0 = \eta|0\rangle\langle 0|, \quad E_1 = \eta|1\rangle\langle 1|, \quad E_\emptyset = (1 - \eta)I. \quad (287)$$

They satisfy

$$E_0 + E_1 + E_\emptyset = I, \quad (288)$$

but, for $0 < \eta < 1$,

$$E_0^2 = \eta^2|0\rangle\langle 0| \neq E_0. \quad (289)$$

So E_0 is not a projector. The apparatus defines a perfectly sensible measurement even though its classical records cannot be represented by a set of mutually orthogonal projectors.

This is the simplest reason to enlarge the measurement formalism. The deeper reason, which we shall see later, is that generalized measurements can also be useful even for a perfectly ideal apparatus.

5.5 POVMs: the most general rule for outcome probabilities

A *positive-operator-valued measure*, or *POVM*, with outcomes r is a collection of operators $\{E_r\}$ satisfying

$$E_r \geq 0, \quad \sum_r E_r = I. \quad (290)$$

The probability of outcome r in the state ρ is

$$p(r) = \text{Tr}(\rho E_r). \quad (291)$$

Positivity guarantees $p(r) \geq 0$. Completeness guarantees normalization:

$$\sum_r p(r) = \sum_r \text{Tr}(\rho E_r) \quad (292)$$

$$= \text{Tr}\left(\rho \sum_r E_r\right) \quad (293)$$

$$= \text{Tr} \rho = 1. \quad (294)$$

A projective measurement is the special case

$$E_r = P_r, \quad P_r P_s = \delta_{rs} P_r. \quad (295)$$

A general POVM does not require orthogonality or idempotence, and it may have more outcomes than the dimension of the Hilbert space.

What a POVM tells us—and what it does not

The effect E_r answers one question only: *how probable is the classical record r ?* The POVM $\{E_r\}$ does not, by itself, determine the quantum state left behind after that record occurs. To describe the disturbance as well as the statistics we need one more layer of structure.

This distinction is easy to miss if the generalized Born rule is presented as a complete measurement theory. It is not. Outcome probabilities and post-measurement states are related, but they are not the same information.

5.6 Measurement operators: the direct generalization of P_r

The projective pure-state rule began with

$$|\tilde{\psi}_r\rangle = P_r|\psi\rangle. \quad (296)$$

For a general measurement, the natural replacement is

$$|\tilde{\psi}_r\rangle = M_r|\psi\rangle. \quad (297)$$

The operator M_r is called a *measurement operator* associated with outcome r .

For a pure input state, the probability of outcome r is the squared norm of the unnormalized branch:

$$p(r) = \langle \tilde{\psi}_r | \tilde{\psi}_r \rangle \quad (298)$$

$$= \langle \psi | M_r^\dagger M_r | \psi \rangle. \quad (299)$$

If $p(r) > 0$, the normalized post-measurement state is

$$|\psi_r\rangle = \frac{M_r|\psi\rangle}{\sqrt{p(r)}}. \quad (300)$$

This is exactly the projective formula with P_r replaced by M_r .

For a mixed input state, the corresponding rules are

$$\tilde{\rho}_r = M_r \rho M_r^\dagger, \quad (301)$$

$$p(r) = \text{Tr}(M_r \rho M_r^\dagger) \quad (302)$$

$$= \text{Tr}(\rho M_r^\dagger M_r), \quad (303)$$

and

$$\rho_r = \frac{M_r \rho M_r^\dagger}{p(r)}. \quad (304)$$

Again, the unnormalized state carries both pieces of information:

$$\begin{array}{l} \text{the trace gives the outcome probability,} \\ \text{while normalization gives the conditional post-measurement state} \end{array} \quad (305)$$

Why the completeness relation must hold

Exactly one of the recorded outcomes must occur, so for every normalized pure state

$$\sum_r p(r) = 1. \quad (306)$$

Using Eq. (299),

$$1 = \sum_r \langle \psi | M_r^\dagger M_r | \psi \rangle \quad (307)$$

$$= \langle \psi | \left(\sum_r M_r^\dagger M_r \right) | \psi \rangle. \quad (308)$$

Since this must hold for every normalized $|\psi\rangle$,

$$\boxed{\sum_r M_r^\dagger M_r = I.} \quad (309)$$

Now define

$$\boxed{E_r = M_r^\dagger M_r.} \quad (310)$$

For every $|\chi\rangle$,

$$\langle\chi|E_r|\chi\rangle = \|M_r|\chi\rangle\|^2 \geq 0, \quad (311)$$

so $E_r \geq 0$, and Eq. (309) gives

$$\sum_r E_r = I. \quad (312)$$

Thus the E_r automatically form a POVM, and

$$p(r) = \text{Tr}(\rho E_r). \quad (313)$$

The hierarchy is therefore

$$M_r \implies \begin{cases} E_r = M_r^\dagger M_r, & \text{outcome probability,} \\ M_r \rho M_r^\dagger, & \text{conditional state change.} \end{cases} \quad (314)$$

For an ideal projective measurement one may take $M_r = P_r$, and the Lüders rule is recovered immediately.

5.7 The same POVM can disturb the system in different ways

The effects do not determine the state update uniquely. Consider the two-outcome POVM

$$E_0 = |0\rangle\langle 0|, \quad E_1 = |1\rangle\langle 1|. \quad (315)$$

One implementation is the ordinary projective measurement

$$M_0 = |0\rangle\langle 0|, \quad M_1 = |1\rangle\langle 1|. \quad (316)$$

A different implementation is

$$M'_0 = |1\rangle\langle 0|, \quad M'_1 = |0\rangle\langle 1|. \quad (317)$$

In both cases

$$M_r^\dagger M_r = M_r'^\dagger M_r' = E_r, \quad (318)$$

so every input state gives exactly the same probabilities for the labels $r = 0, 1$. But after outcome 0 the first apparatus leaves the system in $|0\rangle$, whereas the second leaves it in $|1\rangle$. The classical record is statistically identical; the physical disturbance is not.

Statistics versus disturbance

E_r tells us how often an outcome occurs. M_r tells us, in addition, what that outcome does to the state. A POVM is therefore the probability part of a measurement, not a complete description of the measurement process.

5.8 Why generalized measurements arise from ordinary unitary dynamics

The M_r formalism becomes much less mysterious when the measuring apparatus is included explicitly. Let S be the system and let A be a small ancillary system representing the part of the apparatus that will carry the classical record. Prepare the ancilla in a fixed state $|0\rangle_A$ and let system and apparatus interact through a unitary U_{SA} .

For a pure system state,

$$|\psi\rangle_S |0\rangle_A \longrightarrow U_{SA} |\psi, 0\rangle. \quad (319)$$

Now measure the ancilla projectively in an orthonormal basis $\{|r\rangle_A\}$. If the apparatus record is r , the corresponding branch of the joint state is

$$(I_S \otimes |r\rangle_A \langle r|) U_{SA} |\psi\rangle_S |0\rangle_A \quad (320)$$

$$= ({}_A \langle r| U_{SA} |0\rangle_A) |\psi\rangle_S \otimes |r\rangle_A. \quad (321)$$

This identifies the effective operator acting on the system:

$$\boxed{M_r \equiv {}_A \langle r| U_{SA} |0\rangle_A.} \quad (322)$$

The system part of the branch is therefore exactly

$$|\tilde{\psi}_r\rangle = M_r |\psi\rangle. \quad (323)$$

Its squared norm gives the probability, and normalizing it gives the post-measurement state. The generalized-measurement rule has emerged from nothing more exotic than unitary evolution on a larger Hilbert space followed by an ordinary projective measurement of the apparatus.

For a mixed input state the initial joint state is

$$\rho_S \otimes |0\rangle_A \langle 0|, \quad (324)$$

and after the unitary interaction

$$\rho'_{SA} = U_{SA} (\rho_S \otimes |0\rangle \langle 0|) U_{SA}^\dagger. \quad (325)$$

The projector corresponding to apparatus record r is

$$\boxed{\Pi_r = I_S \otimes |r\rangle_A \langle r|.} \quad (326)$$

Projecting onto that record and tracing out the apparatus gives the unnormalized conditional system state

$$\boxed{\tilde{\rho}_{S|r} = \text{Tr}_A (\Pi_r \rho'_{SA} \Pi_r) = M_r \rho_S M_r^\dagger.} \quad (327)$$

Therefore

$$p(r) = \text{Tr}(M_r \rho_S M_r^\dagger), \quad (328)$$

and

$$\rho_{S|r} = \frac{M_r \rho_S M_r^\dagger}{p(r)}. \quad (329)$$

Finally, unitarity and completeness of the ancilla basis give

$$\sum_r M_r^\dagger M_r = \sum_r {}_A \langle 0| U_{SA}^\dagger |r\rangle_A {}_A \langle r| U_{SA} |0\rangle_A \quad (330)$$

$$= {}_A \langle 0| U_{SA}^\dagger U_{SA} |0\rangle_A \quad (331)$$

$$= I_S. \quad (332)$$

Thus the completeness relation is itself a consequence of the unitarity of the larger system–apparatus dynamics.

5.9 A fully explicit two-qubit unitary

It is useful to see the construction with an interaction that is manifestly unitary on the *entire* system–apparatus Hilbert space. Let both S and A be qubits and define

$$c = \sqrt{1 - \kappa}, \quad s = \sqrt{\kappa}, \quad 0 \leq \kappa \leq 1. \quad (333)$$

Introduce the ancilla rotation

$$R_\kappa = \begin{pmatrix} c & -s \\ s & c \end{pmatrix}, \quad R_\kappa^\dagger R_\kappa = I_A, \quad (334)$$

and the controlled unitary

$$U_{SA} = |0\rangle\langle 0|_S \otimes I_A + |1\rangle\langle 1|_S \otimes R_\kappa. \quad (335)$$

Because the two system projectors are orthogonal,

$$U_{SA}^\dagger U_{SA} = |0\rangle\langle 0|_S \otimes I_A + |1\rangle\langle 1|_S \otimes R_\kappa^\dagger R_\kappa \quad (336)$$

$$= I_S \otimes I_A. \quad (337)$$

So there is no hidden assumption: Eq. (335) is explicitly unitary.

Its action on the four computational-basis states is

$$|00\rangle \longrightarrow |00\rangle, \quad (338)$$

$$|01\rangle \longrightarrow |01\rangle, \quad (339)$$

$$|10\rangle \longrightarrow c|10\rangle + s|11\rangle, \quad (340)$$

$$|11\rangle \longrightarrow -s|10\rangle + c|11\rangle. \quad (341)$$

The apparatus begins in $|0\rangle_A$, so only the first and third lines are needed in the actual experiment. For

$$|\psi\rangle_S = \alpha|0\rangle + \beta|1\rangle, \quad (342)$$

we obtain

$$U_{SA}|\psi, 0\rangle = \alpha|0\rangle_S|0\rangle_A + \beta c|1\rangle_S|0\rangle_A + \beta s|1\rangle_S|1\rangle_A \quad (343)$$

$$= (\alpha|0\rangle + \beta c|1\rangle)|0\rangle_A + \beta s|1\rangle|1\rangle_A. \quad (344)$$

Reading off the two ancilla branches,

$$M_0 = |0\rangle\langle 0| + \sqrt{1 - \kappa}|1\rangle\langle 1|, \quad M_1 = \sqrt{\kappa}|1\rangle\langle 1|. \quad (345)$$

Indeed,

$$M_0^\dagger M_0 + M_1^\dagger M_1 = I. \quad (346)$$

The probabilities are

$$p(0) = |\alpha|^2 + (1 - \kappa)|\beta|^2, \quad (347)$$

$$p(1) = \kappa|\beta|^2, \quad (348)$$

and the conditional states are

$$|\psi_0\rangle = \frac{\alpha|0\rangle + \sqrt{1 - \kappa}\beta|1\rangle}{\sqrt{|\alpha|^2 + (1 - \kappa)|\beta|^2}}, \quad (349)$$

while outcome 1 leaves the system in $|1\rangle$.

The parameter κ makes the physics transparent:

- $\kappa = 0$: the apparatus learns nothing and $M_0 = I$;
- $0 < \kappa < 1$: the apparatus gains partial information and partially disturbs the state;
- $\kappa = 1$: $M_0 = |0\rangle\langle 0|$ and $M_1 = |1\rangle\langle 1|$, giving an ordinary projective measurement.

The example continuously interpolates between no measurement and a fully projective measurement using a genuine unitary interaction.

5.10 Physical examples of POVMs

Example 1: the inefficient detector revisited

For the inefficient detector of Eq. (287), one possible set of measurement operators is

$$M_0 = \sqrt{\eta} |0\rangle\langle 0|, \quad M_1 = \sqrt{\eta} |1\rangle\langle 1|, \quad M_\emptyset = \sqrt{1-\eta} I. \quad (350)$$

Then $M_r^\dagger M_r = E_r$ for all three records. If the detector reports “0” or “1”, the state is projected onto the corresponding basis state. If it reports no click,

$$\rho_\emptyset = \frac{(1-\eta)\rho}{1-\eta} = \rho, \quad (351)$$

so this particular model of detector failure leaves the state unchanged. A different physical detector could have the same no-click effect $E_\emptyset$ but a different measurement operator and therefore disturb the state even when it fails to click.

Example 2: unambiguous discrimination of two non-orthogonal states

Consider two equally likely qubit states

$$|\psi_1\rangle = |0\rangle, \quad |\psi_2\rangle = \cos\theta |0\rangle + \sin\theta |1\rangle, \quad 0 < \theta < \frac{\pi}{2}. \quad (352)$$

They are not orthogonal:

$$\langle\psi_1|\psi_2\rangle = \cos\theta \neq 0. \quad (353)$$

No measurement can always identify the input without error. But a generalized measurement can adopt a different strategy: whenever it gives a definite answer it is never wrong, while a third outcome “?” is allowed to mean *inconclusive*.

Define

$$|\psi_1^\perp\rangle = |1\rangle, \quad |\psi_2^\perp\rangle = \sin\theta |0\rangle - \cos\theta |1\rangle, \quad (354)$$

and choose

$$E_1 = c |\psi_2^\perp\rangle\langle\psi_2^\perp|, \quad (355)$$

$$E_2 = c |\psi_1^\perp\rangle\langle\psi_1^\perp|, \quad (356)$$

$$E_? = I - E_1 - E_2, \quad (357)$$

with

$$c = \frac{1}{1 + \cos\theta}. \quad (358)$$

For this value $E_? \geq 0$, so the three effects form a POVM. Moreover,

$$\langle\psi_2|E_1|\psi_2\rangle = 0, \quad \langle\psi_1|E_2|\psi_1\rangle = 0. \quad (359)$$

Thus outcome 1 never occurs when the input was $|\psi_2\rangle$, and outcome 2 never occurs when the input was $|\psi_1\rangle$. A conclusive answer is always correct. The success probability for either input is

$$p_{\text{succ}} = c \sin^2 \theta = 1 - \cos \theta. \quad (360)$$

Here the task concerns only the classical answer produced by the apparatus, so the POVM effects are enough. They do *not* specify what state remains after a conclusive or inconclusive result. To answer that further question we would have to choose measurement operators realizing these effects.

Example 3: what a photodetector may actually report

A mathematical two-outcome polarization measurement may correspond in the laboratory to records such as

$$D_0, \quad D_1, \quad \text{no click}, \quad \text{double click}. \quad (361)$$

Loss, finite detector efficiency, background counts, and multi-photon events change the effective measurement away from a pair of ideal projectors. The POVM language describes the probabilities of these actual records. If we also need the conditional state of the remaining optical system after a particular record, the corresponding measurement operators or instrument must be specified as well.

5.11 A final refinement: quantum instruments

So far we have associated one measurement operator M_r with each recorded outcome. Sometimes a single macroscopic record r can arise through several microscopic alternatives that the apparatus does not resolve. Label those alternatives by α . Then the unnormalized state associated with the record r is

$$\boxed{\mathcal{I}_r(\rho) = \sum_{\alpha} M_{r\alpha} \rho M_{r\alpha}^{\dagger}}. \quad (362)$$

The collection $\{\mathcal{I}_r\}$ is called a *quantum instrument*. Its probability is

$$p(r) = \text{Tr}[\mathcal{I}_r(\rho)], \quad (363)$$

and the normalized conditional state is

$$\rho_r = \frac{\mathcal{I}_r(\rho)}{p(r)}. \quad (364)$$

The corresponding POVM effect is

$$E_r = \sum_{\alpha} M_{r\alpha}^{\dagger} M_{r\alpha}. \quad (365)$$

The completeness condition is

$$\sum_{r,\alpha} M_{r\alpha}^{\dagger} M_{r\alpha} = I. \quad (366)$$

This is the most general state-update language we need here. The simpler one- M_r -per-outcome formalism should remain the default picture unless the unresolved microscopic alternatives matter.

5.12 Generalized local measurements on an entangled state

Generalized measurements become especially illuminating when applied to one part of an entangled system. Let the joint state of Alice and Bob be ρ_{AB} , and suppose Bob performs a local measurement described by $\{M_r\}$ on B . If Bob obtains outcome r , the unnormalized joint state is

$$\tilde{\rho}_{AB}^{(r)} = (I_A \otimes M_r) \rho_{AB} (I_A \otimes M_r^\dagger). \quad (367)$$

The probability of that result is

$$p(r) = \text{Tr}_{AB} \tilde{\rho}_{AB}^{(r)}, \quad (368)$$

and the normalized joint state is

$$\rho_{AB}^{(r)} = \frac{\tilde{\rho}_{AB}^{(r)}}{p(r)}. \quad (369)$$

Alice's state conditioned on Bob's known outcome is therefore

$$\rho_{A|r} = \text{Tr}_B \rho_{AB}^{(r)}. \quad (370)$$

Different values of r can give different conditional states for Alice. This is the general mathematical form of the phenomenon often called *steering*: by choosing a measurement and then sorting the data according to the outcome, Bob can prepare different conditional ensembles for Alice. The crucial words are *sorting according to the outcome*. Alice needs to know which outcome Bob obtained before she can know which conditional ensemble is relevant.

5.13 Local filtering: generalized measurements can concentrate entanglement

Generalized measurement brings something genuinely new to the operational study of entanglement. A local measurement can probabilistically transform a partially entangled state into a more strongly entangled one.

Consider

$$|\psi\rangle_{AB} = \sqrt{\lambda} |00\rangle + \sqrt{1-\lambda} |11\rangle, \quad \frac{1}{2} \leq \lambda < 1. \quad (371)$$

Alice performs a two-outcome generalized measurement with

$$M_s = \sqrt{\frac{1-\lambda}{\lambda}} |0\rangle\langle 0| + |1\rangle\langle 1|, \quad (372)$$

$$M_f = \sqrt{\frac{2\lambda-1}{\lambda}} |0\rangle\langle 0|. \quad (373)$$

The subscripts mean “success” and “failure”. First check that this is a valid measurement:

$$M_s^\dagger M_s = \frac{1-\lambda}{\lambda} |0\rangle\langle 0| + |1\rangle\langle 1|, \quad (374)$$

$$M_f^\dagger M_f = \frac{2\lambda-1}{\lambda} |0\rangle\langle 0|, \quad (375)$$

so

$$M_s^\dagger M_s + M_f^\dagger M_f = I. \quad (376)$$

If the successful outcome occurs,

$$(M_s \otimes I) |\psi\rangle = \sqrt{1-\lambda} |00\rangle + \sqrt{1-\lambda} |11\rangle \quad (377)$$

$$= \sqrt{1-\lambda} (|00\rangle + |11\rangle). \quad (378)$$

The success probability is the squared norm,

$$p_s = 2(1 - \lambda), \quad (379)$$

and after normalization the pair is in the Bell state

$$|\psi_s\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \quad (380)$$

Thus a local generalized measurement has converted a partially entangled pair into a maximally entangled pair, but only probabilistically. If the failure outcome occurs, the remaining state is proportional to $|00\rangle$ and is not entangled.

This does not mean that local operations create entanglement for free. They can concentrate entanglement into selected successful branches only at the price of a success probability smaller than one. More generally, local operations and classical communication cannot increase entanglement on average. This example is the first glimpse of the resource-theory viewpoint that will return when we discuss teleportation, dense coding, and quantum networks.

5.14 Why generalized measurements still cannot signal faster than light

The formalism now lets us prove the no-signalling statement at its natural level of generality. Let Bob perform any local generalized measurement $\{M_r\}$ on B , but suppose Alice is not told which outcome occurred. Since she does not know r , the relevant joint state is the average over all branches,

$$\rho'_{AB} = \sum_r (I_A \otimes M_r) \rho_{AB} (I_A \otimes M_r^\dagger). \quad (381)$$

We shall show that every local observable available to Alice has exactly the same expectation value as before.

Let O_A be any observable on Alice's subsystem. Then

$$\text{Tr}[\rho'_{AB}(O_A \otimes I_B)] = \sum_r \text{Tr}[(I \otimes M_r) \rho_{AB} (I \otimes M_r^\dagger) (O_A \otimes I)] \quad (382)$$

$$= \sum_r \text{Tr}[\rho_{AB}(O_A \otimes M_r^\dagger M_r)] \quad (383)$$

$$= \text{Tr}\left[\rho_{AB} \left(O_A \otimes \sum_r M_r^\dagger M_r\right)\right] \quad (384)$$

$$= \text{Tr}[\rho_{AB}(O_A \otimes I_B)]. \quad (385)$$

The second line uses cyclicity of the trace and the fact that operators acting on different tensor factors commute. The last line uses

$$\sum_r M_r^\dagger M_r = I_B. \quad (386)$$

Since every Alice observable has the same expectation value before and after Bob's unannounced local measurement, Alice's reduced state is unchanged:

$$\rho'_A = \rho_A. \quad (387)$$

There is no contradiction with the conditional states $\rho_{A|r}$ in Eq. (370). If Bob tells Alice which outcome occurred, she can sort her data according to r and use the corresponding conditional state. Without that classical message she must average over all outcomes, and that average reproduces the original local state.

Entanglement, steering, and no signalling

Entanglement allows Bob's different measurement outcomes to define different conditional states for Alice. Generalized measurements make this statement very broad and precise. But the outcome is random, and if Bob does not communicate it, Alice sees the same reduced density matrix as before. Quantum correlations can therefore be stronger than any Bell-local hidden-variable model without providing a controllable faster-than-light communication channel.

5.15 What generalized measurement adds to our understanding of entanglement

The formalism developed in this chapter does more than describe imperfect detectors. It sharpens the operational meaning of entanglement in three ways:

1. **Conditional states.** A local measurement outcome on one side can select a different conditional state on the other side. Entanglement therefore organizes not only static correlations but also the ensembles that can be remotely prepared once the outcome is communicated.
2. **Local filtering.** Non-projective measurements can probabilistically concentrate entanglement into selected branches, as in Eq. (380).
3. **No signalling.** Although the conditional branches can be very different, averaging over the unknown local outcome leaves the distant reduced state unchanged.

This is the first appearance of a theme that will recur throughout quantum information theory: entanglement is a resource, but what can be done with it is constrained by which operations are local and by which classical outcomes are communicated.

5.16 Logical summary

The measurement theory can now be read as one continuous generalization:

$$\begin{array}{c}
 \text{projective outcome } r : P_r \\
 \Downarrow \\
 P_r|\psi\rangle, \quad P_r\rho P_r \\
 \Downarrow \\
 \text{general outcome } r : M_r \\
 \Downarrow \\
 M_r|\psi\rangle, \quad M_r\rho M_r^\dagger \\
 \Downarrow \\
 E_r = M_r^\dagger M_r \quad \text{when only probabilities are needed.}
 \end{array} \tag{388}$$

The larger-system construction

$$M_r = {}_A\langle r|U_{SA}|0\rangle_A \tag{389}$$

shows where these operators come from physically. Generalized measurements are not an extra postulate grafted onto quantum mechanics; they are what ordinary unitary dynamics and ordinary projective measurement look like when part of the apparatus is treated as an ancilla and then ignored.

We are now ready to return to the laboratory. Bell's theorem told us which correlations distinguish quantum mechanics from local hidden-variable models; the present chapter has supplied the measurement language needed to describe realistic outcomes and conditional state changes. The next chapter asks how Bell's distinction was actually tested experimentally.

6 Bell's Theorem Enters the Laboratory: Testing the Quantum World

Chapter 4 established Bell's theorem, and Chapter 5 broadened the measurement language from ideal projectors to realistic generalized measurements. We now return to Bell's central experimental challenge. If measurement outcomes are produced by a local hidden description of the Bell type, certain combinations of correlations are bounded; quantum mechanics predicts correlations that can cross that bound. The difference is numerical, not merely philosophical. This immediately raises a new question:

Can a laboratory really distinguish these two descriptions of nature?

That question changed the status of quantum foundations. For decades, the Einstein–Bohr debate had seemed to concern how one should *interpret* a formalism whose experimental success was already overwhelming. Bell's theorem made a more demanding statement: there are situations in which a broad class of classical-looking explanations and quantum mechanics predict *different experimental statistics*. Once this is understood, a Bell experiment is not a ceremonial confirmation that quantum mechanics “works.” It is a test of what kinds of underlying description are compatible with the observed world.

This chapter follows that transition from foundational question to laboratory practice. It is partly historical, but the history is used for a pedagogical purpose. Each generation of experiments teaches us a new physical lesson: what exactly must be controlled before a violation of a Bell inequality can be regarded as compelling evidence against a local hidden-variable account?

6.1 From Schrödinger's diagnosis to Bell's experimental question

The conceptual story begins in 1935. Einstein, Podolsky, and Rosen asked whether the quantum state could be a complete description of physical reality [6]. Schrödinger, responding to the same debate, emphasized the peculiar structure that we now call *entanglement*: after two systems have interacted, the best description of the whole need not reduce to independent state descriptions of the parts [7]. That is exactly the phenomenon with which Chapter 1 began.

At this stage one could still imagine two broad possibilities. Perhaps quantum mechanics is complete, and the failure of definite local properties is genuinely part of nature. Or perhaps the quantum state is incomplete and some additional local variables—hidden from us—supply the definite facts that the wavefunction does not display.

For many years these possibilities were difficult to separate experimentally. Both camps agreed on a huge range of ordinary quantum predictions. Bell's achievement was to find a situation in which the alternatives could no longer hide behind the same numbers. His 1964 theorem showed that local hidden-variable models obey restrictions on correlations that quantum mechanics can violate [1].

The conceptual change introduced by Bell

Before Bell, the completeness debate could look like a dispute over the meaning of a successful formalism. After Bell, locality plus hidden variables became an experimentally constrained hypothesis. A foundational question had been turned into a quantitative laboratory question.

The direct intellectual trigger for John Clauser's later experimental programme was Bell's paper. Clauser has recalled that finding Bell's work while he was a graduate student redirected his career toward the foundations of quantum mechanics [14]. It is nevertheless useful to see the longer line of thought: Schrödinger recognized entanglement as the decisive new structure; Bell found a testable consequence of attempts to supplement quantum mechanics with local hidden variables; Clauser asked whether that test could actually be performed.

6.2 From Bell's ideal argument to an experiment one could build

The CHSH inequality derived in Chapter 4 was introduced in 1969 precisely to make Bell's insight experimentally practical [2]. This is an important point about the relation between theory and experiment. A theorem may be exact and still fail to specify a laboratory protocol. Real detectors miss events, polarizers have finite efficiencies, sources produce particles at finite rates, and one must formulate inequalities in terms of quantities that can actually be counted.

Clauser, Horne, Shimony, and Holt therefore recast Bell's logic into a form that could be tested with real correlation measurements. The laboratory question became:

Prepare many pairs, choose measurement settings independently on the two sides,
record the outcomes, build the correlation functions from the observed counts, and
ask whether the corresponding Bell combination remains inside the local bound.

This sounds straightforward only after the conceptual work has been done. A Bell experiment must simultaneously control the source, define two or more measurement settings on each side, identify which detection events belong to the same emitted pair, and gather enough trials that statistical fluctuations cannot plausibly imitate the predicted violation.

6.3 Freedman and Clauser, 1972: the first decisive optical test

The first widely recognized experimental Bell test was performed by Stuart Freedman and John Clauser in 1972 [8]. Their source used an atomic cascade in calcium. An excited calcium atom could emit two photons in succession, and the polarizations of those photons carried the quantum correlations to be tested.

A useful way to picture the experiment is:

$$\begin{array}{c} \boxed{\text{excited calcium atom}} \longrightarrow \boxed{\text{two correlated photons}} \\ \longrightarrow \boxed{\text{two distant polarization analyzers}} \longrightarrow \boxed{\text{coincidence counts}}. \end{array} \quad (390)$$

Each analyzer was set to a chosen orientation. The experiment was repeated many times, and the rate at which the two sides registered correlated events was measured for several pairs of analyzer settings.

The result agreed with the quantum prediction and violated the relevant local hidden-variable bound [8]. The significance was not that another atomic transition had been measured accurately. The experiment confronted a family of local hidden-variable descriptions with data and found the quantum correlations on the other side of the Bell bound.

Why doing the experiment was scientifically nontrivial

It is easy, with hindsight, to imagine that everyone must have welcomed such a fundamental test. Clauser's recollections describe almost the opposite atmosphere. He has said that many physicists regarded another test of quantum mechanics as unnecessary because the theory was already spectacularly successful. In a later Caltech interview he recalled that Richard Feynman objected strongly to the project and argued that quantum mechanics was already known to be correct and did not need this sort of retesting [15]. Clauser's Nobel interview similarly emphasizes how unfashionable the foundational question appeared at the time [14].

The scientific lesson is subtle. Feynman's confidence in the predictive power of quantum mechanics was amply justified. But Bell had changed the question. The proposed experiment was not merely asking whether quantum mechanics would again predict an atomic spectrum or scattering rate correctly. It was asking whether an entire alternative class of local hidden descriptions could survive. A theory can be extraordinarily successful and still motivate a new experiment when a newly discovered theorem identifies a previously untested distinction.

A useful lesson about foundational experiments

A foundational experiment is most valuable when competing conceptual pictures have been converted into different operational predictions. The point is not to test philosophy by opinion; it is to identify assumptions that constrain measurable probabilities and then test those constraints.

6.4 What is a loophole?

The word *loophole* appears frequently in discussions of Bell tests and can sound more mysterious than it is. A loophole is simply a mismatch between an assumption used in the theoretical inference and what the apparatus has actually guaranteed.

Suppose a theorem says:

If every emitted pair is sampled fairly, and if one laboratory cannot learn the other laboratory's setting in time to influence its outcome, then all local hidden-variable models satisfy a certain inequality.

An experiment can violate the inequality. But if the detector records only a small and potentially biased subset of the emitted pairs, or if the two sides have enough time to exchange information, then a critic can ask whether the assumptions of the theorem were truly realized.

Two loopholes therefore became especially important.

The detection, or fair-sampling, loophole

Real detectors do not register every photon. Imagine that the hidden variable of a pair influenced not only the measurement outcome but also whether that pair was likely to be detected. Then the detected subset might display stronger correlations than the complete ensemble. If one simply assumes that the detected events are representative of all emitted events, one has made a *fair-sampling assumption*.

Closing the detection loophole means arranging the experiment and the analysis so that the Bell conclusion does not depend on such an assumption.

The locality loophole

Bell's locality assumption requires that Alice's outcome not be influenced by Bob's setting, and vice versa. In a laboratory this becomes a spacetime question. The setting choice and measurement on one side should occur so quickly, and the two stations should be so well separated, that no signal travelling at or below the speed of light could carry the setting information to the other side in time to affect its result.

Thus the phrase *spacelike separated measurement events* has a concrete meaning: there is simply not enough time for an ordinary causal signal to connect the relevant events.

How to read a loophole

A loophole does not mean that quantum mechanics has failed. It means that the experimental data rule out local hidden-variable models only after one or more additional assumptions are made. Improving a Bell experiment means removing those extra assumptions one by one.

6.5 Aspect's experiments: changing the settings while the photons fly

Alain Aspect and collaborators carried out a sequence of increasingly refined Bell tests in the early 1980s. The most famous version used time-varying analyzers [9]. The central experimental idea can be stated without any optical jargon:

Do not allow the source to know, when the photons are emitted, which analyzer setting each photon will ultimately encounter.

In the 1982 experiment, rapidly operating optical switches directed each photon toward different analyzer orientations on a timescale short compared with the photon's flight through the apparatus. The measured correlations violated the Bell inequality and agreed with quantum mechanics [9].

Why was this improvement important? With fixed analyzer settings, an exotic local model could try to exploit information that was already present in the apparatus when the pair left the source. Rapidly changing the measurement configuration after emission sharply reduced this possibility and attacked an important locality-related loophole.

Aspect's work did *not* close every loophole. That qualification matters. A good experimental history should not describe science as a sequence of perfect final answers. Instead, Aspect transformed the Bell test into a much more stringent spacetime experiment and made clear what still had to be improved.

6.6 How an actual Bell experiment becomes a number

The conceptual derivation in Chapter 4 used ideal expectation values such as

$$E(a, b) = \langle A_a B_b \rangle.$$

In a real experiment this quantity is built from counts. Suppose each side produces outcomes $+1$ or -1 . For fixed settings a and b , let $N_{++}, N_{+-}, N_{-+}, N_{--}$ be the numbers of corresponding joint outcomes. Then the experimental correlation estimator is

$$E(a, b) = \frac{N_{++} + N_{--} - N_{+-} - N_{-+}}{N_{++} + N_{--} + N_{+-} + N_{-+}}. \quad (391)$$

The CHSH combination is then assembled from four such experimentally measured correlations,

$$\mathcal{S} = E(a_0, b_0) + E(a_0, b_1) + E(a_1, b_0) - E(a_1, b_1). \quad (392)$$

This equation is the bridge between Chapter 4 and the laboratory. The symbol $E(a, b)$ is not an abstract object supplied by nature. It is inferred from a large table of detector records. Statistical uncertainty on the counts becomes uncertainty on E and hence on $\mathcal{S}$.

A Bell violation must therefore be quoted together with a statistical significance or an equivalent hypothesis-test statement. One asks not merely whether the measured central value is larger than the local bound, but whether the observed excess is too large to be plausibly attributed to finite-sample fluctuations under the specified local model.

6.7 2015: closing the major loopholes simultaneously

By 2015 experimental techniques had advanced enough to close the major detection and locality loopholes in the same Bell test. Hensen and collaborators used electron spins associated with defects in diamond at laboratories separated by about 1.3 km, with photons used to establish entanglement between the distant matter systems [10]. In the same year, independent photonic experiments by Giustina and collaborators and by Shalm and collaborators reported loophole-free Bell violations with entangled photons [11, 12].

The conceptual significance is worth stating carefully. Earlier Bell tests already gave strong and increasingly persuasive agreement with quantum mechanics. The 2015 experiments were important because the inference against local hidden-variable models no longer required the traditional combination of fair-sampling and locality assumptions that earlier optical experiments had needed.

This illustrates how experimental physics progresses. The basic quantum prediction did not change between 1972 and 2015. What changed was the strength of the inference that could be drawn from the data.

6.8 The freedom-of-choice question and cosmic Bell tests

There is a still more radical logical escape route. Bell experiments assume that the choices of measurement settings are sufficiently independent of the hidden variables carried by the particle pair. If a common cause in the past were allowed to coordinate both the emitted pair and the supposedly random setting choices, the standard Bell inference could be undermined. This is often called the *freedom-of-choice* or *measurement-independence* loophole.

No finite experiment can eliminate every logically imaginable common cause without making some assumptions about causal structure. But the possible common cause can be pushed dramatically into the past. In a cosmic Bell test, astronomical photons are used to help choose the measurement settings. A 2018 experiment involving Anton Zeilinger and collaborators used light from distant quasars whose photons had been travelling for billions of years [13]. Any local-realist mechanism attempting to coordinate those astronomical setting choices with the laboratory pair source would therefore have to reach far back into cosmic history.

The point is not that quasars possess a mystical kind of randomness. The point is causal separation. The farther into the past the setting information is anchored, the more contrived a local common-cause explanation must become.

6.9 Zeilinger: from testing entanglement to controlling it

The later stages of the story change the central question once again. Bell experiments ask whether the quantum correlations are real in the operational sense that they defeat local hidden-variable bounds. Quantum information asks:

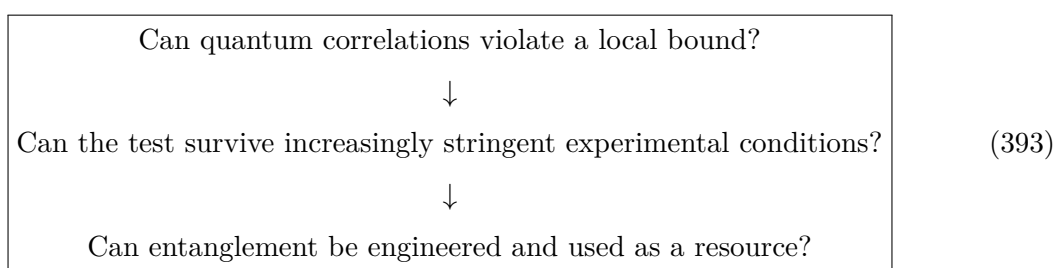
If these correlations are a genuine physical resource, can we deliberately prepare, transform, distribute, and consume them for useful tasks?

Anton Zeilinger and collaborators played a central role in this transition. Their experiments developed high-quality entangled-photon sources, long-distance Bell tests, quantum teleportation, and entanglement swapping. The teleportation and swapping ideas will be developed mathematically in the next two chapters; the laboratory implementations will be revisited later.

The historical progression is therefore not merely

Clauser $\rightarrow$ Aspect $\rightarrow$ Zeilinger.

It is a progression of scientific questions:



6.10 Why the 2022 Nobel Prize was about both foundations and technology

The 2022 Nobel Prize in Physics was awarded jointly to Alain Aspect, John Clauser, and Anton Zeilinger for experiments with entangled photons, establishing violations of Bell inequalities, and pioneering quantum information science [16]. The wording captures the two halves of the story.

The first half is foundational. Bell inequalities made local hidden-variable descriptions experimentally vulnerable, and progressively refined experiments found the correlations predicted by quantum mechanics.

The second half is technological. Once individual quantum systems and entangled states could be prepared and controlled reliably, entanglement ceased to be only a conceptual puzzle. It became a resource for teleportation, networks, cryptography, sensing, and quantum computation.

This is why the new experimental chapter belongs precisely here. Chapter 4 established the mathematical obstruction to local classical explanations. The present chapter showed how that obstruction became an experimental programme. The next chapter begins the resource programme: what can separated observers actually accomplish once shared entanglement is available?

6.11 Optional perspective: Bell tests are not the only tests of classical-looking reality

The Bell programme is one member of a larger family of experiments that probe which classical intuitions can coexist with quantum predictions.

Contextuality tests. The optional part of Chapter 4 used the Peres–Mermin square to show that one cannot consistently assign context-independent predetermined values to all observables while preserving the functional relations among compatible measurements. Experiments inspired by Kochen–Specker contextuality test this kind of noncontextual hidden-variable description. Spatial separation is not the essential ingredient; the key issue is whether the outcome assigned to an observable can be independent of the compatible measurement context in which it appears.

Leggett–Garg tests. Another family of inequalities asks whether a system can be regarded as possessing definite macroscopic properties at all times while those properties can, in principle, be measured without disturbing their later evolution. The assumptions are usually called *macrorealism* and *non-invasive measurability*. Quantum predictions can violate the corresponding Leggett–Garg inequalities. The logic resembles Bell's strategy—translate a classical-looking package of assumptions into an experimentally testable bound—but the physical organization is temporal rather than spatial.

These extensions are mentioned only to locate Bell's theorem in a wider landscape. The main line of these notes now turns from testing entanglement to using it.

6.12 A final map of the experimental story

It is useful to compress the chapter into one timeline:

Period	Step	What changed conceptually?
1935	EPR and Schrödinger	Entanglement and completeness become central foundational questions.
1964–1969	Bell and CHSH	Local hidden-variable explanations acquire experimentally testable bounds.
1972	Freedman–Clauser	A real optical experiment violates a Bell inequality.
1981–1982	Aspect and collaborators	Rapidly changing analyzer settings attack an important locality-related loophole.
2015	Hensen; Giustina; Shalm and collaborators	Major detection and locality loopholes are closed simultaneously.
2018	Cosmic Bell tests	Possible common causes exploiting setting choices are pushed billions of years into the past.
1990s onward	Zeilinger and many others	Entanglement becomes a controllable resource for information processing and networks.
2022	Nobel Prize	The foundations-to-technology arc is recognized as one connected scientific achievement.

The message is not that experiment has “proved quantum mechanics once and for all” in some interpretation-independent metaphysical sense. It is more precise and more powerful: increasingly assumption-light experiments have ruled out broad classes of local hidden-variable explanations and have confirmed the specific correlation structure predicted by quantum theory. The same experimental control that made those tests possible is what allows entanglement to be used in the chapters that follow.

7 Entanglement as a Resource: LOCC, No-Cloning, Teleportation, and Dense Coding

The preceding chapters established both the structure and the empirical status of entanglement. Chapters 1–3 developed the state-based description; Chapter 4 showed why Bell correlations cannot be reduced to local classical ignorance; and Chapter 6 showed how that distinction became an increasingly stringent experimental programme. We now turn from testing entanglement to using it. The question is:

What can two distant observers actually *do* with shared entanglement?

This question requires us to be precise about the physical setting. If Alice can simply hand her qubit to Bob, then transferring a quantum state is trivial: she sends the qubit. Teleportation becomes interesting only when Alice and Bob are already separated and are not allowed to exchange quantum systems during the protocol. They may act on systems in their own laboratories, and they may send ordinary classical messages. This operational restriction is called *LOCC*.

7.1 The operational setting: what does LOCC mean?

Suppose Alice controls one laboratory and Bob controls another. Alice may have several qubits in her laboratory, and Bob may have several qubits in his. The phrase *local operation* means local with respect to the division *Alice versus Bob*; it does *not* mean that Alice is restricted to single-qubit gates.

LOCC: local operations and classical communication

An *LOCC protocol* is any protocol built from the following ingredients:

- (i) Alice may perform arbitrary quantum operations on systems located in Alice’s laboratory: unitary gates, joint measurements, the introduction of local ancillas, and the discarding of local systems.
- (ii) Bob may independently perform arbitrary quantum operations on systems located in Bob’s laboratory.
- (iii) Alice and Bob may send ordinary *classical* messages to one another and may choose later local operations according to the messages received.

What is *not* included is the physical transmission of a qubit from one laboratory to the other, or a quantum gate acting jointly on an Alice system and a Bob system while they are spatially separated.

The last point is essential. A Bell measurement on two qubits both held by Alice is a perfectly legitimate *local* operation in the LOCC sense, even though the measurement acts jointly on two qubits and may itself involve entangling gates. What would be non-local is a gate acting on one of Alice’s qubits and one of Bob’s distant qubits.

Why introduce this restriction? Because it separates two resources that are otherwise easy to confuse:

- a *quantum channel*, through which an actual quantum system can be transmitted;
- previously shared *entanglement*, together with local actions and ordinary classical communication.

If a quantum channel from Alice to Bob were freely available, Alice could simply send the unknown qubit. Teleportation asks a more subtle question:

Can a previously shared entangled pair, together with LOCC, reproduce the *effect* of sending an unknown qubit?

The answer is yes. That is precisely what quantum teleportation does.

There is also a fundamental reason why the entangled pair counts as a genuine resource. LOCC cannot create entanglement across the Alice–Bob separation if none was present initially. The reason is already contained in the definition of separability from Chapter 3. If Alice performs a local measurement with outcome k and Bob responds locally after receiving k , then conditioned on that classical record the two laboratories still possess an ordinary product of local states. If the classical record is ignored, the final state has the form

$$\rho'_{AB} = \sum_k p_k \rho_A^{(k)} \otimes \rho_B^{(k)}, \quad (394)$$

which is precisely a separable state. More elaborate LOCC protocols simply repeat this logic in several rounds. Thus local operations can rearrange quantum information inside each laboratory, and classical messages can create classical correlations between the laboratories, but LOCC cannot manufacture entanglement across the spatial cut from nothing.

LOCC is therefore not a new dynamical law of quantum mechanics. It is the resource restriction appropriate to two already separated laboratories. It allows us to ask cleanly what pre-existing entanglement can accomplish when no new quantum system is transmitted between the parties.

Key idea

Quantum teleportation is a one-way LOCC protocol supplied with one pre-shared entangled pair. Alice performs a local joint measurement, sends Bob two classical bits, and Bob performs a local correction. No quantum system crosses from Alice to Bob during the teleportation protocol itself.

7.2 Why Alice cannot simply “read the state” and send the answer

Before constructing the protocol, it is useful to understand why the obvious classical strategies fail.

Suppose Alice is handed a single qubit in an unknown state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (395)$$

A measurement of one specimen gives one measurement outcome. It does not reveal the two amplitudes α and β . Full state tomography requires many identically prepared copies. Alice therefore cannot, from one unknown qubit, simply determine a classical description of $|\psi\rangle$ and email that description to Bob.

Could she first make many perfect copies and then perform tomography on the copies? Quantum mechanics forbids this as well.

The no-cloning theorem: first an intuitive example

Assume that a universal copying device exists. Let the second qubit begin in a fixed blank state $|0\rangle$. If the device copies the computational-basis states, it must satisfy

$$U|0\rangle|0\rangle = |0\rangle|0\rangle, \quad U|1\rangle|0\rangle = |1\rangle|1\rangle. \quad (396)$$

Now feed it the superposition

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}. \quad (397)$$

Linearity gives

$$U|+\rangle|0\rangle = \frac{1}{\sqrt{2}} (U|0\rangle|0\rangle + U|1\rangle|0\rangle) \quad (398)$$

$$= \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \quad (399)$$

But a perfect copy of $|+\rangle$ would instead be

$$|+\rangle|+\rangle = \frac{1}{2} (|00\rangle + |01\rangle + |10\rangle + |11\rangle), \quad (400)$$

which is a different state. A machine that copies $|0\rangle$ and $|1\rangle$ therefore cannot also copy their arbitrary superpositions.

The general proof is equally short. A universal copier would have to satisfy

$$U(|\psi\rangle|s\rangle) = |\psi\rangle|\psi\rangle, \quad U(|\phi\rangle|s\rangle) = |\phi\rangle|\phi\rangle \quad (401)$$

for arbitrary states $|\psi\rangle$ and $|\phi\rangle$. Unitary evolution preserves inner products, so

$$\langle\psi|\phi\rangle = (\langle\psi|\phi\rangle)^2. \quad (402)$$

For two distinct non-orthogonal states this is impossible. Thus there is no universal perfect quantum copier [17].

Remark

The word *unknown* matters. If Alice already knows a state, she may tell Bob how to prepare it. Teleportation addresses a different task: Alice is given one physical system in a state that the protocol must transfer correctly for *every* α and β , without either party learning those amplitudes.

No-cloning therefore blocks the strategy “copy first, learn the state, and send a classical description.” Teleportation succeeds by transferring the state while destroying the original local copy.

7.3 The teleportation task and its resources

Alice possesses an input qubit, labelled 1, in the unknown state

$$|\psi\rangle_1 = \alpha|0\rangle_1 + \beta|1\rangle_1. \quad (403)$$

Before the protocol begins, Alice and Bob have already distributed and stored a Bell pair

$$|\Phi^+\rangle_{23} = \frac{|00\rangle_{23} + |11\rangle_{23}}{\sqrt{2}}. \quad (404)$$

Alice holds qubits 1 and 2; Bob holds qubit 3. The spatial arrangement is therefore

$$\underbrace{(1, 2)}_{\text{Alice's laboratory}} \quad \Big| \quad \underbrace{(3)}_{\text{Bob's laboratory}}. \quad (405)$$

The vertical bar reminds us of the LOCC separation. During teleportation no qubit is allowed to cross this bar.

The total initial state is

$$|\Omega\rangle_{123} = |\psi\rangle_1 \otimes |\Phi^+\rangle_{23}. \quad (406)$$

The goal is that, at the end of the protocol, Bob's qubit 3 should be in the state $|\psi\rangle$ for arbitrary α and β .

One maximally entangled pair of qubits is said to contain one *ebit* of entanglement. Teleportation begins with one ebit already shared between Alice and Bob.

7.4 Why the Bell basis appears

Alice must somehow perform a measurement on the two qubits in her laboratory such that the result tells Bob what local correction to make. The Bell basis is exactly adapted to this task.

The four Bell states are

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \quad |\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}}, \quad (407)$$

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \quad |\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}. \quad (408)$$

They form an orthonormal basis for the two-qubit Hilbert space of Alice's qubits 1 and 2. Therefore Alice is allowed to perform a four-outcome projective measurement in this basis.

Checkpoint: organize the state before reading on

The three-qubit state is already known:

$$|\psi\rangle_1 |\Phi^+\rangle_{23}.$$

Before looking at the derivation, expand it in the computational basis and circle the first two qubit labels in every term. Those are Alice's qubits. The point of the calculation that follows is not to change the state, but to rewrite precisely those two labels in the basis in which Alice will measure.

At first sight the key teleportation identity can look mysterious. We now derive it slowly. Nothing miraculous happens in this derivation: we simply rewrite the *same* three-qubit vector in a basis suited to Alice's measurement.

Step 1: expand the initial state in the computational basis

Using Eqs. (403) and (404),

$$|\Omega\rangle_{123} = (\alpha|0\rangle_1 + \beta|1\rangle_1) \frac{|00\rangle_{23} + |11\rangle_{23}}{\sqrt{2}} \quad (409)$$

$$= \frac{1}{\sqrt{2}} (\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle). \quad (410)$$

We now group the first two qubits together, because those are the two qubits Alice will measure:

$$\begin{aligned} |\Omega\rangle_{123} = \frac{1}{\sqrt{2}} [& |00\rangle_{12} \alpha|0\rangle_3 + |01\rangle_{12} \alpha|1\rangle_3 \\ & + |10\rangle_{12} \beta|0\rangle_3 + |11\rangle_{12} \beta|1\rangle_3]. \end{aligned} \quad (411)$$

Step 2: rewrite Alice's computational states in the Bell basis

The Bell-state definitions can be inverted:

$$|00\rangle = \frac{|\Phi^+\rangle + |\Phi^-\rangle}{\sqrt{2}}, \quad |11\rangle = \frac{|\Phi^+\rangle - |\Phi^-\rangle}{\sqrt{2}}, \quad (412)$$

$$|01\rangle = \frac{|\Psi^+\rangle + |\Psi^-\rangle}{\sqrt{2}}, \quad |10\rangle = \frac{|\Psi^+\rangle - |\Psi^-\rangle}{\sqrt{2}}. \quad (413)$$

Substituting these four relations into Eq. (411) and collecting equal Bell states gives

$$\begin{aligned} |\Omega\rangle_{123} = \frac{1}{2} & \left[|\Phi^+\rangle_{12}(\alpha|0\rangle + \beta|1\rangle)_3 \right. \\ & + |\Phi^-\rangle_{12}(\alpha|0\rangle - \beta|1\rangle)_3 \\ & + |\Psi^+\rangle_{12}(\beta|0\rangle + \alpha|1\rangle)_3 \\ & \left. + |\Psi^-\rangle_{12}(\alpha|1\rangle - \beta|0\rangle)_3 \right]. \end{aligned} \quad (414)$$

The factor $1/2$ has a simple origin: one factor $1/\sqrt{2}$ came from the shared Bell pair and a second factor $1/\sqrt{2}$ came from rewriting a computational-basis vector in the Bell basis.

Step 3: recognize Bob's four possible states

Introduce the Pauli operators

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle, \quad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle. \quad (415)$$

Then

$$\alpha|0\rangle + \beta|1\rangle = |\psi\rangle, \quad (416)$$

$$\alpha|0\rangle - \beta|1\rangle = Z|\psi\rangle, \quad (417)$$

$$\beta|0\rangle + \alpha|1\rangle = X|\psi\rangle, \quad (418)$$

$$\alpha|1\rangle - \beta|0\rangle = XZ|\psi\rangle. \quad (419)$$

Therefore Eq. (414) becomes

$$\begin{aligned} |\psi\rangle_1 |\Phi^+\rangle_{23} = \frac{1}{2} & \left[|\Phi^+\rangle_{12} |\psi\rangle_3 + |\Phi^-\rangle_{12} Z|\psi\rangle_3 \right. \\ & \left. + |\Psi^+\rangle_{12} X|\psi\rangle_3 + |\Psi^-\rangle_{12} XZ|\psi\rangle_3 \right]. \end{aligned} \quad (420)$$

How to read the teleportation identity

Equation (420) is *not* a dynamical process and it does not say that the state has already travelled to Bob. It is one vector written in a new basis. Its usefulness is that every possible Bell outcome in Alice's laboratory is paired with a Bob state that differs from the original $|\psi\rangle$ only by one of four known Pauli operations.

This is why the Bell basis was chosen. It converts the problem of recovering an arbitrary unknown state into the much simpler problem of identifying which one of four discrete corrections is needed. Alice never learns α or β .

Now the algebra becomes a measurement statement. Before Alice measures, the right-hand side of Eq. (420) is a coherent superposition of four branches. The Bell states carried by Alice are mutually orthogonal, so her Bell measurement projects onto exactly one of them. For example, if Alice obtains $|\Phi^-\rangle_{12}$, the post-measurement state is proportional to

$$|\Phi^-\rangle_{12} \otimes Z|\psi\rangle_3. \quad (421)$$

Thus Bob's qubit is then $Z|\psi\rangle$. Nothing has to be guessed: the state multiplying Alice's observed Bell vector is precisely Bob's conditional state. The other three outcomes are read in exactly the same way.

A Bell measurement is also experimentally concrete. If one has already learned quantum gates, it can be implemented by applying the inverse of the standard Bell-state preparation circuit—a controlled-NOT followed by a Hadamard on Alice's first qubit—and then measuring Alice's two qubits in the computational basis. This entire circuit is local in the LOCC sense because both qubits remain inside Alice's laboratory.

Notice also that each Bell branch has amplitude $1/2$ multiplying a normalized state of Bob. Hence every Bell outcome occurs with probability

$$p(\Phi^+) = p(\Phi^-) = p(\Psi^+) = p(\Psi^-) = \frac{1}{4}, \quad (422)$$

independently of the unknown state $|\psi\rangle$.

7.5 The teleportation protocol as an LOCC protocol

We can now separate the algebra from the physical actions. The protocol has three active steps.

Step 1. Alice performs a Bell measurement on qubits 1 and 2. Both qubits are in Alice's laboratory, so this is a *local operation* in the LOCC sense. The measurement has four possible outcomes.

Step 2. Alice sends the outcome to Bob as two classical bits. Four possible outcomes require two bits to label:

$$\Phi^+ \leftrightarrow 00, \quad \Phi^- \leftrightarrow 01, \quad \Psi^+ \leftrightarrow 10, \quad \Psi^- \leftrightarrow 11. \quad (423)$$

These two bits do *not* contain α and β . They identify only which of the four Pauli-related branches Bob occupies.

Step 3. Bob performs a local Pauli correction. After receiving Alice's message, Bob applies the corresponding unitary to qubit 3. His qubit is then exactly in the state $|\psi\rangle$.

The protocol is summarized in figure 5. The vertical dashed line emphasizes the Alice–Bob separation: the only communication that crosses it during the protocol is the ordinary two-bit classical message. The green line denotes entanglement that was distributed *before* the protocol began.

The correction table is

Alice's Bell outcome	Message m	Bob's conditional state	Bob applies
$ \Phi^+\rangle$	00	$ \psi\rangle$	I
$ \Phi^-\rangle$	01	$Z \psi\rangle$	Z
$ \Psi^+\rangle$	10	$X \psi\rangle$	X
$ \Psi^-\rangle$	11	$XZ \psi\rangle$	XZ

The Pauli matrices are their own inverses up to an irrelevant global phase, so the listed correction recovers $|\psi\rangle$ in every case.

The resource accounting is therefore

$$\boxed{1 \text{ ebit} + 2 \text{ classical bits} + \text{LOCC} \longrightarrow \text{one perfectly transferred qubit state}} \quad (424)$$

The entangled pair is consumed in the process.

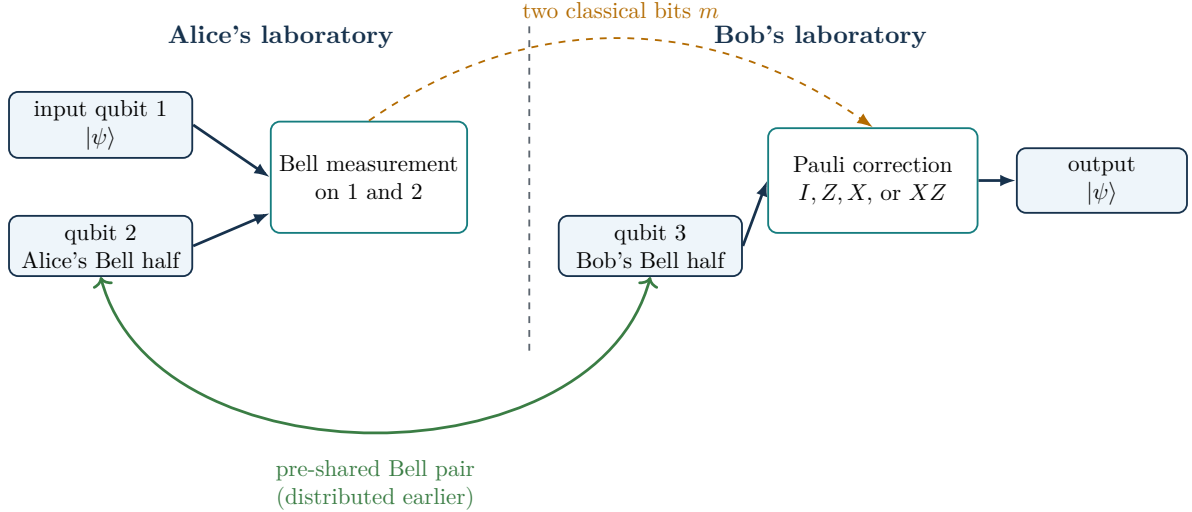


Figure 5: Quantum teleportation viewed explicitly as an LOCC protocol. Alice’s Bell measurement is local because both measured qubits are in her laboratory. The shared entangled pair was distributed beforehand. During teleportation itself, only two classical bits cross from Alice to Bob; Bob then performs a local correction.

7.6 Why the classical message is fundamentally necessary

At this point one might ask: if entanglement is the non-classical resource, why does Alice still have to send an ordinary message? Why can Bob not simply use his half of the Bell pair immediately after Alice’s measurement?

The answer is contained in Eq. (420). After Alice’s Bell measurement, Bob’s state is one of

$$|\psi\rangle, \quad Z|\psi\rangle, \quad X|\psi\rangle, \quad XZ|\psi\rangle, \quad (425)$$

with equal probability $1/4$. If Bob does not know Alice’s result, he must average over these four possibilities. Writing $\rho_\psi = |\psi\rangle\langle\psi|$, his local density matrix is

$$\begin{aligned} \rho_B^{\text{no message}} &= \frac{1}{4} \left(\rho_\psi + Z\rho_\psi Z + X\rho_\psi X + XZ\rho_\psi (XZ)^\dagger \right) \\ &= \frac{I}{2}. \end{aligned} \quad (426)$$

The result is independent of α and β . Before Alice’s classical message arrives, *every* experiment Bob can perform on his qubit has statistics that are completely independent of the state Alice was given.

This equation expresses the no-signalling principle in the teleportation protocol. Entanglement creates correlations, but it does not by itself provide a controllable message. Alice cannot choose which Bell outcome occurs. The two classical bits tell Bob which correlated branch has occurred and hence which local correction he should apply.

Common misconception

The two classical bits do not encode the quantum state. A generic qubit depends on continuous parameters α and β ; Alice’s message contains only one of four discrete labels. The detailed quantum information is carried by the pre-existing entangled correlations. The classical message merely tells Bob how to decode those correlations locally.

This also explains why teleportation cannot be used for faster-than-light communication. Bob cannot recover any information about $|\psi\rangle$ until the ordinary classical message reaches him.

7.7 Why pre-shared entanglement is fundamentally necessary

There is a particularly sharp way to see that the Bell pair is not a cosmetic addition. Imagine that Alice’s input qubit 1 is itself maximally entangled with a reference system R that Alice never touches. A perfect teleportation protocol must preserve this entanglement: after teleportation, R must be maximally entangled with Bob’s output qubit.

Now suppose Alice and Bob initially shared no entanglement and were allowed only LOCC. Across the bipartition

$$(R + \text{Alice}) \mid \text{Bob}, \quad (427)$$

there would initially be no entanglement. But perfect teleportation would leave R entangled with Bob, thereby creating entanglement across this cut. LOCC cannot do that from an initially separable state. Therefore perfect teleportation of an arbitrary state requires a non-local resource to have been supplied in advance: the shared entangled pair.

Key idea

The Bell pair is what makes the quantum transfer possible; the classical message is what makes the transfer usable by Bob. Both resources are necessary, and they play fundamentally different roles.

7.8 What exactly has been teleported?

The word “teleportation” can be misleading if taken too literally.

1. **No matter is teleported.** Bob’s physical qubit was already in Bob’s laboratory. What is transferred is the complete quantum state of the input system.
2. **Alice never learns the state.** Her Bell outcome is one of four random labels and contains no information about α and β .
3. **The original copy is destroyed.** Alice’s Bell measurement consumes the input together with her half of the entangled resource. At the end there is one usable copy at Bob’s location, not two. No-cloning is respected.
4. **Teleportation preserves entanglement with external systems.** If the input qubit is entangled with a reference R , then after the protocol the same correlations are reproduced between R and Bob’s output qubit. This is a stronger statement than merely reproducing the probabilities of a few measurements.

The last point gives the cleanest operational definition of perfect quantum state transfer: teleportation acts as the *identity quantum channel* from Alice’s input to Bob’s output, even when the input is part of a larger entangled state.

7.9 Superdense coding: the complementary use of an ebit

Teleportation uses one shared ebit and two classical bits to simulate the transfer of one qubit. Superdense coding uses the same shared ebit in the opposite direction: Alice physically sends one qubit and thereby communicates two classical bits [19].

Alice and Bob begin with $|\Phi^+\rangle$. Alice encodes a two-bit message by applying one of four local unitaries to her half:

Message	Alice applies	Shared Bell state becomes
00	I	$ \Phi^+\rangle$
01	Z	$ \Phi^-\rangle$
10	X	$ \Psi^+\rangle$
11	XZ	$ \Psi^-\rangle$

Alice then sends her qubit through a *quantum channel* to Bob. Bob now holds both qubits and performs a Bell measurement, which identifies the two-bit message.

It is useful to contrast the two protocols:

	Teleportation	Dense coding
Pre-shared resource	one ebit	one ebit
What crosses?	two classical bits	one physical qubit
LOCC protocol?	yes	no
Net effect	one qubit state transferred	two classical bits sent

Thus

$$1 \text{ ebit} + 2 \text{ classical bits} \longrightarrow 1 \text{ qubit of quantum communication}, \quad (428)$$

$$1 \text{ ebit} + 1 \text{ transmitted qubit} \longrightarrow 2 \text{ classical bits}. \quad (429)$$

These protocols make the resource character of entanglement concrete. Entanglement is not a signal and it is not a substitute for communication. Rather, once distributed, it can be combined with specified communication resources to change what distant parties are able to accomplish.

8 Entanglement Swapping and the Architecture of Quantum Networks

Chapter 7 taught us how to *use* entanglement once Alice and Bob already share it. Quantum teleportation, for example, begins with a Bell pair whose two halves are already present in the two laboratories. A communication network immediately raises the harder question:

How do we create useful entanglement between two laboratories that may be hundreds or thousands of kilometres apart?

Sending one member of a Bell pair directly across the entire distance is the simplest idea, but photons are lost in fibres and free-space links. Over a sufficiently long distance the probability that the transmitted photon arrives becomes extremely small. The central idea of a quantum network is therefore not to insist on one heroic long transmission. Instead we first create entanglement over shorter, more manageable links and then *join those links*. The operation that performs the joining is called *entanglement swapping*.

This section develops that idea carefully. We shall first understand a single swap at the level of four qubits. Only after the quantum-mechanical logic is clear will we turn to quantum repeaters and the engineering needed to make many such swaps work reliably.

Key idea

A quantum repeater does not receive an unknown qubit, copy it, amplify it, and send it onward. Instead it distributes short-distance entanglement, stores successful links, and joins them by entanglement swapping until the distant end nodes share entanglement.

8.1 The basic problem: two short links, but no long link

Consider four qubits, labelled 1, 2, 3, 4. Qubits 1 and 2 share a Bell pair, and qubits 3 and 4 share another Bell pair:

$$|\Phi^+\rangle_{12} \otimes |\Phi^+\rangle_{34}, \quad |\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \quad (430)$$

We imagine the following geography:

- qubit 1 is held by Alice at a left-hand node;
- qubits 2 and 3 are brought to the *same intermediate station*;
- qubit 4 is held by Bob at a right-hand node.

Initially Alice is entangled with the intermediate station through the pair 1–2, while Bob is entangled with the intermediate station through the pair 3–4. Alice and Bob are *not* entangled with each other.

The goal is to convert

$$(1-2) + (3-4) \longrightarrow 1-4, \quad (431)$$

without ever sending qubit 1 to Bob or qubit 4 to Alice.

The geometry is shown in figure 6. The important detail is that qubits 2 and 3 are co-located. The intermediate station can therefore perform an ordinary *local* two-qubit Bell measurement on them.

8.2 Deriving the swapping identity slowly

The statement that a measurement on 2 and 3 can entangle 1 and 4 may sound mysterious. The algebra becomes much less mysterious if we do it in small steps.

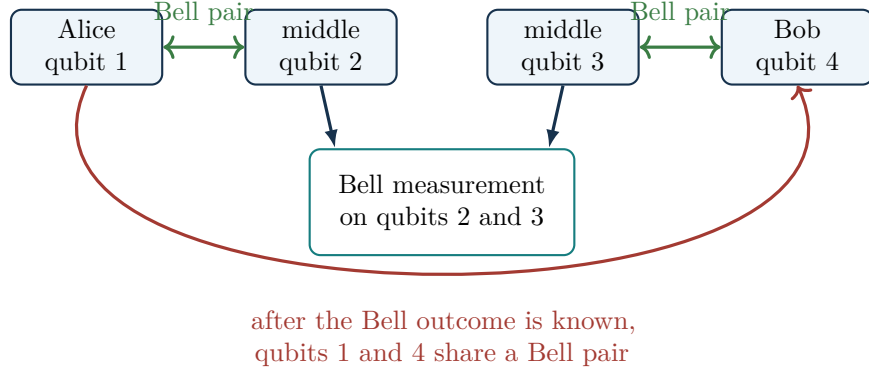


Figure 6: The logic of entanglement swapping. The intermediate station holds qubits 2 and 3 and performs a Bell measurement on them. Conditioned on that measurement outcome, the distant qubits 1 and 4 are projected into a Bell state even though they have never interacted directly.

Step 1: expand the two Bell pairs

Using the definition of $|\Phi^+\rangle$, the initial state is

$$\begin{aligned} |\Phi^+\rangle_{12}|\Phi^+\rangle_{34} &= \frac{1}{2}(|00\rangle_{12} + |11\rangle_{12})(|00\rangle_{34} + |11\rangle_{34}) \\ &= \frac{1}{2}(|0000\rangle + |0011\rangle + |1100\rangle + |1111\rangle), \end{aligned} \quad (432)$$

where the order of the qubits is 1, 2, 3, 4.

At this stage the expression is organized according to the original pairs (1, 2) and (3, 4). But the measurement will be performed on the *middle* qubits 2 and 3. We should therefore regroup the same four basis vectors according to the pair (2, 3) and the remaining pair (1, 4). Reading each four-bit string in that order gives

$$\begin{aligned} |\Phi^+\rangle_{12}|\Phi^+\rangle_{34} &= \frac{1}{2}(|00\rangle_{23}|00\rangle_{14} + |01\rangle_{23}|01\rangle_{14} \\ &\quad + |10\rangle_{23}|10\rangle_{14} + |11\rangle_{23}|11\rangle_{14}). \end{aligned} \quad (433)$$

Nothing physical has happened in going from equation (432) to equation (433). We have only rearranged the labels so that the pair we are about to measure appears first.

Checkpoint: predict the structure of the answer

The middle pair (2, 3) is about to be measured in the Bell basis. Before performing the substitution, ask what form the answer must have if every Bell outcome at the middle station is to leave the outer pair (1, 4) in a definite conditional state. The calculation below will determine which Bell state of (1, 4) accompanies each Bell state of (2, 3).

Step 2: rewrite the middle pair in the Bell basis

The Bell states obey

$$\begin{aligned} |00\rangle &= \frac{|\Phi^+\rangle + |\Phi^-\rangle}{\sqrt{2}}, & |11\rangle &= \frac{|\Phi^+\rangle - |\Phi^-\rangle}{\sqrt{2}}, \\ |01\rangle &= \frac{|\Psi^+\rangle + |\Psi^-\rangle}{\sqrt{2}}, & |10\rangle &= \frac{|\Psi^+\rangle - |\Psi^-\rangle}{\sqrt{2}}. \end{aligned} \quad (434)$$

Substituting these relations into equation (433), the cross terms cancel pairwise and we obtain

$$|\Phi^+\rangle_{12}|\Phi^+\rangle_{34} = \frac{1}{2} \left(|\Phi^+\rangle_{23}|\Phi^+\rangle_{14} + |\Phi^-\rangle_{23}|\Phi^-\rangle_{14} + |\Psi^+\rangle_{23}|\Psi^+\rangle_{14} + |\Psi^-\rangle_{23}|\Psi^-\rangle_{14} \right). \quad (435)$$

This equation is the swapping analogue of the teleportation identity derived in Chapter 7. It should be read in exactly the same way: it is *not* a time-evolution equation. It is the same four-qubit state written in a basis adapted to the measurement that the intermediate station intends to perform.

Key idea

Before the Bell measurement, the state already contains a perfect correlation between two statements:

“which Bell state will the middle pair 2, 3 be found in?”

and

“which Bell state will the outer pair 1, 4 then occupy?”

The Bell measurement reveals one branch of this pre-existing global correlation.

8.3 What the Bell measurement actually does

Now the intermediate station measures qubits 2 and 3 in the Bell basis. There are four possible outcomes. From equation (435), each occurs with probability 1/4.

Suppose, for example, that the result is $|\Psi^+\rangle_{23}$. Projection onto that outcome removes the other three terms in equation (435); after normalization the remaining state is

$$|\Psi^+\rangle_{23} \otimes |\Psi^+\rangle_{14}. \quad (436)$$

The middle qubits have been measured, while the distant pair 1, 4 is now in the Bell state $|\Psi^+\rangle_{14}$.

The same reasoning applies to every outcome:

Bell outcome on 2, 3	State of remote pair 1, 4	Correction to obtain $ \Phi^+\rangle_{14}$
$ \Phi^+\rangle_{23}$	$ \Phi^+\rangle_{14}$	I
$ \Phi^-\rangle_{23}$	$ \Phi^-\rangle_{14}$	Z
$ \Psi^+\rangle_{23}$	$ \Psi^+\rangle_{14}$	X
$ \Psi^-\rangle_{23}$	$ \Psi^-\rangle_{14}$	XZ

The Pauli correction may be applied to either member of the remote pair. For concreteness we may imagine Bob applying it to qubit 4.

Notice what has happened. Qubits 1 and 4 have never met. Nevertheless, a joint measurement on the two intermediate qubits has changed the *pattern* of entanglement: two short entangled links have been consumed and a longer entangled link has been produced.

Common misconception

Entanglement swapping does not create entanglement from nothing. The initial state already contains two Bell pairs. The Bell measurement consumes the two short links and, conditioned on its outcome, leaves one Bell pair between the outer nodes. It is better to think of the protocol as *rerouting* or *reconnecting* pre-existing entanglement.

8.4 Where LOCC enters

The discussion of LOCC in Chapter 7 is especially useful here. There are now three laboratories: Alice, an intermediate station, and Bob. The intermediate station holds both qubits 2 and 3, so its Bell measurement is a perfectly ordinary *local* quantum operation at that station. After the measurement, the station sends its two-bit outcome to the distant nodes by an ordinary classical channel. Bob may then perform the appropriate local Pauli correction.

Thus the swapping protocol consists of

$$\boxed{\text{local Bell measurement at the middle node} + \text{classical announcement} + \text{local correction.}} \quad (437)$$

It is therefore an LOCC protocol among the network nodes, *given the two short Bell pairs as pre-existing resources*.

There is an important geographical qualification. A Bell measurement is “local” here only because qubits 2 and 3 are physically available in the same station. If those two qubits were themselves held at distant sites, one could not simply declare a nonlocal Bell measurement to be an LOCC operation. One would first need to bring quantum systems together or use additional nonlocal resources.

This observation explains the architecture of a repeater node: it is a place where the two quantum links meeting from the left and right are made locally accessible so that a Bell-state measurement can join them.

8.5 Why the classical outcome is indispensable

Why must the intermediate station announce its Bell-measurement result? Why can Alice and Bob not simply use their newly created pair immediately?

Checkpoint: conditioned versus unconditioned

If you know that the middle outcome was, say, $|\Psi^+\rangle_{23}$, the outer pair is pure and entangled. What should you expect if the classical record is completely erased and all four outcomes are equally likely? Compute the mixture before reading the next line.

If the outcome is unknown, Alice and Bob must average over all four possible Bell states. Their state is then

$$\begin{aligned} \rho_{14}^{\text{unconditioned}} &= \frac{1}{4} (|\Phi^+\rangle\langle\Phi^+| + |\Phi^-\rangle\langle\Phi^-| + |\Psi^+\rangle\langle\Psi^+| + |\Psi^-\rangle\langle\Psi^-|) \\ &= \frac{I_{14}}{4} = \frac{I_1}{2} \otimes \frac{I_4}{2}. \end{aligned} \quad (438)$$

This state is completely mixed and separable. It contains no usable entanglement between Alice and Bob.

The entangled state exists *conditionally*: Alice and Bob must know which Bell branch occurred. The classical message is therefore not bookkeeping added after the quantum protocol; it is part of what makes the distributed entanglement operationally available.

This also makes causality transparent. The intermediate measurement cannot instantaneously give Alice and Bob a usable shared Bell pair. Until the ordinary classical announcement arrives, their local and joint operational statistics are exactly those of the unconditioned state in equation (438).

The word *heralded* is used for precisely this situation. A heralding signal tells the users that a desired quantum event has occurred and, when necessary, tells them which conditional state was produced.

8.6 Swapping as teleportation of half of an entangled pair

There is a useful way to remember the entire protocol. Regard qubit 2 as Alice's half of the Bell pair 1–2. Now use the pair 3–4 as the teleportation resource and teleport the state of qubit 2 to qubit 4. Because qubit 2 was entangled with qubit 1, teleportation must preserve that external entanglement. Consequently the entanglement that originally joined 1 to 2 is transferred so that it joins 1 to 4.

In this sense,

entanglement swapping is teleportation applied to one half of an entangled pair.

This is why the algebra of swapping looks so similar to the algebra of teleportation, and it is also why teleportation must preserve correlations with external reference systems, a point emphasized in Chapter 7.

8.7 From one swap to a network: why direct transmission fails

We now turn from the four-qubit thought experiment to the engineering problem. Suppose Alice and Bob are separated by a distance L . In a lossy optical channel, the transmission probability has the generic form

$$\eta(L) \sim e^{-L/L_{\text{att}}}, \quad (439)$$

where L_{att} is an attenuation length determined by the physical channel. The important point is the exponential dependence: making the route twice as long can make successful direct transmission dramatically less likely.

A classical repeater solves an analogous problem by detecting a weak signal, regenerating it, and sending a fresh copy onward. That strategy is unavailable for an arbitrary unknown quantum state. Measuring the state generally disturbs it, and the no-cloning theorem forbids a device that makes a perfect copy of every unknown input state.

The quantum solution changes the task. Instead of trying to preserve one unknown carrier over the whole distance, divide the route into shorter segments,

$$A - - - R_1 - - - R_2 - - - \dots - - - R_N - - - B, \quad (440)$$

and establish entanglement on those short segments first. Once neighbouring segments are ready, entanglement swapping joins them into longer ones [22].

8.8 A quantum repeater as a waiting-and-joining machine

The easiest way to understand a repeater is to follow one cycle in the order in which events actually occur. The basic architecture is illustrated in figure 7.

1. Attempt to create each elementary link

Neighbouring nodes repeatedly try to create an entangled pair across a short segment. A single attempt may fail because the photon is lost or a detector does not click. Failure is therefore expected; the protocol is designed to try again.

2. Herald which attempts succeeded

A successful detector pattern produces a classical heralding signal. The nodes now know that a particular elementary link contains entanglement. This is essential: an unheralded “perhaps there is a Bell pair” is not a useful network resource.

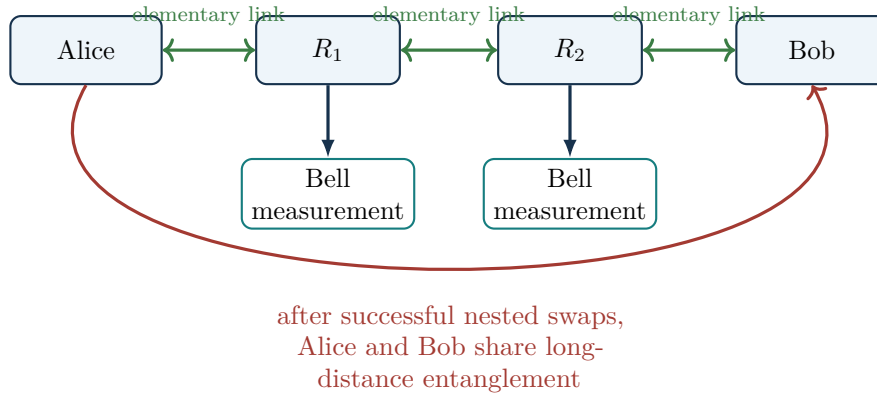


Figure 7: The elementary repeater idea. Short entangled links are generated first. Successful links are stored, and Bell measurements at intermediate nodes join neighbouring links. Repeating the procedure in stages eventually creates entanglement between the end nodes.

3. Store a successful link while waiting for the others

Different links do not usually succeed in the same trial. Perhaps the Alice– R_1 link succeeds now while the R_1 – R_2 link fails ten more times. If the first successful pair were immediately lost, the network would need all links to succeed simultaneously, which would be extremely unlikely.

A *quantum memory* solves this synchronization problem. It stores the qubit belonging to a successful link while neighbouring links continue to be attempted. A useful memory must preserve not only a classical bit value but quantum coherence and entanglement.

Key idea

Quantum memories are not merely convenient storage devices. They change the probability problem. They allow successful elementary links created at different times to be accumulated and later joined, instead of demanding that all links succeed in the same clock cycle.

4. Perform a Bell measurement when two neighbouring links are ready

Once the links on the two sides of a repeater node have both succeeded, the node performs a Bell-state measurement on its two stored qubits. This is exactly the four-qubit swapping protocol derived above. A classical message announces the outcome, and the two outer ends of those links now share a longer entangled pair, up to a known Pauli correction.

5. Repeat the process in nested levels

The longer links produced by the first round can themselves be joined. For example, two links of length L_0 become one of length approximately $2L_0$; two such links can then become one of length approximately $4L_0$, and so on. The procedure resembles building a bridge from prefabricated segments rather than trying to cast the entire bridge in one piece.

The final output of the repeater chain is not necessarily a qubit that has physically travelled from Alice to Bob. The fundamental output is a high-quality Bell pair shared between the end nodes. Alice and Bob may then consume that Bell pair for teleportation, cryptography, distributed computation, or another network task.

8.9 What makes a real repeater difficult?

The ideal description above consists of a few lines of algebra. A laboratory must make several probabilistic and imperfect processes work together. It is useful to separate the main engineering

requirements and understand why each one is present.

Photon loss and detector efficiency

Most link attempts fail simply because the transmitted quantum carrier never produces the required detection event. Better sources, lower-loss channels, large collection efficiencies, and efficient detectors all increase the rate at which elementary entanglement can be heralded. Detector dark counts create the opposite problem: they can falsely announce an event that did not really occur, reducing the fidelity of the supposed Bell pair.

Quantum-memory lifetime

A stored qubit must remain coherent while the network waits for other links and for classical heralding signals to arrive. If the memory decoheres before its neighbours succeed, the advantage of storing a successful link is lost. The relevant memory time is therefore set by the stochastic waiting time of the network, not merely by the duration of a local gate.

Indistinguishability and synchronization

Many photonic Bell measurements rely on quantum interference. Photons arriving from independent links must therefore be sufficiently indistinguishable in frequency, polarization, temporal wave packet, and spatial mode. They must also arrive at the interferometer at the correct time. A repeater network is consequently a timing and mode-matching problem as well as a quantum-information problem.

Imperfect local operations

Bell measurements, local gates, memory readout, and detectors are never perfect. Errors accumulate as more swapping levels are performed. A long network can therefore have an excellent probability of producing *some* pair while the resulting pair has poor fidelity. Success probability and state quality must be optimized together.

Classical latency

Heralding and feed-forward require ordinary classical signals. A station cannot know immediately that a distant elementary link succeeded; it must wait at least for the appropriate light-travel time through the classical channel. This latency is another reason why long-lived memories are important.

Multiplexing

One way to improve rates is to attempt many elementary links in parallel, for example in different time bins, frequencies, spatial modes, or memory cells. This is called *multiplexing*. It does not remove loss, but it gives the network more chances per unit time to obtain at least one successful link.

8.10 Fidelity, purification, and the rate–quality tradeoff

An experimentally distributed pair is described by a density matrix rather than an ideal Bell vector. If the target is $|\Phi^+\rangle$, a simple measure of quality is the Bell-state fidelity

$$F = \langle \Phi^+ | \rho_{AB} | \Phi^+ \rangle. \quad (441)$$

A network must care about at least two distinct quantities:

- **fidelity**: how close each delivered pair is to the desired Bell state;
- **rate**: how many useful pairs are delivered per unit time.

These goals compete. A stringent protocol may deliver exceptionally clean pairs very rarely, whereas a fast protocol may produce many pairs whose noise is too large for the intended application.

One strategy for dealing with noise is *entanglement purification* or *distillation* [23]. Alice and Bob begin with several imperfect but still entangled pairs. They apply coordinated local operations, measure and discard some pairs, and compare classical outcomes. Conditioned on suitable outcomes, the remaining pairs can have higher fidelity.

Purification is deliberately wasteful: it trades several low-quality pairs for fewer high-quality ones. It also illustrates again the logic of LOCC. Local operations and classical communication cannot manufacture entanglement from a separable input; they can only select and concentrate entanglement that was already present in noisy form.

Remark

Modern repeater proposals do not all use the same architecture. Some rely heavily on heralding, quantum memories, and two-way purification; others use quantum error-correcting codes and more nearly one-way operation. The details belong to a more advanced quantum-information course. The conceptual core, however, is common: loss and noise must be prevented from compounding catastrophically as the network distance grows.

8.11 What a quantum network is meant to distribute

A classical network moves classical messages. A quantum network may also transmit quantum states directly, but its more fundamental task is often to *establish entanglement on demand between remote nodes*. Once a high-quality Bell pair is available, it can be consumed as a resource for

- teleportation between separated quantum processors;
- entanglement-based quantum key distribution;
- distributed quantum computation;
- networked sensing, clocks, and interferometry;
- tests of quantum correlations over large separations.

The phrase *quantum internet* can therefore be misleading if it suggests a replacement for the classical internet. A useful quantum network needs a classical layer at every stage: to herald successful links, synchronize clocks, announce Bell outcomes, route requests, perform feed-forward corrections, and coordinate error handling. The quantum and classical layers are complementary.

We can now summarize the architecture in one sentence:

short links create entanglement; memories wait; Bell measurements join the links; classical messages herald what happened; and the process is repeated until distant end nodes share a usable Bell pair.

The next question is no longer conceptual but physical: what objects serve as qubits, how are Bell pairs generated, how are Bell measurements implemented, and how can the resulting quantum signals be sent through fibres, free space, and satellite links? We turn to those questions in the next chapter.

9 From the Blackboard to the Laboratory and to Space

Up to this point we have deliberately spoken the language of ideal quantum information: qubits, Bell pairs, Bell measurements, Pauli corrections, quantum channels, and LOCC. A laboratory does not contain any object literally labelled “qubit” or “Bell measurement.” It contains lasers, crystals, mirrors, fibres, beam splitters, wave plates, detectors, electronic clocks, and computers.

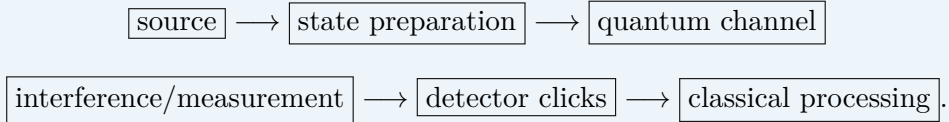
The purpose of this section is to build a dictionary between these two languages. We shall repeatedly ask the same question:

What physical piece of apparatus is playing the role of the mathematical object that appeared in the protocol?

This is also where some of the idealizations made earlier become visible. Perfect Bell pairs become noisy photon pairs. A perfect measurement becomes a set of detector clicks. A lossless quantum channel becomes an optical fibre in which most sufficiently distant photons never arrive. A deterministic Bell measurement may become probabilistic. The abstract protocol remains the same, but an experiment must keep track of the probability that each physical step actually succeeds.

Key idea

A useful way to think about an optical quantum experiment is as a chain



Every abstract quantum-information protocol must eventually be translated into such a physical chain.

9.1 First bridge: what is a photonic qubit?

A photon has many physical degrees of freedom: polarization, position, momentum, frequency, arrival time, and so on. A *degree of freedom* simply means an independent physical feature whose state can vary. A single photon may, for example, arrive early or late, travel along one path or another, and have one polarization or another. We do not try to use all of these possibilities at once. To make a qubit, we choose two reliably controllable and distinguishable alternatives and call them $|0\rangle$ and $|1\rangle$.

The important point is that the qubit is not “the photon” as a whole. It is one chosen two-dimensional degree of freedom carried by the photon. Other degrees of freedom are still physically present, and in a real experiment they must often be controlled so that they do not accidentally reveal which quantum alternative occurred.

Polarization encoding

The most intuitive choice is polarization:

$$|0\rangle \equiv |H\rangle, \quad |1\rangle \equiv |V\rangle, \quad (442)$$

where H and V mean horizontal and vertical polarization. A general qubit is therefore

$$|\psi\rangle = \alpha|H\rangle + \beta|V\rangle, \quad |\alpha|^2 + |\beta|^2 = 1. \quad (443)$$

This equation has a very concrete meaning. It does *not* say that a classical photon is secretly either horizontal or vertical and that we do not know which. It says that, before measurement,

the polarization state is a coherent superposition. Wave plates rotate this state, and a polarizing beam splitter converts the two polarization alternatives into two different output paths that can be detected separately.

Path encoding

Instead of using two polarizations, we may use two possible paths through an interferometer:

$$|\psi\rangle = \alpha|\text{upper}\rangle + \beta|\text{lower}\rangle. \quad (444)$$

A beam splitter creates or recombines the two alternatives, while a phase shifter controls their relative phase. Path encoding makes the idea of superposition almost visible: the two basis states literally correspond to two spatial routes through the apparatus.

Time-bin encoding

For long optical fibres, a particularly useful encoding uses two possible arrival times,

$$|\psi\rangle = \alpha|e\rangle + \beta e^{i\varphi}|l\rangle, \quad (445)$$

where $|e\rangle$ denotes an early wavepacket and $|l\rangle$ a late one. A *wavepacket* is simply a photon pulse localized within a finite time window, rather than an infinitely long monochromatic wave. An *unbalanced interferometer* is an interferometer whose two arms have different lengths: the short arm produces the early arrival and the long arm produces the late arrival. A second interferometer can later bring the two possibilities back together so that their relative phase can be measured.

Because the early and late pulses travel through almost the same physical fibre, slow environmental fluctuations tend to affect them similarly. This is one reason time-bin encoding is attractive for communication. The engineering term therefore has a simple quantum meaning: the computational basis is stored in *when* the photon arrives.

Remark

There is no universally best photonic qubit. Polarization is conceptually simple, path encoding is natural on an optical chip, and time-bin encoding is well suited to fibre. The abstract state $\alpha|0\rangle + \beta|1\rangle$ is the same mathematics in all three cases; what changes is the hardware used to prepare, manipulate, and measure it.

9.2 What an optical experiment actually looks like

It is useful to translate a few objects from Chapters 6 and 7 into laboratory language:

Blackboard object	Typical optical realization
Qubit	polarization, path, or time-bin state of a photon
State preparation	laser pulses, single-photon source, wave plates, interferometers, modulators
Bell pair	correlated photon pair produced by a nonlinear source or another entangled emitter
Unitary rotation	wave plate, phase shifter, electro-optic modulator
Projective measurement	optical basis rotation followed by a beam splitter or polarizing beam splitter and photon detectors
Bell measurement	interference of two photons followed by a characteristic multi-detector click pattern
Classical message	ordinary electrical or optical digital signal
Feed-forward	electronics uses a detector outcome to control a later modulator or gate
Quantum channel	optical fibre or free-space optical link

This table is worth keeping in mind when reading an experimental paper. The paper may contain a complicated photograph of an optical bench, but the logic underneath is often just a physical implementation of one of the simple circuits drawn earlier.

9.3 How an entangled photon pair is made

The name of the standard optical process sounds more complicated than the physical idea. We send light of relatively high photon energy into a specially chosen crystal. Very rarely, the interaction converts one quantum of the incoming optical field into a pair of lower-energy photons. If the apparatus is designed so that two alternative ways of producing the pair are indistinguishable, those alternatives can be added coherently and the outgoing photons can be entangled.

A common source based on this idea is *spontaneous parametric down-conversion* (SPDC). The incoming laser beam is traditionally called the *pump*. The crystal is called *nonlinear* because its optical response is not simply proportional to the applied electric field; that nonlinearity allows different optical frequencies to couple to one another. With a small probability, the interaction produces two lower-energy photons, traditionally called the *signal* and *idler*. These are historical labels for the two daughter photons, not names for two different particle species.

At the level of frequencies, energy conservation requires approximately

$$\omega_p = \omega_s + \omega_i, \quad (446)$$

and the crystal geometry is chosen so that the corresponding momentum condition is also satisfied. In optics this is usually called the *phase-matching condition*. Physically, contributions to the pair-production amplitude are generated throughout the crystal. For the desired outgoing directions and frequencies, those contributions should add constructively rather than cancelling one another. Phase matching is the condition that makes this coherent build-up possible. The phrase therefore encodes ordinary wave interference plus momentum conservation inside the optical medium.

The word *spontaneous* matters. A pump pulse normally produces no pair at all. Occasionally it produces one pair; more rarely, it produces two or more. For a weak source one may schematically write

$$|\text{output}\rangle \approx |\text{vac}\rangle + \sqrt{p}|1_s 1_i\rangle + O(p), \quad p \ll 1. \quad (447)$$

The notation in Eq. (447) is deliberately schematic. $|\text{vac}\rangle$ means that no daughter photons were produced in the two selected output modes; $|1_s 1_i\rangle$ means one signal photon and one idler photon. Because the pair-production probability p is small, the amplitude for one pair is of order $\sqrt{p}$, while terms involving two or more pairs are smaller and are hidden in the $O(p)$ notation.

The vacuum term explains why most pump pulses produce no useful detector event. The higher-pair terms explain why simply increasing the laser power eventually creates errors: two independent pairs can imitate the detector pattern expected from one entangled pair. This rate–quality tradeoff is one of the recurring lessons of experimental quantum optics.

Where does the entanglement come from?

The nonlinear crystal by itself does not magically label a pair “entangled.” The apparatus must create two physically indistinguishable ways to obtain the same final detection event. Suppose the two alternatives are

$$\text{signal } H, \quad \text{idler } V, \quad (448)$$

$$\text{signal } V, \quad \text{idler } H. \quad (449)$$

If no measurement can reveal which alternative occurred, quantum mechanics adds their *amplitudes*, not their probabilities. The emitted state can then take the form

$$|\Psi\rangle_{si} = \frac{1}{\sqrt{2}} \left(|H\rangle_s |V\rangle_i + e^{i\phi} |V\rangle_s |H\rangle_i \right). \quad (450)$$

The relative phase ϕ is controlled by optical path lengths and compensating optical elements. When $\phi = \pi$, for example, this becomes the polarization version of the singlet state.

This gives a useful physical lesson:

Entanglement is produced when different alternatives leading to the same physical event are made coherent and indistinguishable.

A real source must therefore control timing, frequency, spatial mode, polarization, and phase well enough that unwanted “which-process” information does not leak into the environment.

9.4 Detector clicks, coincidences, and heralding

A photodetector usually does not tell us a complex amplitude. It produces a classical event: a *click* in a particular time window. If two detectors click within an agreed coincidence window, the electronics records a *coincidence*.

This is where the generalized-measurement language of Chapter 5 becomes physical. An ideal two-outcome polarization measurement may be represented by two projectors, but a real apparatus can also report “no click,” a dark-count event, or more than one click. Those laboratory records are naturally represented by different POVM outcomes. The formalism was not an abstract generalization added for mathematical completeness; it was designed precisely to describe measurements of this kind.

This is how many optical quantum experiments are read. One does not watch a wavefunction collapse on a screen. One repeats the experiment thousands or millions of times and builds statistics from time-tagged detector clicks.

Suppose the detection of photon s tells us that its partner i has probably been created. Then the click of s *heralds* the presence of i . This is the laboratory meaning of the word *heralded* that appeared in the quantum-repeater discussion. A heralding signal does not make the event certain in advance; it tells us, after the relevant detector pattern occurs, which trials should be kept as successful ones.

Dark counts, imperfect efficiency, detector dead time, and accidental coincidences all complicate this simple picture. They are not changes to quantum mechanics; they are imperfections in the physical implementation of the measurement.

9.5 How can two photons perform a Bell measurement?

The Bell measurement used in teleportation and entanglement swapping sounds abstract: “measure two qubits in the Bell basis.” With photons, the central physical trick is *two-photon interference* at a beam splitter.

Before using operators, consider the physical alternatives. One identical photon enters from the left and one from the right. For the event “one photon leaves by each output port,” there are two indistinguishable quantum ways for this to happen: both photons can be transmitted, or both can be reflected. At a symmetric 50:50 beam splitter the amplitudes for these two alternatives have opposite relative phase. They cancel. The event with one photon in each output is therefore suppressed, and the photons emerge together from the same port. This is the essence of the Hong–Ou–Mandel effect.

We can now encode that argument compactly. A *spatial mode* means one distinguishable optical route through the apparatus. The creation operator $a^\dagger$ means “add one photon to mode a ”; similarly for the other modes. Thus two input spatial modes are described by creation operators $a^\dagger$ and $b^\dagger$, and the two output modes by $c^\dagger$ and $d^\dagger$. A symmetric 50:50 beam splitter may be represented, up to phase conventions, by

$$a^\dagger \longrightarrow \frac{c^\dagger + d^\dagger}{\sqrt{2}}, \quad b^\dagger \longrightarrow \frac{c^\dagger - d^\dagger}{\sqrt{2}}. \quad (451)$$

Now send in one otherwise identical photon through each input port. The input contains the factor $a^\dagger b^\dagger$. Using Eq. (451),

$$a^\dagger b^\dagger \longrightarrow \frac{1}{2}(c^\dagger + d^\dagger)(c^\dagger - d^\dagger) \quad (452)$$

$$= \frac{1}{2}[(c^\dagger)^2 - (d^\dagger)^2]. \quad (453)$$

The cross term $c^\dagger d^\dagger$ has cancelled. Thus the two identical photons tend to leave through the *same* output port rather than one through each. This is the famous Hong–Ou–Mandel interference effect.

Why is this useful for Bell measurements? Different two-photon Bell states behave differently when the two photons are exchanged. This property is called *exchange symmetry*. The optical network converts that symmetry into a pattern of detector clicks: some states favour both photons leaving together, whereas an antisymmetric polarization state can lead to one photon in each output. A *coincidence pattern* simply means that specified detectors click within the same short time window. By recording which coincidences occur, the apparatus can therefore distinguish some Bell-state possibilities from others.

For this interference to be sharp, the two photons must be indistinguishable in every degree of freedom that the apparatus is not intentionally measuring. A small mismatch in arrival time, spectrum, or spatial mode gives the apparatus partial information about “which photon was which” and weakens the interference.

Common misconception

A Bell measurement on photonic qubits is not automatically deterministic. Using ordinary *passive linear optics*—beam splitters, phase shifters, and wave plates that redirect or mix light without creating an optical nonlinearity—one cannot unambiguously distinguish all four Bell states with unit probability unless additional resources are supplied. An *ancilla* here means an extra prepared quantum system added to help the measurement; an *entangled ancilla* is an extra entangled photonic resource. Many photonic teleportation and swapping experiments therefore succeed only for particular detector patterns. The successful events are perfectly valid; the price is a reduced success rate.

Once a successful Bell outcome has been identified, the result is an ordinary classical record. Fast electronics can send that record to Bob, and an electro-optic modulator or another controlled element can implement the Pauli correction required by the teleportation protocol. This dependence of a later operation on an earlier detector result is called *feed-forward*.

9.6 What would I actually see in a teleportation experiment?

It is useful to imagine standing beside the optical table during one attempted teleportation event.

1. A source prepares an entangled pair. One photon is routed toward Alice's Bell-measurement apparatus; the other travels toward Bob.
2. A separate preparation stage creates the input qubit $|\psi\rangle$ that Alice wishes to teleport.
3. Alice's input photon and her half of the Bell pair arrive simultaneously at the Bell-measurement interferometer.
4. A particular coincidence pattern appears in Alice's detectors. If it is one of the accepted Bell patterns, the trial is declared successful.
5. Alice's electronics converts that pattern into the corresponding classical Bell label and sends the required bits to Bob.
6. Bob's electronics drives a modulator implementing the appropriate Pauli correction.
7. Bob finally measures the output photon. Repeating the experiment many times allows the output state to be reconstructed statistically.

Notice how closely this follows the LOCC protocol of Section 4. The quantum part is the previously shared entanglement and Alice's local Bell measurement. The message from Alice to Bob is a completely ordinary classical signal. The laboratory has not changed the logic of teleportation; it has supplied physical hardware for each step.

9.7 How do experimenters know that teleportation worked?

There is a subtle conceptual point here. If the input is truly an unknown single qubit, the experimentalist cannot first measure it completely to find out what it is and then test whether the same state appears at Bob. Such a measurement would itself destroy the unknown state.

Instead, the experimentalist plays two different roles. During calibration and testing, the preparation device is instructed to produce many known test states, such as several points spread over the Bloch sphere. The teleportation apparatus is run many times for each test state. Bob then measures many output photons in different bases and reconstructs the output density matrix by *quantum-state tomography*. The word *tomography* means reconstruction from many projections, just as medical tomography reconstructs an interior image from many measured views.

For a qubit the reconstruction can be written explicitly. Every density matrix has the Bloch form

$$\rho = \frac{1}{2} (I + r_x \sigma_x + r_y \sigma_y + r_z \sigma_z), \quad (454)$$

where

$$r_x = \langle \sigma_x \rangle, \quad r_y = \langle \sigma_y \rangle, \quad r_z = \langle \sigma_z \rangle. \quad (455)$$

The experimentalist therefore prepares the same test state many times, measures separate subensembles in the x , y , and z bases, estimates these three expectation values from the observed

frequencies, and substitutes them into Eq. (454). That is qubit state tomography in its simplest form.

Thus the density matrix introduced in Chapter 3 reappears here in its experimental role: it is the object inferred from a sufficiently rich collection of measurement statistics.

For a target pure state $|\psi\rangle$ and reconstructed output state ρ_{out} , a natural figure of merit is the fidelity

$$F_\psi = \langle \psi | \rho_{\text{out}} | \psi \rangle. \quad (456)$$

If $F_\psi = 1$, the output is exactly the desired state. If the output is orthogonal to the target, the fidelity is zero.

For a uniformly distributed unknown pure qubit, the best strategy with no entanglement is to measure the input, make the best classical guess, and prepare that guessed state at the output. Its optimal average fidelity is $2/3$. Thus an average teleportation fidelity above $2/3$ is an important benchmark: it shows that the observed performance cannot be reproduced by the best classical measure-and-prepare strategy for that ensemble.

The number F is not the only experimental quantity that matters. A protocol that attains high fidelity once per day may be less useful than one with slightly lower fidelity that succeeds millions of times per second. A serious experiment therefore reports both *quality* and *rate*, together with losses, dark counts, accidental coincidences, and the probability that the Bell measurement succeeds.

The first optical proof-of-principle teleportation experiments were reported in 1997 [24]. Teleportation has since been demonstrated with photons, trapped ions, neutral atoms, atomic ensembles, solid-state spins, and superconducting circuits. The division of labour is instructive: matter qubits are often easier to store and process, whereas photons are natural carriers between distant locations.

As one example of progress toward metropolitan networking, a 2023 experiment reported teleportation of time-bin photonic qubits through a 64 km fibre system at a rate of about 7 Hz, with reported single-photon fidelities above the classical benchmark [25].

9.8 Why distance is difficult: photons get lost

Imagine sending one photon down a fibre. Even if the photon remains perfectly coherent whenever it survives, there is still a more basic problem: it may simply be absorbed or scattered before reaching the receiver.

Optical-fibre attenuation is commonly quoted in *decibels* (dB) per kilometre. A decibel is a logarithmic way of describing a ratio. A loss of 10 dB means that only one tenth of the optical power remains; a loss of 20 dB means one hundredth remains; a loss of about 3 dB corresponds to roughly one half. Using logarithms is convenient because losses from successive segments simply add in dB even though the physical transmission probabilities multiply.

If the attenuation coefficient is α dB/km and the fibre length is L , the total loss is αL dB and the transmission probability is approximately

$$\eta(L) = 10^{-\alpha L/10}. \quad (457)$$

At a good telecommunication wavelength one may have roughly $\alpha \simeq 0.2$ dB/km. Then

$$L = 50 \text{ km} : \quad \eta \sim 10^{-1} = 10\%, \quad (458)$$

$$L = 100 \text{ km} : \quad \eta \sim 10^{-2} = 1\%, \quad (459)$$

$$L = 200 \text{ km} : \quad \eta \sim 10^{-4} = 0.01\%. \quad (460)$$

These estimates immediately explain why direct single-photon transmission becomes inefficient over continental distances. The problem is exponential: adding another fixed length of fibre multiplies the survival probability by another fixed factor.

A classical optical network solves loss by detecting and regenerating the signal. Section 5 explained why an unknown quantum state cannot simply be measured, copied, and amplified in this way. This is precisely the motivation for entanglement distribution, heralding, quantum memories, and quantum repeaters.

9.9 Fibre or free space?

Fibre confines the light to a glass waveguide. It is stable, protected from weather, and naturally compatible with existing telecommunications infrastructure. Its weakness is the accumulated attenuation just described.

A free-space optical link removes the glass. The beam instead propagates through air and, for a satellite link, through near-vacuum for most of its journey. This avoids continuous absorption in fibre, but creates a different set of problems:

- the beam diffracts and spreads;
- atmospheric turbulence makes it wander and distort;
- the transmitter and receiver must point at each other very accurately;
- clouds can block the link completely;
- sunlight and other background photons create detector noise.

Ground-based experiments have teleported photonic states across long free-space links; a 2012 experiment used active feed-forward over 143 km between two Canary Islands [26].

Neither fibre nor free space is universally superior. Fibre is ideal for many fixed metropolitan links. Free space is attractive when terrain, oceans, or very long distances make laying fibre difficult, and it is the natural route to a satellite.

9.10 Why a satellite can help

The basic comparison is shown in figure 8. The drawing is schematic, but it captures the crucial geometric fact. In a fibre, loss acts all along the path. In a satellite link, most of the optical path is through near-vacuum. The difficult part is then concentrated in launching and collecting a narrow beam and in crossing the atmosphere near the ground.

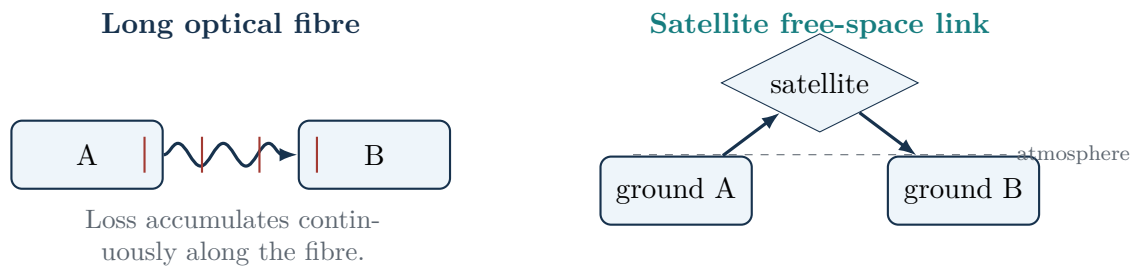


Figure 8: Why satellite links can become attractive over very long distances. In fibre, attenuation accumulates throughout the route. In a satellite link, most propagation occurs through near-vacuum, although diffraction, pointing, atmospheric turbulence, and detector noise remain serious limitations.

A useful everyday analogy is a flashlight. Even in perfectly transparent air, a distant wall intercepts only part of the spreading beam. A satellite optical link faces the same geometric problem at a much more demanding level: a narrow beam must be kept aligned with a moving telescope hundreds or thousands of kilometres away, and only a small fraction of the transmitted light is collected.

This immediately explains several pieces of satellite engineering. The technical vocabulary is easier to remember if each term is tied to the question it answers: *where is the other terminal?*, *where must I aim?*, *how do I keep aiming while it moves?*, and *how do I reject photons that did not come from my signal?*

1. **Acquisition, pointing, and tracking.** Before quantum signals can be exchanged, the ground station and satellite must find one another and keep a narrow optical beam aligned while the satellite moves rapidly across the sky.
2. **Timing.** The receiver opens very narrow detector time windows around the predicted photon arrival times. Good synchronization suppresses background counts that occur at unrelated times.
3. **Reference frames.** If polarization carries the qubit, the meaning of “horizontal” and “vertical” changes as the satellite moves. The optical system must track or compensate this rotation.
4. **Background rejection.** Telescopes, *spectral filters* that accept only a narrow range of wavelengths, and narrow detector time windows help reject sunlight and other stray photons.
5. **Efficient collection.** A telescope’s *aperture* is the diameter of the opening through which it collects light. Large apertures and precise optics are needed because the receiver captures only a tiny part of the expanded beam.

Uplink versus downlink

There is an important asymmetry. In an *uplink*, the beam encounters the densest turbulent atmosphere immediately after leaving the ground. Early angular distortions then have a long distance in which to make the beam spread. In a *downlink*, the beam has already travelled through near-vacuum before it encounters the thick lower atmosphere near the end of the journey. This is one reason downlinks are often more favourable, although the best architecture depends on the precise task and hardware.

9.11 Three satellite experiments that sound similar but are not

The phrase “satellite quantum communication” is too broad by itself. Three physically different tasks have been demonstrated, and it is important to know which resource is being transmitted.

1. Satellite-to-ground quantum key distribution

The satellite prepares quantum optical signals and sends them to a ground station. The goal is not to teleport a qubit but to establish correlated classical data from which a secret key can be distilled. The detailed QKD protocols are postponed to Chapters 9 and 10, so two terms are translated here in advance. A *sifted key* is the preliminary string remaining after Alice and Bob discard transmission rounds in which their basis choices were incompatible. *Decoy-state QKD* is a practical technique in which the sender varies the optical intensity in order to detect attacks that exploit multi-photon pulses; Chapter 11 will derive the logic carefully.

The *Micius* mission demonstrated decoy-state QKD from low Earth orbit over links up to about 1,200 km, with kilohertz-scale sifted key rates reported in the experiment [27]. “Kilohertz-scale” simply means that the preliminary sifted string was produced at a rate of order thousands of bits per second under the reported link conditions.

2. Satellite distribution of entanglement

Here the satellite contains an entangled-pair source and sends one member of a pair toward each of two ground stations. In 2017, entangled photons were distributed to ground stations separated by 1,203 km, and the measured correlations violated a Bell inequality [28].

Conceptually this is closer to Section 5: the satellite is helping two distant nodes acquire a shared entangled resource.

3. Ground-to-satellite teleportation

Here an unknown photonic state is present on the ground. The ground station performs a Bell measurement between that state and one half of an entangled pair. The other half of the pair has been sent to the satellite. Classical information about the Bell outcome is then used to identify the corresponding state at the satellite. A 2017 experiment reported teleportation of six test states from a ground station to a low-Earth-orbit satellite over distances up to 1,400 km, with average fidelity 0.80 ± 0.01 [29].

The distinction is therefore:

Task	What is established?	Main quantum resource
QKD	secret classical key	nonorthogonal quantum states or measured quantum correlations
Entanglement distribution	shared Bell-like state	an entangled photon pair distributed between distant sites
Teleportation	transfer of an unknown quantum state	shared entanglement together with classical communication

9.12 From flagship missions to smaller hardware

Early satellite demonstrations required large specialized payloads and major ground infrastructure. A 2025 experiment reported real-time QKD using a microsatellite payload of approximately 23 kg and portable ground stations of approximately 100 kg. The experiment shared up to 1.07 million secure key bits during a single pass and used a satellite-relayed key for encrypted images between sites in China and South Africa separated by more than 12,900 km [30].

This is an impressive engineering step, but the phrase “global quantum internet” should be used carefully. Many satellite QKD demonstrations use a *trusted-node* architecture. The phrase means exactly what it says: the intermediate node is assumed to behave honestly and to keep its stored key material secret. In the satellite example, the spacecraft establishes one key with station A and another with station B and then helps combine them classically. Because the satellite has access to the relevant key information, if the satellite itself is compromised, the end-to-end secrecy can be lost. Entanglement-based links and future repeater networks seek to reduce such trust assumptions, but they demand better sources, memories, interfaces, and measurements.

9.13 The laboratory lesson

We can now translate the abstract network language into physical roles:

- **photons** are excellent flying qubits;
- **atoms, ions, spins, or superconducting circuits** can serve as stationary qubits and memories;
- **beam splitters, wave plates, interferometers, and gates** implement quantum operations;

- **detectors** turn quantum measurement outcomes into classical records;
- **ordinary electronics and communication links** provide heralding, synchronization, routing, and feed-forward.

A future network will therefore not be “quantum instead of classical.” It will be a hybrid machine in which a delicate quantum layer is continuously supported by an ordinary classical control layer.

Key idea

The blackboard protocol and the laboratory experiment are the same story told at two different levels. On the blackboard we say “prepare a Bell pair, perform a Bell measurement, and send two classical bits.” In the laboratory that sentence becomes sources, interferometers, detector coincidences, classical electronics, and a great deal of engineering devoted to making the idealized steps happen often enough and accurately enough to be useful.

This physical picture prepares us for the next chapter. Quantum key distribution uses exactly these ingredients—single-photon states, optical basis choices, detectors, lossy channels, and authenticated classical communication—but arranges them for a different task: creating a secret key.

10 Quantum Cryptography and Quantum Key Distribution: Core Principles

The preceding chapters developed a sequence of increasingly operational ideas. We learned what entanglement is, why Bell correlations cannot be reduced to local classical ignorance, what shared entanglement can accomplish, how it can be distributed through a network, and how the abstract qubit language is implemented in optical hardware. Quantum key distribution now rearranges many of those same ingredients around a different goal.

Alice and Bob do not necessarily want to transmit an unknown quantum state. They want to establish a shared *classical* secret in the presence of an eavesdropper, Eve. Quantum mechanics does not encrypt a message automatically. Its role is subtler and more useful: carefully chosen quantum states allow Alice and Bob to generate correlated random data while testing how much an adversary could have learned from the quantum channel.

Key idea

QKD generates a shared secret *classical key*. The quantum part of the protocol supplies experimentally testable constraints on an eavesdropper's information; classical post-processing converts those constraints into a shorter key suitable for cryptographic use.

This chapter develops the conceptual core using ideal single-photon BB84. We will keep three descriptions connected throughout: the qubit states, the optical operations of Chapter 9, and the classical information processing that turns detector records into a key. Real sources, imperfect detectors, finite-key security, and more advanced protocols are deliberately postponed to Chapter 11 so that the central logic is visible before engineering complications are added.

10.1 Why a secret key is useful: the one-time pad

Suppose Alice wishes to send an n -bit message

$$M = (M_1, \dots, M_n), \quad (461)$$

and Alice and Bob already share an independent uniformly random n -bit key

$$K = (K_1, \dots, K_n). \quad (462)$$

Alice forms the ciphertext

$$C = M \oplus K, \quad (463)$$

where $\oplus$ means bitwise addition modulo two. Bob recovers the message by using the same key:

$$C \oplus K = (M \oplus K) \oplus K = M. \quad (464)$$

The security of the one-time pad can be seen directly. Fix a message m and a ciphertext c . The equation $c = m \oplus k$ determines exactly one key, namely $k = c \oplus m$. Since all 2^n keys are equally probable,

$$P(C = c \mid M = m) = P(K = c \oplus m) = 2^{-n}, \quad (465)$$

which is independent of m . Therefore observing $C = c$ does not change Eve's probability distribution for the message:

$$P(M = m \mid C = c) = P(M = m). \quad (466)$$

This is perfect information-theoretic secrecy.

The difficulty is obvious. Alice and Bob must somehow obtain the shared key K in the first place. Sending it through an ordinary public channel would simply move the secrecy problem elsewhere. QKD is designed to solve this *key-distribution problem*.

There is one important qualification from the beginning. QKD requires an *authenticated classical channel*. Eve may listen to Alice and Bob's public discussion, but she must not be able to impersonate Alice to Bob and Bob to Alice. Otherwise she could perform two separate protocols, one with Alice and one with Bob, and mount a man-in-the-middle attack. Authentication can be bootstrapped from a short pre-shared secret and then refreshed from later QKD keys.

10.2 The BB84 signal alphabet in polarization

Chapter 9 developed the physical dictionary for photonic qubits, wave plates, polarizing beam splitters, and detectors. We now reuse that toolbox for one specific information-processing task. BB84 can be implemented in several degrees of freedom—polarization, optical phase, time bins, or paths—but polarization makes the logic especially transparent because the four signal states can be pictured directly as four directions in a two-dimensional Hilbert space.

We choose horizontal and vertical polarization as the computational basis:

$$|0\rangle \equiv |H\rangle, \quad |1\rangle \equiv |V\rangle. \quad (467)$$

The conjugate, or diagonal, basis is

$$|+\rangle \equiv |D\rangle = \frac{|H\rangle + |V\rangle}{\sqrt{2}}, \quad |-\rangle \equiv |A\rangle = \frac{|H\rangle - |V\rangle}{\sqrt{2}}. \quad (468)$$

Here D and A stand for diagonal and anti-diagonal linear polarization.

A general linearly polarized photon at angle θ to the horizontal may be written

$$|\theta\rangle = \cos \theta |H\rangle + \sin \theta |V\rangle. \quad (469)$$

Thus

$$|H\rangle = |0^\circ\rangle, \quad |V\rangle = |90^\circ\rangle, \quad |D\rangle = |45^\circ\rangle, \quad |A\rangle = |-45^\circ\rangle. \quad (470)$$

What does a polarization filter do?

An ideal linear polarizer oriented at angle ϕ transmits the component parallel to

$$|\phi\rangle = \cos \phi |H\rangle + \sin \phi |V\rangle. \quad (471)$$

Mathematically, transmission corresponds to the projector

$$P_\phi = |\phi\rangle\langle\phi|. \quad (472)$$

If the incoming photon is $|\theta\rangle$, the probability that it passes is

$$p_{\text{pass}} = \langle\theta|P_\phi|\theta\rangle = |\langle\phi|\theta\rangle|^2 \quad (473)$$

$$= |\cos \phi \cos \theta + \sin \phi \sin \theta|^2 = \cos^2(\theta - \phi). \quad (474)$$

This is Malus' law, now read as a single-photon probability law.

A single absorbing polarizer is not the best way to perform a two-outcome measurement because one outcome is simply discarded. A *polarizing beam splitter* (PBS) is more useful. Ideally it sends $|H\rangle$ into one output port and $|V\rangle$ into the other. With a detector on each port, the PBS implements the projective measurement

$$\{P_H, P_V\} = \{|H\rangle\langle H|, |V\rangle\langle V|\}. \quad (475)$$

A click in the H detector is the outcome 0; a click in the V detector is the outcome 1.

10.3 How Alice actually prepares the four BB84 states

The ideal theory simply says that Alice prepares one of

$$|H\rangle, |V\rangle, |D\rangle, |A\rangle. \quad (476)$$

In an optical laboratory there are several ways to do this.

A conceptually simple preparation begins with a photon in a known polarization, say $|H\rangle$, and rotates it with a half-wave plate (HWP). Ignoring an overall phase, an ideal HWP whose fast axis is at angle ϑ to the horizontal acts in the $\{|H\rangle, |V\rangle\}$ basis as

$$U_{\text{HWP}}(\vartheta) = \begin{pmatrix} \cos 2\vartheta & \sin 2\vartheta \\ \sin 2\vartheta & -\cos 2\vartheta \end{pmatrix}. \quad (477)$$

Acting on $|H\rangle$ gives

$$U_{\text{HWP}}(\vartheta)|H\rangle = \cos 2\vartheta |H\rangle + \sin 2\vartheta |V\rangle. \quad (478)$$

Therefore the settings

HWP angle ϑ	output state	BB84 label
0°	$ H\rangle$	Z basis, bit 0
45°	$ V\rangle$	Z basis, bit 1
22.5°	$ D\rangle$	X basis, bit 0
-22.5°	$ A\rangle$	X basis, bit 1

produce the four ideal signal states from a horizontally polarized input. In high-speed systems a rapidly switched electro-optic modulator may replace a mechanically rotated wave plate, but the Hilbert-space operation is the same: Alice chooses one of four controlled polarization transformations.

A literal polarizing filter can also prepare a state: if a photon passes a polarizer oriented at θ , the transmitted photon is in $|\theta\rangle$. Such filtering is conceptually simple but intrinsically probabilistic because photons in the orthogonal component are rejected. Wave plates or electro-optic modulators are therefore preferable when a known input polarization is already available.

10.4 How Bob measures in the two BB84 bases

The Z -basis measurement is immediate: Bob sends the photon directly to a PBS. The two outputs correspond to H and V .

For the X basis Bob first rotates the diagonal basis into the horizontal–vertical basis. Setting a HWP at 22.5° gives

$$U_{\text{HWP}}(22.5^\circ)|D\rangle = |H\rangle, \quad U_{\text{HWP}}(22.5^\circ)|A\rangle = |V\rangle. \quad (479)$$

A PBS placed after the wave plate can therefore distinguish $|D\rangle$ from $|A\rangle$ perfectly.

Bob must choose the basis independently of Alice. There are two common idealized ways to represent this physically:

- **Active basis choice:** a random bit controls a fast modulator. For a Z measurement Bob applies no rotation; for an X measurement he applies the 22.5° HWP rotation before the PBS.
- **Passive basis choice:** a nominally 50:50 ordinary beam splitter randomly sends the incoming photon to one of two analyzer arms. One arm contains a Z analyzer and the other an X analyzer.

In either idealization, the final detectors merely record which output port clicked. The quantum measurement is implemented by the optics before the photodetection event.

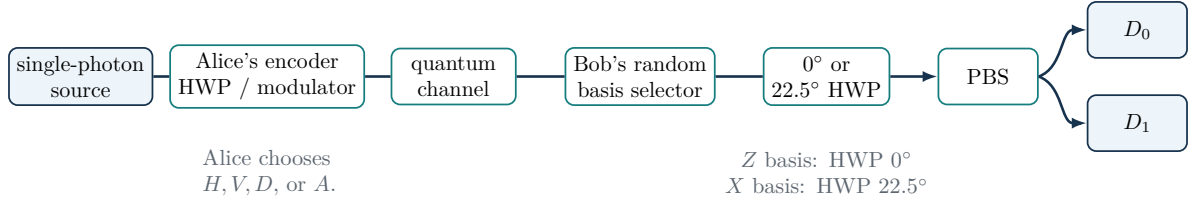


Figure 9: An idealized polarization implementation of BB84. Alice encodes a random bit in one of two polarization bases. Bob independently chooses a basis, rotates that basis into the H/V basis if necessary, and a PBS plus two detectors performs the final two-outcome measurement.

10.5 Why the two bases are incompatible

The security mechanism begins with a simple geometric fact in Hilbert space. Within each basis the states are orthogonal:

$$\langle H|V \rangle = 0, \quad \langle D|A \rangle = 0. \quad (480)$$

Therefore a measurement in the correct basis distinguishes the encoded bit perfectly.

Across the two bases, however,

$$|\langle H|D \rangle|^2 = |\langle H|A \rangle|^2 = |\langle V|D \rangle|^2 = |\langle V|A \rangle|^2 = \frac{1}{2}. \quad (481)$$

The Z and X bases are therefore *mutually unbiased*. If a photon is prepared in one basis and measured in the other, the outcome is completely random.

For example, if Alice sends $|D\rangle$, then a Z measurement gives

$$P(H|D) = |\langle H|D \rangle|^2 = \frac{1}{2}, \quad (482)$$

$$P(V|D) = |\langle V|D \rangle|^2 = \frac{1}{2}. \quad (483)$$

Nothing in Bob's H/V outcome tells him which diagonal state was sent.

It is also useful to average over Alice's four equally likely preparations. The ensemble density matrix is

$$\rho_{\text{sig}} = \frac{1}{4} (|H\rangle\langle H| + |V\rangle\langle V| + |D\rangle\langle D| + |A\rangle\langle A|) \quad (484)$$

$$= \frac{1}{4} (I + I) = \frac{I}{2}. \quad (485)$$

Thus before Alice later announces her basis, there is no single measurement that can read off a pre-existing classical “basis label” and “bit label” from a perfectly distinguishable set of four states: the four states are not mutually orthogonal.

10.6 The ideal BB84 protocol, one round at a time

We now formulate the protocol precisely. For each transmission round i :

1. Alice chooses a uniformly random bit

$$a_i \in \{0, 1\} \quad (486)$$

and independently chooses a uniformly random basis

$$\theta_i \in \{Z, X\}. \quad (487)$$

2. She prepares

$$|\psi_{a_i, \theta_i}\rangle = \begin{cases} |H\rangle, & (a_i, \theta_i) = (0, Z), \\ |V\rangle, & (a_i, \theta_i) = (1, Z), \\ |D\rangle, & (a_i, \theta_i) = (0, X), \\ |A\rangle, & (a_i, \theta_i) = (1, X), \end{cases} \quad (488)$$

and sends the photon to Bob.

3. Bob independently chooses a random basis $\phi_i \in \{Z, X\}$ and measures in that basis. He records a bit b_i if a valid detection event occurs.
4. Only after the quantum transmission stage is complete do Alice and Bob announce their basis choices θ_i and ϕ_i on an authenticated public channel. They do *not* announce the secret bits.
5. They retain the rounds for which $\theta_i = \phi_i$ and discard the rest. This is called *sifting*.

If Alice and Bob choose the bases uniformly, the probability that their bases match is

$$P(\theta = \phi) = P(Z, Z) + P(X, X) \quad (489)$$

$$= \frac{1}{4} + \frac{1}{4} = \frac{1}{2}. \quad (490)$$

Thus even in a perfect lossless experiment, standard symmetric BB84 keeps on average only half of the detected rounds. More efficient variants deliberately bias the basis choices, but the symmetric version is conceptually cleaner.

Conditioned on a matching basis and an ideal channel,

$$P(b = a \mid \theta = \phi) = 1. \quad (491)$$

Conditioned on a mismatched basis,

$$P(b = 0 \mid \theta \neq \phi) = P(b = 1 \mid \theta \neq \phi) = \frac{1}{2}. \quad (492)$$

The mismatched rounds contain no reliable key information and are discarded.

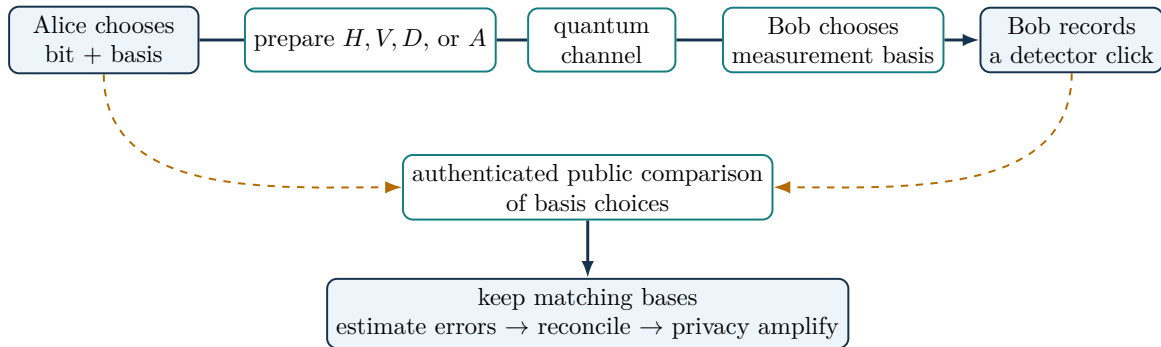


Figure 10: Logical stages of ideal BB84. The public discussion reveals basis choices, not the raw key values. The retained data are not yet a secret final key: they must still pass parameter estimation, error correction, and privacy amplification.

10.7 A worked optical round

Suppose Alice chooses bit 1 in the X basis. She prepares

$$|A\rangle = \frac{|H\rangle - |V\rangle}{\sqrt{2}}. \quad (493)$$

If Bob also chooses the X basis, his 22.5° HWP maps $|A\rangle \mapsto |V\rangle$, so the V output of the PBS clicks with probability one. Bob records the bit 1.

If Bob instead chooses the Z basis, he sends $|A\rangle$ directly to the PBS. Then

$$P(H|A) = P(V|A) = \frac{1}{2}. \quad (494)$$

His result is random. Later, Alice and Bob announce that their bases differed, so this round is discarded regardless of which detector clicked.

This simple example is worth keeping in mind: *nothing mysterious happens at the sifting stage*. The quantum measurement has already occurred. Sifting is only the later classical decision about which recorded data are useful.

10.8 A minimal information-theory dictionary

Before quantifying what Eve learns, we need three elementary notions from classical information theory. They are the classical counterparts of ideas already encountered in the von Neumann entropy of Chapter 3.

Let a classical random variable X take values x with probabilities $p(x)$. Its *Shannon entropy* is

$$H(X) = - \sum_x p(x) \log_2 p(x). \quad (495)$$

It measures uncertainty in bits. A perfectly known bit has entropy zero; an unbiased random bit has entropy one.

If another variable Y is known, the remaining uncertainty about X is the *conditional entropy*

$$H(X|Y) = \sum_y p(y) H(X|Y = y). \quad (496)$$

The information about X carried by Y is measured by the *mutual information*

$$I(X : Y) = H(X) - H(X|Y) = H(Y) - H(Y|X). \quad (497)$$

Thus $I(X : Y) = 1$ bit when two unbiased bits are perfectly correlated, whereas $I(X : Y) = 0$ when they are statistically independent.

There is a useful bridge to Chapter 3. If a quantum state is diagonal in a basis,

$$\rho = \sum_x p(x) |x\rangle\langle x|, \quad (498)$$

then its von Neumann entropy is exactly the Shannon entropy of the classical probabilities:

$$S(\rho) = H(X). \quad (499)$$

The quantum theory of information therefore extends, rather than replaces, these classical notions.

For the intercept–resend attack below, Eve’s relevant record is classical, so $H(A|E)$ and $I(A : E)$ have the elementary meanings just defined. In a full security proof Eve may retain a quantum memory rather than a classical record; that stronger situation will be revisited in Chapter 11.

Checkpoint: two extremes

Let A be an unbiased bit. If Eve knows A perfectly, verify that $H(A|E) = 0$ and $I(A : E) = 1$. If Eve's record is independent of A , verify that $H(A|E) = 1$ and $I(A : E) = 0$. The intercept–resend attack will sit exactly halfway between these two extremes.

10.9 Why a naive eavesdropper is detected: intercept and resend

Consider the simplest attack. Eve intercepts every photon. She does not know Alice's basis, so she chooses Z or X at random, measures the photon, prepares a new photon in the state she found, and sends that replacement to Bob.

We calculate the error rate only on the *sifted* rounds, because these are the rounds Alice and Bob would otherwise use for a key. On a sifted round, Bob's basis equals Alice's basis.

There are two cases.

Case 1: Eve chose Alice's basis. This occurs with probability $1/2$. Eve obtains Alice's bit with certainty and resends exactly the corresponding state. Bob then gets the correct bit with certainty. This branch causes no error.

Case 2: Eve chose the wrong basis. This also occurs with probability $1/2$. Eve's outcome is random. The state she resends is an eigenstate of the wrong basis. Bob measures it in Alice's original basis, so his outcome is again random. Therefore he disagrees with Alice with probability $1/2$.

The quantum bit error rate (QBER) on the sifted key is therefore

$$Q = P(\text{Eve wrong basis})P(\text{Bob wrong} \mid \text{Eve wrong basis}) \quad (500)$$

$$= \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}. \quad (501)$$

Thus a full intercept–resend attack produces a 25% error rate in the sifted data.

The same calculation can be written as a probability tree. Per transmitted round, only one half of the rounds survive sifting; among those surviving rounds, one quarter are wrong. Thus the probability of a transmitted round both surviving and being wrong is $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$. Dividing by the survival probability $1/2$ again gives a sifted-key error rate of $1/4$.

For this particular attack Eve's information is also easy to quantify. After the basis announcement she knows exactly those sifted bits for which she had chosen the correct basis, which happens half the time. On the other half her measurement result is independent of Alice's bit. Hence the conditional entropy is

$$H(A|E) = \frac{1}{2} \times 0 + \frac{1}{2} \times 1 = \frac{1}{2} \quad (502)$$

and, since Alice's sifted bit is uniformly random,

$$I(A : E) = H(A) - H(A|E) = 1 - \frac{1}{2} = \frac{1}{2} \quad (503)$$

bit of information per sifted bit for this simple attack.

Common misconception

The 25% calculation is not a complete security proof. It analyzes one very specific attack. A real proof must allow Eve to attach arbitrary quantum ancillas, keep quantum memory, act coherently on many signals, wait for the public discussion, and optimize her final measurement. The value of the intercept–resend calculation is pedagogical: it makes the

basic information– disturbance mechanism visible.

Why can Eve not simply make a perfect copy of every photon, forward one copy to Bob, and wait until Alice reveals the basis before measuring her stored copy? The no-cloning theorem of Chapter 7 forbids precisely this strategy. The non-orthogonality of the BB84 states and the impossibility of cloning them are two faces of the same obstruction.

10.10 From a raw string to a secret key

After sifting, Alice and Bob possess correlated strings, but these strings are not yet suitable for cryptographic use. Three logically distinct tasks remain:

1. **Parameter estimation:** estimate how noisy the quantum channel was.
2. **Information reconciliation:** correct Alice–Bob disagreements using authenticated public communication.
3. **Privacy amplification:** shorten the reconciled string enough that Eve’s remaining information about the final string is negligible.

10.10.1 Quantum bit error rate

If Alice and Bob publicly compare m randomly chosen sifted bits and observe k disagreements, the sample QBER is

$$Q_{\text{sample}} = \frac{k}{m}. \quad (504)$$

In the asymptotic idealization of infinitely many signals, this sample frequency approaches the underlying bit-error probability. In a finite experiment, statistical confidence intervals must be included; practical finite-key analysis is more conservative.

The test bits are sacrificed permanently. Once Alice publicly reveals a test bit, it obviously cannot be part of the secret key.

10.10.2 Why error correction costs about $nh_2(Q)$ bits

Suppose the remaining sifted string has length n and Bob’s string differs from Alice’s in approximately nQ positions. The number of possible error patterns of this weight is

$$\binom{n}{nQ}. \quad (505)$$

For large n , Stirling’s approximation gives

$$\binom{n}{nQ} \approx 2^{nh_2(Q)}, \quad (506)$$

where

$$h_2(Q) = -Q \log_2 Q - (1 - Q) \log_2 (1 - Q) \quad (507)$$

is the binary entropy.

Equation (506) has a simple interpretation. Bob needs roughly $nh_2(Q)$ bits of syndrome information to identify which one of the typical error patterns occurred. In an ideal Shannon-efficient reconciliation procedure this is the minimum public information that must be revealed. A real error-correcting protocol uses somewhat more; one often writes

$$\text{leak}_{\text{EC}} \simeq f_{\text{EC}} nh_2(Q), \quad f_{\text{EC}} \geq 1. \quad (508)$$

All public error-correction messages are assumed known to Eve and therefore must be subtracted from the secrecy budget.

10.10.3 Privacy amplification

Error correction makes Alice and Bob agree; it does not make Eve forget what she may already know. Alice and Bob therefore apply a publicly chosen random hash function to their common reconciled string and keep only a shorter output. This is *privacy amplification*.

The important point is counterintuitive: the hash function need not be secret. Eve may know exactly which hash function was chosen. Security comes from the fact that she does not know the entire input string. Compressing that partly unknown input can produce a much shorter output about which her information is arbitrarily small.

At the level needed here, the operational message is enough: privacy amplification deliberately trades length for secrecy. A rigorous finite-key statement must also quantify Eve's possible *quantum* side information and statistical uncertainty. We postpone that refinement to Chapter 11, after the ideal protocol is conceptually complete.

10.11 The ideal asymptotic BB84 secret fraction

There is a remarkably compact formula that summarizes the ideal single-photon BB84 security argument. Let Q_{bit} denote the bit-error rate and Q_{ph} the corresponding phase-error rate in the entanglement-based security proof. The asymptotic secret fraction per sifted bit obeys

$$r \geq 1 - h_2(Q_{\text{bit}}) - h_2(Q_{\text{ph}}). \quad (509)$$

Why do two entropy terms appear? The first is the information that must be spent to reconcile ordinary bit errors. The second is the amount that must be sacrificed to remove Eve's information associated, in the equivalent entanglement picture, with phase errors. This is the conceptual content of the Shor–Preskill proof [32].

For symmetric ideal BB84, the complementary-basis test lets Alice and Bob bound Q_{ph} by the same value Q that characterizes the bit errors. Then

$$\boxed{r \geq 1 - 2h_2(Q)}. \quad (510)$$

The right-hand side vanishes when

$$h_2(Q) = \frac{1}{2}, \quad (511)$$

which gives

$$Q \simeq 0.1100. \quad (512)$$

Thus the familiar roughly 11% threshold belongs to the asymptotic ideal single-photon BB84 protocol with one-way post-processing. It is not a universal threshold for every implementation or every security proof.

As a consistency check, the intercept–resend attack gives $Q = 1/4$, well above this ideal secure-key threshold.

10.12 Entanglement-based BB84: prepare and measure seen from another angle

The prepare-and-measure protocol has an equivalent entanglement-based interpretation. Let Alice create

$$|\Phi^+\rangle_{AB} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \quad (513)$$

The same state can also be written in the X basis:

$$|\Phi^+\rangle_{AB} = \frac{|++\rangle + |--\rangle}{\sqrt{2}}. \quad (514)$$

If Alice measures her half in the Z basis, outcome 0 leaves Bob in $|0\rangle$ and outcome 1 leaves Bob in $|1\rangle$. If instead she measures in the X basis, outcome $+$ leaves Bob in $|+\rangle$ and outcome $-$ leaves Bob in $|-\rangle$. Therefore Alice's local measurement can be viewed as remotely preparing exactly one of the four BB84 states for Bob.

This equivalence is more than a mathematical curiosity. It explains why entanglement purification and quantum error-correction ideas enter security proofs of a protocol that, in the laboratory, may never explicitly create an entangled pair.

10.13 The conceptual core of BB84

We can now see the whole ideal protocol without appealing to any advanced security machinery:

- encode a random bit in one of two incompatible bases
- transmit and measure before the basis is revealed
- sift matching bases and estimate disturbances
- reconcile Alice–Bob errors
- compress away Eve's bounded information
- shared secret classical key.

(515)

Two points connect this chapter to the rest of the notes. First, the no-cloning theorem of Chapter 7 prevents Eve from storing perfect copies while leaving Bob's signals untouched. Second, the entanglement-based reformulation shows that even prepare-and-measure BB84 can be understood through the same correlation structure that motivated Chapters 1–4. The next chapter keeps this core intact but removes the idealizations one by one.

11 Realistic and Advanced Quantum Key Distribution

The ideal BB84 chapter deliberately separated the essential information logic from the imperfections of a real experiment. We now restore what was left out. There are two complementary directions. One can change the protocol so that entanglement or Bell correlations play a more explicit role, and one can ask how security survives realistic sources, lossy channels, imperfect detectors, finite statistics, and partially untrusted devices. The underlying principles do not change, but the bookkeeping becomes more demanding because an adversary is allowed to exploit every physical degree of freedom not ruled out by the security assumptions.

11.1 E91: using Bell correlations inside a key-distribution protocol

Ekert’s E91 protocol uses entangled pairs directly [33]. A source sends one member of each pair to Alice and the other to Bob. On different rounds they choose among several measurement settings. Some setting pairs are used to obtain strongly correlated raw key bits; other setting pairs are used to estimate a Bell parameter.

The Bell theorem needed to understand this test has already been developed in Chapter 4. There we derived the CHSH bound

$$|S| \leq 2$$

for local hidden-variable models and the quantum singlet correlations that can reach $2\sqrt{2}$. There is no reason to repeat that derivation here. What is new in E91 is the *division of labour* between measurement rounds:

- **key-generation rounds** produce correlated random data from which a secret key may be distilled;
- **test rounds** estimate whether the shared correlations are compatible with the entangled resource and security assumptions of the protocol.

The crucial lesson is therefore operational rather than foundational. A Bell test is not performed merely to demonstrate that nature is nonclassical; its observed statistics become part of the evidence from which Alice and Bob bound an adversary’s predictive power. Modern device-independent QKD, discussed below, pushes this logic much further. E91 should not be conflated with a fully device-independent security proof, but it makes the conceptual bridge between entanglement, Bell certification, and cryptography exceptionally transparent.

For polarization-entangled photons the measurement settings are implemented by the same wave plates, polarizing elements, and detectors introduced in Chapter 9. Thus E91 introduces no new Bell algebra: it gives the Bell correlations of Chapter 4 a new job.

11.2 Real light sources: why an attenuated laser is not a single-photon source

The ideal BB84 story speaks of “one photon per pulse.” In practice, many systems use strongly attenuated laser pulses because deterministic high-rate single-photon sources are technologically demanding.

The first conceptual point is crucial: *turning down a laser does not make it emit exactly one photon*. It reduces the *mean* photon number. Pulse by pulse, a weak laser still sometimes produces no photon, sometimes one, and sometimes two or more. Quantum optics describes an ideal laser mode by a *coherent state*. The expression below is simply the decomposition of that state into alternatives with definite photon number.

An ideal single optical mode in a coherent state is

$$|\alpha\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle. \quad (516)$$

Here $|n\rangle$ means a state containing exactly n photons in the chosen optical mode, and the complex number α sets both the mean field amplitude and the photon-number distribution. If

$$\mu = |\alpha|^2 \quad (517)$$

is the mean photon number, the photon-number probability is Poissonian:

$$P_\mu(n) = e^{-\mu} \frac{\mu^n}{n!}. \quad (518)$$

Therefore

$$P_\mu(0) = e^{-\mu}, \quad (519)$$

$$P_\mu(1) = \mu e^{-\mu}, \quad (520)$$

$$P_\mu(n \geq 2) = 1 - e^{-\mu}(1 + \mu). \quad (521)$$

The word *Poissonian* is not new quantum jargon; it names the familiar probability distribution written above. Its physical message is that reducing μ suppresses multi-photon pulses but simultaneously increases the fraction of empty pulses. Even when $\mu < 1$, multiphoton pulses are not absent. For example, at $\mu = 0.5$,

$$P(n \geq 2) \simeq 0.0902. \quad (522)$$

The coherent state also contains phase coherence between different photon numbers. In standard decoy-state security analysis Alice deliberately randomizes the overall optical phase from pulse to pulse. Why does this turn the state into a classical mixture of photon-number sectors? A phase shift by ϕ multiplies the n -photon amplitude by $e^{in\phi}$. When the phase is uniformly randomized, an off-diagonal term $|n\rangle\langle m|$ acquires the average factor

$$\int_0^{2\pi} \frac{d\phi}{2\pi} e^{i(n-m)\phi} = \delta_{nm}. \quad (523)$$

Thus the coherences between different photon numbers disappear, while their probabilities remain. The emitted pulse is therefore described by the classical mixture

$$\rho_\mu = \sum_{n=0}^{\infty} P_\mu(n) |n\rangle\langle n|. \quad (524)$$

This photon-number decomposition is the basis of the standard decoy-state analysis. It lets Alice and Bob reason about the vacuum, one-photon, and multi-photon components separately.

11.3 The photon-number-splitting problem

Suppose a pulse contains two or more photons carrying the same polarization. A sufficiently powerful Eve could, in principle, keep one photon and allow another to continue toward Bob. She could then wait until Alice announces the basis and measure her stored photon in the correct basis. Unlike ordinary intercept-resend, this can reveal information without necessarily introducing a polarization error on Bob's photon.

This is the essence of a photon-number-splitting (PNS) vulnerability. Merely attenuating the laser more strongly does not solve the problem efficiently, because it also increases the already large fraction of vacuum pulses.

11.4 Decoy states: turning source imperfection into measurable data

Decoy-state BB84 solves the PNS problem by having Alice randomly vary the mean photon number [36, 35]. Some pulses are ordinary signal pulses, some are weaker decoy pulses, and often some are near-vacuum decoys. Alice reveals the intensity choice only after Bob has reported his detections.

The strategy is easier to understand as an inference problem. Alice cannot look inside each outgoing pulse and label it “this one contained exactly one photon.” But she *can* send several ensembles with known, different Poisson weights. Bob then observes different overall detection rates. Because each intensity is a different weighted mixture of the same underlying n -photon behaviours, the observed data give several equations for those unknown behaviours.

Before writing them, it helps to translate the standard notation:

- Y_n , the *yield*, is the probability that Bob obtains a detection event given that exactly n photons were sent;
- e_n is the error fraction within those detected n -photon events;
- Q_μ , the *gain*, is the overall probability of a detection event when Alice uses mean intensity μ ;
- E_μ is the observed error fraction among detections at that intensity, so $E_\mu Q_\mu$ is the probability that a transmitted pulse both produces a detection and produces an erroneous bit.

The terminology is specialized, but the mathematics is just the law of total probability.

For a phase-randomized source, an n -photon pulse is simply the Fock state $|n\rangle$ regardless of whether it arose from a signal or decoy intensity. Eve therefore cannot consistently make the behavior of an n -photon pulse depend on the later intensity label. Let

$$Y_n = P(\text{Bob detects} \mid n \text{ photons were sent}) \quad (525)$$

be the *yield* of the n -photon component, and let e_n be its error rate. For a source intensity μ , the directly observed total detection probability, or gain, is

$$Q_\mu = \sum_{n=0}^{\infty} P_\mu(n) Y_n, \quad (526)$$

and the observed error gain is

$$E_\mu Q_\mu = \sum_{n=0}^{\infty} P_\mu(n) Y_n e_n. \quad (527)$$

Equation (526) is simply a weighted average: first nature selects a photon number n with probability $P_\mu(n)$, and then Bob detects it with conditional probability Y_n . Summing over the unobserved photon number gives the total detection probability. The error-gain equation is the same logic with the additional conditional probability e_n that the detected bit is wrong.

By measuring Q_μ and E_μ for several values of μ , Alice and Bob obtain several linear constraints on the same unknown quantities Y_n and e_n . They can therefore bound the crucial single-photon quantities Y_1 and e_1 instead of pessimistically treating every multiphoton event as fully compromised.

A commonly encountered asymptotic decoy-state key-rate expression has the form

$$R \geq q \left[-Q_\mu f_{\text{EC}} h_2(E_\mu) + Q_1 (1 - h_2(e_1)) \right], \quad (528)$$

where q is the basis-sifting factor, $Q_1 = P_\mu(1)Y_1$ is the single-photon gain, and f_{EC} measures reconciliation inefficiency. The formula is much easier to read term by term than as one block:

- q says that only the chosen key-generating basis combinations are retained;
- $Q_\mu f_{\text{EC}} h_2(E_\mu)$ is the public-information cost of making Alice's and Bob's noisy strings agree;
- $Q_1[1 - h_2(e_1)]$ is the useful contribution associated with the characterized single-photon events.

The precise finite-key formula depends on the protocol and security proof, but the structure is physically transparent: public error correction costs key, while well-characterized single-photon events contribute secure key.

11.5 Losses, dark counts, and what a detector actually reports

A real photodetector does not simply implement the mathematical instruction “measure a qubit.” In many QKD receivers the device is a *threshold detector*: it distinguishes approximately between “no photons registered” and “one or more photons registered,” but it may not reliably count the exact photon number. Several additional laboratory outcomes are therefore possible. This is the same physical point that motivated POVMs in Chapter 5: the actual outcome alphabet of a detector can be larger than the two eigenvalues of an ideal qubit observable.

- **No click:** the pulse may have contained no photon, the photon may have been lost in the channel, or the detector may have failed to register it.
- **Dark count:** a detector may click even though no signal photon arrived.
- **Double click:** in a multi-photon event or in the presence of background light, both threshold detectors may click.
- **Dead time and efficiency mismatch:** after a detector clicks it may be temporarily unable to register another event; this recovery interval is its *dead time*. Two nominally equivalent detectors may also have different efficiencies that depend on time, polarization, or intensity. Such mismatches can become exploitable side channels.

These effects change both the observed gain and the QBER. Security proofs for imperfect devices must explicitly account for such deviations [37]. More importantly, if a security proof assumes ideal detectors while the hardware behaves differently, Eve may attack the discrepancy rather than the abstract protocol. This is why implementation security is a distinct subject.

11.6 Measurement-device-independent QKD

A *side channel* is information or control available through some physical feature that the idealized protocol forgot to model. For example, if a detector responds differently to slightly different arrival times, an attacker may exploit that timing dependence rather than attack the abstract qubit measurement itself. Detector side channels motivated measurement-device-independent QKD (MDI-QKD) [34].

The phrase *trust model* means the list of devices and behaviours that the security proof assumes are correctly characterized. MDI-QKD deliberately removes the measurement station from that trusted list. Alice and Bob each prepare quantum states and send them to an untrusted middle station, conventionally Charlie. Charlie performs a Bell-state measurement and publicly announces which events succeeded.

The remarkable point is that Charlie and his detectors need not be trusted. Even if Eve controls the entire measurement station, the protocol can still be secure provided Alice's and Bob's sources satisfy the assumed characterization. Conceptually, MDI-QKD is closely related

to a time-reversed entanglement-swapping experiment: the middle Bell measurement establishes correlations between data held at the two ends.

MDI-QKD does not remove every device assumption. It specifically removes measurement-device vulnerabilities from the trust model; source flaws must still be controlled or incorporated into the proof.

11.7 Device-independent QKD

Device-independent QKD goes further. The aim is to certify secrecy directly from observed loophole-free Bell correlations, without trusting a detailed model of the internal measurement devices. “Device independent” does *not* mean “assumption free.” The users still need assumptions such as secure laboratories, sufficiently independent setting choices, correct identification of which systems belong to Alice and Bob, and no uncontrolled classical information leaking out of the laboratories. What is removed is the need to trust a detailed internal quantum model of the measurement boxes.

The CHSH structure derived once and for all in Chapter 4 now acquires an operational role: a sufficiently strong observed violation constrains any adversary compatible with the protocol assumptions and therefore bounds her predictive power. The Bell algebra is not new here; the security interpretation is.

This is conceptually beautiful but experimentally demanding. Detection losses, locality and memory effects, finite statistics, random-number quality, and unwanted information leakage must all be controlled. Device-independent QKD is therefore an advanced topic rather than the simplest laboratory route to QKD.

11.8 Finite keys, quantum side information, and composable security

The Shannon quantities introduced in Chapter 10 were sufficient for the simple intercept–resend example because Eve’s relevant information there was an ordinary classical record. A security proof must be more pessimistic. Eve may entangle an ancilla with the transmitted signals, keep that quantum system coherent, wait until all public communication is complete, and only then choose her optimal measurement. Finite experiments introduce a second complication: Alice and Bob infer channel properties from samples rather than from infinitely many trials.

The right entropy notion must now answer a more operational question than ordinary Shannon entropy: *after exploiting all of her quantum side information, how well could Eve optimally guess Alice’s string?* Conditional min-entropy is closely tied to this optimal guessing probability. Roughly, larger $H_{\min}(X|E)$ means that even the best measurement on Eve’s quantum system leaves her with less ability to predict X . The adjective *smooth* allows one to ignore events whose total probability is extremely small, which is essential for robust finite-size statements.

The second important word is *composable*. A key is composablely secure if its security guarantee remains valid when the key is later used as one component inside a larger cryptographic task. One does not want a key that is “secure in isolation” but becomes unsafe merely because it is plugged into an encryption or authentication protocol.

A modern composable statement is therefore expressed in terms of conditional smooth min-entropy. Schematically, if X is Alice’s reconciled string and E denotes Eve’s quantum side information, a privacy-amplified final length is chosen approximately as

$$\ell \lesssim H_{\min}^{\varepsilon}(X|E) - \text{leak}_{\text{EC}} - 2 \log_2 \frac{1}{\varepsilon_{\text{PA}}}, \quad (529)$$

where the ε parameters quantify tolerated failure probabilities. The precise theorem requires the smooth-entropy formalism and a careful finite-size analysis, which belong to a dedicated quantum-information course. For these notes the conceptual content is more important than the technical machinery:

privacy amplification converts a quantitative lower bound on Eve’s uncertainty about the reconciled data—including quantum side information and finite-size uncertainty—into a shorter key that is close to uniform and independent of Eve in the composable sense.

This is the rigorous descendant of the simple Chapter 10 intuition that “compressing what Eve only partly knows can produce something she essentially does not know.”

11.9 Satellite QKD

The free-space links of Chapter 9 can be used to distribute keys over distances for which direct fibre transmission becomes extremely lossy.

In a *trusted-node* architecture, a satellite separately establishes a key K_A with Alice and a key K_B with Bob. The satellite can publicly broadcast

$$K_A \oplus K_B. \quad (530)$$

Alice, who knows K_A , can recover K_B , and Bob can similarly recover K_A . An outside listener who knows neither key learns neither one from the XOR alone. The satellite itself, however, knows both keys. End-to-end security therefore requires trusting the satellite.

An entanglement-distribution satellite can instead send one photon of each entangled pair toward Alice and the other toward Bob. The users can then run an entanglement-based protocol. Such an architecture can reduce the trust placed in the satellite, but the coincidence rate is harder to maintain because both lossy optical links must succeed in the same entangled-pair trial.

Satellite QKD does not abolish engineering limitations. Clouds, daylight background, pointing error, finite orbital-pass duration, detector noise, and finite-key statistics all remain relevant.

11.10 What QKD security does and does not mean

A mathematically precise security statement is conditional on explicitly stated assumptions. It is therefore better to avoid the casual word “unhackable.”

- QKD does not prevent denial of service. Eve can always block the channel and stop key generation.
- QKD does not remove the need for authentication.
- Security of an ideal protocol does not automatically imply security of a particular source, modulator, detector, clock, or control circuit.
- QKD normally distributes a classical key; it does not normally send the secret message itself as a quantum state.
- The final key is shorter than the raw data because test bits, error-correction leakage, and privacy amplification all consume data.

The achievement is nevertheless profound. Alice and Bob do not have to infer secrecy merely from the assumed computational difficulty of a mathematical problem. They can use experimentally observed quantum statistics to place a quantitative upper bound on what an adversary could know.

11.11 The conceptual chain

Across Chapters 9 and 10, the logic of QKD can now be compressed into one sequence:

$$\begin{array}{l}
 \text{non-orthogonal quantum encoding} \longrightarrow \text{measurement disturbance and no-cloning} \\
 \longrightarrow \text{observable error statistics} \longrightarrow \text{bound Eve's information} \\
 \longrightarrow \text{error correction + privacy amplification} \longrightarrow \text{shared secret classical key.}
 \end{array} \quad (531)$$

The physical optics and the information theory are two descriptions of the same protocol. A wave plate or modulator prepares a controlled direction in a two-dimensional Hilbert space; the detector apparatus implements the relevant projective measurement or, for realistic outcomes, a POVM; incompatible encodings prevent an adversary from extracting arbitrary information without consequences; and classical post-processing converts experimentally bounded uncertainty into a quantitatively secure cryptographic resource. Bell correlations, when the protocol uses them, enter through Chapter 4 rather than through a second foundational derivation here.

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